

On the Cardinality of the Walsh Support of a Boolean Function

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Abstract. We provide a complete characterization of the possible cardinalities of Walsh supports of Boolean functions. Our approach begins with a detailed study of Siegenthaler’s construction and its properties, which allow us to derive relations between admissible support sizes in successive numbers of variables. We then introduce new notions such as Walsh space, reduction, and equivalence on supports, which form the structural framework of our analysis. For $n = 6$, we perform an experimental enumeration of affine-equivalence classes, and we analyze the geometric structure of supports of small cardinalities, proving uniqueness for sizes 10 and 13 and obtaining partial results for size 16. By combining these findings with a sieving method, we rule out twelve impossible cardinalities and establish constructive methods that transform a support of size s into one of size $4s + r$ for different values of r , sufficient to obtain every admissible cardinality for $n \geq 7$. As a consequence, we provide a complete characterization and resolve several open problems.

Keywords: Boolean Functions, Walsh Transform.

1 Introduction

Functions from $\mathbb{F}_2^n$ to $\mathbb{F}_2$ are called Boolean functions and are central to cryptography, coding theory, and several branches of discrete mathematics. In this extended abstract, we focus on the Walsh transform of a Boolean function, where the Walsh transform of the Boolean function f at the point $a \in \mathbb{F}_2^n$ is defined as $W_f(a) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) \oplus a \cdot x}$ where $a \cdot x$ denotes the usual inner product in $\mathbb{F}_2^n$. More precisely, we focus on the support of the Walsh transform of a function f $\text{Wsupp}_f = \{a \in \mathbb{F}_2^n \mid W_f(a) \neq 0\}$, and in particular the attainable cardinalities of that set. For an integer $n \geq 1$ we denote by $\mathcal{C}_n \subset \mathbb{N}$ the set of cardinalities of the Walsh support of Boolean functions in n variables.

The Walsh transform is the principal analytic tool for evaluating cryptographic indicators of Boolean functions. Specifically, the nonlinearity of f is given by $2^{n-1} - \frac{1}{2} \max_{a \in \mathbb{F}_2^n} |W_f(a)|$, while the order of resilience equals the largest integer m for which $W_f(a) = 0$ whenever $1 \leq w_H(a) \leq m$, where w_H denotes the Hamming weight. Beyond these global parameters, the geometry of the Walsh support itself has recently been exploited in the characterization of plateaued functions [HPWZ19] and of spectra with only a few distinct values [HHP21, WF22].

The Walsh support encodes resilience properties and provides insight into whether affine-equivalent functions may exhibit improved resilience. A motivation to study this support arises from [DMR23], which investigates the existence and significance of Boolean functions combining optimal algebraic immunity with high resilience. On the negative side, the structure of the Walsh support is used in [DMR23] to rule out the existence of 1-resilient functions within the affine equivalence class of majority functions in an odd number of variables. On the positive side, a deeper understanding of the Walsh support may offer a foundation for constructing functions that simultaneously achieve optimal algebraic immunity and high resilience. These parameters are critical for instantiating Goldreich’s PseudoRandom Generator [Gol00] (PRG) within the class NC^0 , as explored in *e.g.* [AL16, CDM⁺18, YGJL22, Üna23]. The existence of a secure PRG in NC^{01} has far-reaching implications in cryptology, including the construction of indistinguishability obfuscation [JLS21].

¹ Nick’s class 0, NC^0 is the class of decision problems decidable by uniform Boolean circuits of constant depth and polynomial size where each gate has bounded fan-in $[a, b]$.

Despite its importance, the structure of the Walsh support remains poorly understood; even the set of its possible cardinalities is not fully determined. To the best of our knowledge, the first systematic study was conducted by Pei and Qin [PQ00], who showed that a Walsh support cannot have cardinality 2, 3, 5, 6, or 7, and fully characterized the cases of cardinalities 4 and 8. In 2004, Carlet and Mesnager [CM05] carried out a more in-depth investigation. Among their results, they enumerated all possible Walsh supports for functions in five variables, and studied in particular those with support $\mathbb{F}_2^n \setminus \{0\}$. They proved that no such support arises for $n \leq 5$, but provided constructions for $n \geq 10$. More recently, Lou and Wang [LW24] resolved the remaining open cases for cardinality $2^n - 1$: they constructed examples for $n \in \{7, 8, 9\}$, and proved by exhaustive search that no such function exists for $n = 6$. They concluded by the open question of the existence of Walsh supports of size $2^n - 2$. Regarding all cardinalities, in 2008, Preneel and Logachev proposed several open questions on Boolean functions including one on the cardinality of the Walsh support [PL08] Problem C3): If $s \in [1, 2^n]$, is there a function $f \in \mathcal{B}_n$ such that $|\text{Wsupp}_f| = s$?

From the enumeration of [CM05], $\mathcal{C}_5 = \{1, 4, 8, 10, 13, 16, 18, 20, 21, 22, 23, 24, 25, 26, 28, 32\}$. Experimental evidence shows that certain elements of $[1, 2^5]$ which do not belong to $\mathcal{C}_5$ appear in $\mathcal{C}_6$ (such as 27, 29, 30, 31), while some elements of $[2^5 + 1, 2^6]$ are absent from $\mathcal{C}_6$ (such as 63). These observations raise two questions: (1) Are there always missing elements in the interval $[2^{n-1} + 1, 2^n]$ for each n ? (2) Which elements not in $\mathcal{C}_n$ eventually belong to some $\mathcal{C}_m$ with $m > n$?

In this work, we answer all the questions mentioned above, showing that for all $n \geq 7$, the set $\mathcal{C}_7$ can be completely characterized as:

$$\mathcal{C}_n = [1, 2^n] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\}.$$

Our full characterization of the possible cardinalities of the Walsh support is performed in various steps:

- Section 2: We recall the necessary preliminaries on Boolean functions and Walsh supports, and introduce the notation and definitions used throughout the paper.
- Section 3: We study the possible sizes of Walsh supports through Siegenthaler’s construction. Using the universality of this approach, we establish connections between the admissible cardinalities in n variables and those in $n + 1$ variables.
- Section 4: We develop new tools and concepts for analyzing Walsh supports, including the notion of Walsh space, a reduction method, and an equivalence relation on supports. We also introduce technical results such as a lemma on plateaued functions. These provide the structural and technical foundation for the remainder of the paper.
- Section 5: We present an experimental study for $n = 6$, enumerating the number of affine-equivalence classes for each support size.
- Section 6: We characterize in detail the geometric structure of Walsh supports of small cardinalities. In particular, we establish uniqueness results for supports of size 10 and 13, and we give partial results for size 16. Using these findings, we systematically derive the admissible small cardinalities and, by combining several arguments, we establish the impossibility of constructing Walsh supports of sizes in the set $2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19$.
- Section 7: We address the global question of determining all possible Walsh support cardinalities. We propose constructions that map a support of size s to one of size $4s + r$ for several r , and we prove that these suffice to generate all admissible cardinalities for $n \geq 7$. This resolves the problem of characterizing the sets $\mathcal{C}_n$ for all dimensions.
- Section 8: We conclude by summarizing our results, resolving open problems raised by Preneel and Logachev [PL08] and by Lou and Wang [LW24], and formulating new open questions for future research.

2 Preliminaries

We denote by $[1, n]$ the set of all integers from 1 to n , i.e. $\{1, \dots, n\}$. We distinguish the addition in $\mathbb{Z}$ and $\mathbb{F}_2$, $\sum$ and $+$ represent the sum in $\mathbb{Z}$ and $\bigoplus$ and $\oplus$ represent the sum in $\mathbb{F}_2$, i.e. the XOR addition. The only vectors considered are from $\mathbb{F}_2^n$, for simplicity we denote by $\sum$ and $+$ the sum of vectors in $\mathbb{F}_2^n$, i.e. the coordinate-wise XOR. The inner product between two vectors $a, b \in \mathbb{F}_2^n$ is denoted by $\cdot$, i.e. $a \cdot b = \bigoplus_{i=1}^n a_i b_i$. We denote by 0_n the vector $(0, \dots, 0)$ when there might be a confusion with the scalar 0.

In this work we consider vector and affine spaces of $\mathbb{F}_2$. Without any further precision, a *space* should be seen as an affine space. Two affine (resp. vector) spaces are said to be isomorphic if there exists an affine (resp. a linear) invertible transformation mapping them. We recall the concept of multiset, that is a set of couples $(x, m(x)) \in X \times \mathbb{N}$ where $m(x)$ is called the multiplicity. Informally, that is a set in which multiplicity matters. The multiset $M = \{(x, m(x)) \mid x \in X\}$ is denoted by $\{x \times m(x) \mid x \in X \text{ and } m(x) > 0\}$. If we consider another multiset $N = \{(x, n(y)) \mid y \in Y\}$ both M and N can be seen as multiset of $X \cup Y$, $\{(x, m(x)) \mid x \in X \cup Y\}$ and $\{(x, n(y)) \mid y \in X \cup Y\}$. Then, $M \cup N = \{(z, m(z) + n(z)) \mid z \in X \cup Y\}$, informally we add up the multiplicities.

2.1 General Definitions on Boolean Functions

We recall basic definitions on Boolean functions and their cryptographic parameters, we refer to e.g. [Car21] for more details.

Definition 1 (Boolean Function). A Boolean function f in n variables (an n -variable Boolean function) is a function from $\mathbb{F}_2^n$ to $\mathbb{F}_2$. The set of all Boolean functions in n variables is denoted by $\mathcal{B}_n$.

Definition 2 (Walsh Transform and Walsh Support). Let $f \in \mathcal{B}_n$ be a Boolean function, its Walsh transform W_f at $a \in \mathbb{F}_2^n$ is defined as:

$$W_f(a) = \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) \oplus a \cdot x}.$$

- The Walsh support is the set $\text{Wsupp}_f = \{a \in \mathbb{F}_2^n \mid W_f(a) \neq 0\}$,
- The absolute Walsh spectrum is the multiset $\text{AWspec}_f = \{|W_f(a)| \mid a \in \mathbb{F}_2^n\}$.

A vector $a \in \text{Wsupp}_f$ is said to be w valued for f if $|W_f(a)| = w$.

Our study mainly focuses on the cardinality of the Walsh supports. To ease some statements formulation, we introduce the set $\mathcal{B}_{n,s}$ of Boolean functions of n variables such that $|\text{Wsupp}_f| = s$. We also define the set of the achievable cardinalities of the Walsh support of n -variable functions, that is the set of s 's such that $\mathcal{B}_{n,s} \neq \emptyset$:

Definition 3 (Possible Walsh Cardinals).

Let $n \in \mathbb{N}^*$, we denote by $\mathcal{C}_n$ the subset of $\mathbb{N}$ of cardinals the Walsh support of an n -variable Boolean can take:

$$\mathcal{C}_n = \{|\text{Wsupp}_f| \mid f \in \mathcal{B}_n\}.$$

Additionally, for $s \in \mathbb{N}$ we denote by $\mathcal{B}_{n,s}$ the set $\{f \in \mathcal{B}_n \mid |\text{Wsupp}_f| = s\}$.

We recall the notion of extended affine equivalent Boolean functions. Various cryptographic properties are invariant in the affine equivalent class, we will see it is also the case for some properties of the Walsh support.

Definition 4 (Extended Affine Equivalence). Let $f, g \in \mathcal{B}_n$, f and g are called *Extended Affine (EA) equivalent* if there exist a linear automorphism L of $\mathbb{F}_2^n$, $a, b \in \mathbb{F}_2^n$ and $c \in \mathbb{F}_2$ such that $g(x) = f(a + L(x)) \oplus b \cdot x \oplus c$.

Property 1 ([CM05] Section 3.1). Let $f, g \in \mathcal{B}_n$ such that $g(x) = f(a + L(x)) \oplus b \cdot x \oplus c$, then we have $\text{Wsupp}_g = b + L^T(\text{Wsupp}_f)$, and in particular for all $u \in \mathbb{F}_2^n$:

$$W_g(u) = (-1)^{a \cdot L^{T^{-1}}(u+b) \oplus c} W_f(L^{T^{-1}}(u+b))$$

where L^T denotes the adjoint operator of L , that is the unique automorphism of $\mathbb{F}_2^n$ such that for all $x, y \in \mathbb{F}_2^n$, we have $x \cdot L(y) = L^T(x) \cdot y$.

Proof.

$$\begin{aligned} W_g(u) &= \sum_{x \in \mathbb{F}_2^n} (-1)^{f(a+L(x)) \oplus b \cdot x \oplus c \oplus u \cdot x} = (-1)^c \sum_{x \in \mathbb{F}_2^n} (-1)^{f(L(x+L^{-1}(a))) \oplus (u+b) \cdot x} \\ &= (-1)^c \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) \oplus (u+b) \cdot L^{-1}(x+a)} = (-1)^{a \cdot L^{T^{-1}}(u+b) \oplus c} \sum_{x \in \mathbb{F}_2^n} (-1)^{f(x) \oplus L^{T^{-1}}(u+b) \cdot x} \\ &= (-1)^{a \cdot L^{T^{-1}}(u+b) \oplus c} W_f(L^{T^{-1}}(u+b)). \end{aligned}$$

□

Remark 1. In [CM05] (Section 3.1), this formula has been used to show that for a Walsh support Wsupp_f there exists a function g such that $\text{Wsupp}_g = d + M(\text{Wsupp}_f)$ for any affine automorphism M . Indeed, applying Property 1 with $a = 0$, $c = 0$, $b = d$ and $L = M^T$, we obtain $g(x) = f(M^T(x)) \oplus d \cdot x$, hence $W_g(u) = W_f(M^{-1}(u+d))$ and thus $\text{Wsupp}_g = d + M(\text{Wsupp}_f)$.

Remark 2 (On the Adjoint Operator). If L is a linear map from $\mathbb{F}_2^m$ to $\mathbb{F}_2^n$, one can define for a given basis B of $\mathbb{F}_2^m$ the associated matrix M_L^B of L . For a fixed basis, $L \mapsto M_L^B$ is known to be an isomorphism from the space of linear maps to space matrices of size $m \times n$. For linear maps on $\mathbb{F}_2$ -vector spaces, L^T is defined such that: $M_{L^T}^B = (M_L^B)^T$, where T is the transpose operator for the matrices. We choose B to be the canonical basis of $\mathbb{F}_2^m$. Since $L(x) = M_L^B \cdot x$ where $\cdot$ is the matrix-to-vector multiplication, one can observe that for all $x \in \mathbb{F}_2^m$ and $y \in \mathbb{F}_2^n$, we have:

$$L(x) \cdot y = (M_L^B \cdot x) \cdot y = x \cdot ((M_L^B)^T \cdot y) = x \cdot L^T(y).$$

We finish this part on general definitions by mentioning the plateaued functions. These functions have been thoroughly studied over the last two decades due to the particularity of their Walsh spectrum (e.g. [ZZ99, CP03, Mes14]). In the article we will show a result on plateaued functions, and rely on the fact that some functions are plateaued to prove one of the main results.

Definition 5 (Plateaued Functions). A function $f \in \mathcal{B}_n$ is said to be *plateaued* if there exists $\lambda \in \mathbb{N}$ and $s \in \mathbb{N}$ such that $\text{AWspec}_f = \{0 \times (2^n - s), \lambda \times s\}$. Additionally, λ and s are necessarily powers of two.

2.2 Relations on the Walsh Transform

Many useful relations involving the Walsh transform can be used to retrieve properties on Wsupp_f or AWspec_f . We present a fraction of them largely useful for this article's proofs.

Property 2 (Inversion Formula, e.g. [Car21] Corollary 4). *Let $f \in \mathcal{B}_n$, then*

$$\forall x \in \mathbb{F}_2^n, (-1)^{f(x)} = \frac{1}{2^n} \sum_{a \in \mathbb{F}_2^n} W_f(a) (-1)^{a \cdot x}. \quad (1)$$

In particular, for $x = 0$ we have $\sum_{a \in \mathbb{F}_2^n} W_f(a) = (-1)^{f(0)} 2^n$.

Property 3 (Parseval's Formula, e.g. [Car21] Corollary 5). *Let $f \in \mathcal{B}_n$, then*

$$\sum_{u \in \mathbb{F}_2^n} W_f(u)^2 = 2^{2n}.$$

Property 4 (Poisson's Formula, e.g. [Car21] Corollary 3). *Let $f \in \mathcal{B}_n$, $E \subset \mathbb{F}_2^n$ a vector space and $d, e \in \mathbb{F}_2^n$, then*

$$\sum_{u \in d+E} W_f(-1)^{u \cdot e} = (-1)^{d \cdot e} |E| \sum_{x \in e+E^\perp} (-1)^{x \cdot d}.$$

2.3 Previous Results on Walsh Support Cardinalities

For $s \leq 8$, the Walsh supports have either been fully characterized or shown to be impossible to construct. First, the function of $\mathcal{B}_{n,1}$ are well-known, they form the class of affine functions. Then, we recall the other known results in the following properties.

Property 5 (Walsh support of Affine Functions). *Let $f \in \mathcal{B}_{n,1}$ with $\text{Wsupp}_f = \{a\}$, then there exists $b \in \mathbb{F}_2$ such that $f(x) = a \cdot x + b$ and $\text{AWspec}_f = \{0 \times (2^n - 1), 2^n \times 1\}$.*

In [PQ00] the Walsh supports of cardinality 4 or 8 have been characterized. They have also shown that it is not possible to construct a Walsh support having a cardinality $s \in \{2, 3, 5, 6, 7\}$. For completeness, we give different proofs in Appendix A.3.

Property 6 ([PQ00] Theorem 1). *Let $f \in \mathcal{B}_{n,s}$, then*

- *If $s = 4$, then Wsupp_f is an affine space of dimension 2. Reciprocally, if H is an affine space of dimension 2, there exists $f' \in \mathcal{B}_{n,4}$ such that $H = \text{Wsupp}_{f'}$.
Moreover $\text{AWspec}_f = \{0 \times (2^n - 4), 2^{n-1} \times 4\}$,*
- *If $s = 8$, then Wsupp_f is an affine space of dimension 3. Reciprocally, if H is an affine space of dimension 3, there exists $f' \in \mathcal{B}_{n,8}$ such that $H = \text{Wsupp}_{f'}$.
Moreover $\text{AWspec}_f = \{0 \times (2^n - 8), 2^{n-2} \times 7, (3 \cdot 2^{n-2}) \times 1\}$.*

Property 7 ([PQ00] Theorem 1). *Let $s \in \{2, 3, 5, 6, 7\}$, then for all $n \in \mathbb{N}$ we have $\mathcal{B}_{n,s} = \emptyset$.*

For n small, the enumeration of all EA equivalent classes of $\mathcal{B}_n$ is feasible. From these classifications, one can determine $\mathcal{C}_n$, it has been done until $n \leq 5$ [CM05].

Property 8. *The following holds for the first sets $\mathcal{C}_n$:*

$$\mathcal{C}_1 = \{1\}, \quad \mathcal{C}_2 = \{1, 4\}, \quad \mathcal{C}_3 = \{1, 4, 8\}, \quad \mathcal{C}_5 = \{1, 4, 8, 10, 16\},$$

$$\mathcal{C}_5 = [1, 2^5] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19, 27, 29, 30, 31\}.$$

2.4 Siegenthaler Construction

We recall the secondary construction of Boolean functions referred as Siegenthaler's construction. This classic construction allows us to build an $n + 1$ variable function from two n -variable functions.

Construction 1 (Siegenthaler's Construction *e.g.* [Sie84]). *Given $f, g \in \mathcal{B}_n$, we define $\text{Sieg}[f, g] \in \mathcal{B}_{n+1}$ as:*

$$\forall (x, y) \in \mathbb{F}_2^n \times \mathbb{F}_2, \text{Sieg}[f, g](x, y) = (1 \oplus y)f(x) \oplus yg(x).$$

Remark 3. This construction is also known as the *concatenation* construction. If we consider the truth tables tt_f and tt_g of f and g respectively, then $\text{Sieg}[f, g]$ is simply the function defined by the truth table $\text{tt}_f | \text{tt}_g$ where $|$ denotes the concatenation.

If $h \in \mathcal{B}_{n+1}$ is defined as a Siegenthaler's construction, we shall usually consider vectors of $\mathbb{F}_2^{n+1}$ as elements of $\mathbb{F}_2^n \times \mathbb{F}_2$. In particular, if no precision is given $(u, v) \in \mathbb{F}_2^{n+1}$ should be understood as $(u, v) \in \mathbb{F}_2^{n+1}$ with $u \in \mathbb{F}_2^n$ and $v \in \mathbb{F}_2$.

Property 9. *Let $f, g \in \mathcal{B}_n$ and denote $h = \text{Sieg}[f, g]$, then:*

$$\forall (u, v) \in \mathbb{F}_2^n \times \mathbb{F}_2, W_h(u, v) = W_f(u) + (-1)^v W_g(u). \quad (2)$$

Since any function $h \in \mathcal{B}_n$ can be seen as Siegenthaler's construction of functions from $\mathcal{B}_{n-1}$ by identifying one variable, Property 9 can be used to study the Walsh spectrum of h using the Walsh support structures of its subfunctions.

Property 10 (Inverse Siegenthaler's Construction). *Let $f \in \mathcal{B}_{n+1}$, let $f_0, f_1 \in \mathcal{B}_n$ defined as $f_0(x) = f(x, 0)$ and $f_1 = f(x, 1)$. Then, for all $u \in \mathbb{F}_2^n$:*

$$W_{f_0}(u) = \frac{1}{2} \left[W_f(u, 0) + W_f(u, 1) \right] \quad \text{and} \quad W_{f_1}(u) = \frac{1}{2} \left[W_f(u, 0) - W_f(u, 1) \right].$$

3 More on Siegenthaler's Construction

In this section we present preliminary results on Siegenthaler's construction, more precisely on the relation it implies on the Walsh supports' sizes of the functions f, g and h . The notations and definitions we introduce in this part will be reused in all the article. The preliminary results are the basis for the machinery both to discard cardinals (Section 6 and determine all the possible ones (Section 7).

We will study for any function $f, g \in \mathcal{B}_n$ the following sets:

$$K(f, g) = \text{Wsupp}_f \cap \text{Wsupp}_g \quad \text{and} \quad \Xi(f, g) = \{a \in K(f, g) \mid |W_f(a)| = |W_g(a)|\}.$$

These definitions are motivated by Property 9. Indeed, if one has information on $K(f, g)$ and $\Xi(f, g)$, then they can deduce information about Wsupp_h and AWspec_h where $h = \text{Sieg}[f, g]$.

Theorem 1. *Let $f \in \mathcal{B}_{n,s}$, $g \in \mathcal{B}_{n,t}$ and $h = \text{Sieg}[f, g]$. We denote by $k = |K(f, g)|$ and $\xi = |\Xi(f, g)|$, then we have:*

$$|\text{Wsupp}_h| = 2(s + t) - 2k - \xi.$$

We say that h is an (s, t, k, ξ) construction.

Proof. We recall that $W_h(u, v) = W_f(u) + (-1)^v W_g(u)$ according to Property 9, thus we deduce that:

$$\forall (u, v) \in \mathbb{F}_2^n \times \mathbb{F}_2, W_h(u, v) = \begin{cases} W_f(u) + (-1)^v W_g(u) & \text{if } u \in K(f, g), \\ W_f(u) & \text{if } u \in \text{Wsupp}_f \setminus K(f, g), \\ (-1)^v W_g(u) & \text{if } u \in \text{Wsupp}_g \setminus K(f, g), \\ 0 & \text{Otherwise.} \end{cases}$$

From lines 2 and 3, we have $2(s - k)$ and $2(t - k)$ support vectors of W_h . We have ξ elements u of the intersection having the same absolute Walsh transform value. Then, for each one of these elements there exists v_u such that $W_h(u, v_u) \neq 0$ and $W_h(u, 1 + v_u) = 0$. Hence, we can count ξ vectors in the Walsh support.

The other elements u of the intersection will be such that $W_h(u, 0) \neq 0$ and $W_h(u, 1) \neq 0$. Hence, we count $2(k - \xi)$ vectors in the support. All the other elements are not in the Walsh support. Hence,

$$|\text{Wsupp}_h| = 2(s - k) + 2(t - k) + \xi + 2(k - \xi) = 2(s + t) - 2k - \xi.$$

□

Remark 4. This kind of approach has already been considered but not formalized as it is done here. In [LW24] (Section III), Lou and Wang have constructed functions such that $s = t = 2^6$ with $k = 2^6$ and $\xi = 1$ to obtain functions having Walsh support of cardinality $2(2^6 + 2^6) - 2 \cdot 2^6 - 1 = 2^7 - 1$.

In the article, we will denote by k the quantity $|K(f, g)|$ and by ξ the quantity $|\Xi(f, g)|$ when there will be no ambiguity on f and g to ease the notations.

To show that a size r is not achievable for a Walsh support, we may first look at the equation $r = 2(s + t) - 2k - \xi$ and find constraints on s, t, k and ξ to eliminate possible values of r , or at least eliminate possible values of r given s and t . In Section 6, we will establish constraints for precise cases, with s and t fixed. The following results give mainly general inequalities between s, t, k and ξ :

Corollary 1. *Let $h \in \mathcal{B}_{n,r}$ such that $h = \text{Sieg}[f, g]$ with $f \in \mathcal{B}_{n-1,s}$, $g \in \mathcal{B}_{n-1,t}$ and $s \leq t$. Then, $\xi \leq k \leq s \leq t \leq r \leq 2(s + t)$. Moreover, $s \geq 2t - r$.*

Proof. According to Theorem 1, we have $r = 2(s + t) - 2k - \xi$ and thus $r \leq 2(s + t)$. By definition, $\Xi(f, g) \subset K(f, g) \subset \text{Wsupp}_f$, thus $\xi \leq k \leq s \leq t$. We deduce that $2(s + t) - 2s - s \leq 2(s + t) - 2k - \xi = r$. Thus, $s \leq r$. We also have $t = \frac{r + \xi}{2} + k - s \leq \frac{r + \xi}{2}$. Since $\xi \leq s \leq r$, we deduce that $t \leq r$. Finally, we have $r = 2(s + t) - 2k - \xi \geq 2(s + t) - 3s = 2t - s$ and thus $s \geq 2t - r$. □

Remark 5. Therefore, one way to construct Walsh support of a given size r is to search for the 4-tuples (s, t, k, ξ) already verifying the inequalities of Corollary 1 and more constraints derived from the Walsh supports' properties.

There are simple examples for the two extremes cases of Corollary 1. Taking $f = g$ gives a case where $\Xi(f, g) = K(f, g) = \text{Wsupp}_f$ and thus $h \in \mathcal{B}_{n+1,s}$. Taking f and g such that $K(f, g) = \emptyset$ is an example of the other extreme case, $h \in \mathcal{B}_{n+1,2(s+t)}$.

We provide a first lemma on the values ξ cannot take depending on the equality between s and t .

Lemma 1. *Let $f \in \mathcal{B}_{n,s}$ and $g \in \mathcal{B}_{n,t}$ such that $s \leq t$.*

- If $s = t = k$, then $\xi \neq k - 1$,
- If $s \neq t$, then we have $\xi \neq s$.

Proof. In the case $s = t = k$, we have $K(f, g) = \text{Wsupp}_f = \text{Wsupp}_g$. We do the proof by contradiction, let assume that $\xi = k - 1$ and denote by a the unique element not in $\Xi(f, g)$. Using Parseval's relation we get (Property 3):

$$\sum_{u \in K(f, g)} W_f(u)^2 = \sum_{u \in K(f, g)} W_g(u)^2.$$

However, we have the following equalities:

$$\begin{aligned} \sum_{u \in K(f, g)} W_f(u)^2 &= \sum_{u \in \Xi(f, g)} W_f(u)^2 + W_f(a)^2 = \sum_{u \in \Xi(f, g)} W_g(u)^2 + W_f(a)^2 \\ &= \sum_{u \in K(f, g)} W_g(u)^2 + W_f(a)^2 - W_g(a)^2. \end{aligned}$$

Therefore, $W_f(a)^2 - W_g(a)^2 = 0$. We deduce that $a \in \Xi(f, g)$, hence $\xi \neq k - 1$.

In the other case, $s \neq t$, if $\xi = s$, then $k = s$ and $\text{Wsupp}_f \subset \text{Wsupp}_g$. Therefore, we deduce the following using Parseval's relation (Property 3).

$$\begin{aligned} \sum_{u \in \text{Wsupp}_g} W_g(u)^2 &= \sum_{u \in \text{Wsupp}_g \setminus \text{Wsupp}_f} W_g(u)^2 + \sum_{u \in \text{Wsupp}_f} W_g(u)^2 \\ &= \sum_{u \in \text{Wsupp}_g \setminus \text{Wsupp}_f} W_g(u)^2 + \sum_{u \in \text{Wsupp}_f} W_f(u)^2 \end{aligned}$$

We deduce that $\sum_{u \in \text{Wsupp}_g \setminus \text{Wsupp}_f} W_g(u)^2 = 0$. Therefore, we should have every term of the sum equal to zero, which is not possible. Therefore, $\xi \neq s$. $\square$

One may remark that we restrict the functions $f \in \mathcal{B}_{n,s}$ and $g \in \mathcal{B}_{n,t}$ to verify $s \leq t$. Actually, if we define $h = \text{Sieg}[f, g]$ and $h' = \text{Sieg}[g, f]$, we can notice that for all $(x, y) \in \mathbb{F}_2^{n+1}$, $h(x, y) = h'(x, y \oplus 1)$ and in particular h and h' are EA equivalent (Definition 4). Thus, according to Property 1, $\text{Wsupp}_h = \text{Wsupp}_{h'}$ and $\text{AWspec}_h = \text{AWspec}_{h'}$. Therefore, we can assume without loss of generality that $s \leq t$.

We introduce the sets of Walsh support sizes that can be built through Siegenthaler's construction using functions with Walsh support's size (upper) bounded. Using the fact than any function in $n > 1$ variables can be written as a Siegenthaler's construction, and that the Walsh support of h cannot be smaller than the one of f nor g (Corollary 1), these sets will allow us to discard some cardinalities for the Walsh supports, step by step.

Definition 6. We define the set $\mathcal{C}^{s,t}$ of all cardinalities obtained through Theorem 1 using $s, t \in \mathbb{N}$ fixed with $s \leq t$ and every possible k, ξ :

$$\mathcal{C}^{s,t} = \{|\text{Wsupp}_h| \mid \exists n \in \mathbb{N}, \exists f \in \mathcal{B}_{n,s}, \exists g \in \mathcal{B}_{n,t}, h = \text{Sieg}[f, g]\}.$$

We also define the set $\mathcal{C}^{\leq r}$ of all cardinalities obtained using $s, t \leq r$, i.e.

$$\mathcal{C}^{\leq r} = \bigcup_{1 \leq s \leq t \leq r} \mathcal{C}^{s,t}.$$

Corollary 1 already states that if $h \in \mathcal{B}_{n,r}$ is an (s, t, k, ξ) construction then we must have $\max\{s, t\} \leq r$. This can be restated using Definition 6 to say that if r cannot be obtained with (s, t, k, ξ) constructions for all $s, t \in [1, r - 1]$, then there is no function having a Walsh support of size r .

Corollary 2. Let $r > 1$, then for all $n \in \mathbb{N}$, there is no function $f \in \mathcal{B}_n$ such that $|\text{Wsupp}_f| = r$ if and only if $r \notin \mathcal{C}^{\leq r-1}$.

Proof. If $r \in \mathcal{C}^{\leq r-1}$ then by definition there exist $n \in \mathbb{N}$ and a function $f \in \mathcal{B}_{n,r}$. Then, we consider the case $r \notin \mathcal{C}^{\leq r-1}$. If there exists a $h \in \mathcal{B}_{n,r}$, we can assume without loss of generality that h is such that $h = \text{Sieg}[f, g]$ with $f \in \mathcal{B}_{n-1,s}$, $g \in \mathcal{B}_{n-1,t}$ and $s \leq t$. However, using Corollary 1, we have $t \leq r$. Therefore, the only possibility is to have $t = r$. However, using Theorem 1's notations, $2r + 2s - 2k - \xi = r$ if and only if $r = t = s = k = \xi$. If we ever have $t = r$, we apply the same reasoning with $h^{(1)} = f$ and repeat the process with the subfunctions.

If for $i < n - 1$, we have $t^{(i)} \neq r$, then $t^{(i)} \leq r - 1$ and $t^{(i)} \geq r$ since $r \notin \mathcal{C}^{\leq r-1}$. We end up with a contradiction, therefore for all $n \in \mathbb{N}$, there is no function $f \in \mathcal{B}_{n,r}$. Otherwise, we can continue until we reach a limited number of variables. We obtain a sequence $h^{(1)}, \dots, h^{(n-1)}$ of Boolean functions of decreasing number of variables such that $r^{(i)} = t^{(i)} = s^{(i)} = k^{(i)} = \xi^{(i)}$. $h^{(n-1)} \in \mathcal{B}_{1,r}$. Boolean function of 1 variable have only one Walsh support size possible, therefore $r = 1$ which is contradictory. Thus, the function h does not exist, i.e. $\mathcal{B}_{n,s} = \emptyset$ for all $n \in \mathbb{N}$. □

4 General Properties on the Walsh Support's Structure

As we have seen in Section 3, knowing properties on $K(f, g)$, that is the intersection of Wsupp_f and Wsupp_g , allows us to understand the construction of Walsh supports of greater cardinalities. Therefore, the main goal of this section is to provide tools and theoretical results to ease the study of the geometrical structures of the Walsh supports.

More precisely, in Section 4.1 we introduce the notion of Walsh space and a notion of equivalence on the Walsh support of Boolean functions. In Section 4.2 we define the notion of reduction of a Boolean function, that allows us to study the Walsh support of small cardinalities in spaces of low dimension, independently of n . We conclude this theoretical part by Section 4.3 where we provide a technical lemma on plateaued functions.

4.1 Equivalence of Geometrical Structures in Walsh Supports

The informal idea of this section is to extend the comparison of Walsh supports geometrical properties beyond extended affine equivalence. Indeed, if f and g are extended affine equivalent, then the Walsh support Wsupp_g is equal to $a + L(\text{Wsupp}_f)$ where $a \in \mathbb{F}_2^n$ and L is a linear automorphism. Consequently, Wsupp_f and Wsupp_g share many geometrical properties such as affine subspaces and parallelism between these subspaces. However, this requires for f and g to have the same number of variables. We begin by introducing the Walsh space which grasps the linear relations of a Walsh support.

Definition 7 (Walsh Space). Let $f \in \mathcal{B}_n$, the Walsh Space is the vector space defined for any $b \in \text{Wsupp}_f$,

$$\mathcal{WS}_f = \text{Span}(\{a + b \mid a \in \text{Wsupp}_f\}).$$

Remark 6. Without being named as such, the Walsh space has been used in [HPWZ19] in Theorem 3.4. Their result also considers the linear kernel of f , that is, the set Λ_f defined by $\Lambda_f = \{a \in \mathbb{F}_2^n, \exists b \in \mathbb{F}_2, \forall x \in \mathbb{F}_2^n, f(x) + f(x+a) = b\}$, which is known to be a vector subspace of $\mathbb{F}_2^n$. They derive the following relationship: $\dim(\mathcal{WS}_f) + \dim(\Lambda_f) = n$. Actually, we have $\mathcal{WS}_f^\perp = \Lambda_f$. Indeed, let $\text{Wsupp}_f \subset a + \mathcal{WS}_f$

with $a \in \text{Wsupp}_f$ and $y \in \mathcal{WS}_f^\perp$, then by Property 2:

$$\begin{aligned} (-1)^{f(x+y)} &= \frac{1}{2^n} \sum_{u \in \text{Wsupp}_f} W_f(u) (-1)^{(x+y) \cdot u} = \frac{1}{2^n} \sum_{u \in \text{Wsupp}_f} W_f(u) (-1)^{x \cdot u + y \cdot (a+a+u)} \\ &= \frac{1}{2^n} \sum_{u \in \text{Wsupp}_f} W_f(u) (-1)^{x \cdot u + y \cdot a} = (-1)^{y \cdot a} (-1)^{f(x)}. \end{aligned}$$

Therefore, $y \in \Lambda_f$. In particular, we observe that $\mathcal{WS}_f \subset \Lambda_f$ and that $\dim(\mathcal{WS}_f^\perp) = n - \dim(\mathcal{WS}_f) = \dim(\mathcal{WS}_f) + \dim(\Lambda_f) - \dim(\mathcal{WS}_f) = \dim(\Lambda_f)$. Thus $\Lambda_f = \mathcal{WS}_f^\perp$.

$\mathcal{WS}_f$ is the direction, *i.e.* the associated vector space to an affine space, of the affine space $\text{Aff}(\text{Wsupp}_f)$, that is the smallest (in terms of inclusion) affine space including Wsupp_f . A classic result in affine geometry is that for a set $S \subset \mathbb{F}_2^n$ the direction of $\text{Aff}(S)$, *i.e.* $\text{Span}(\{x + y \mid y \in S\})$ for $x \in S$, does not depend on the choice of $x \in S$. For the sake of completeness, we provide a proof that $\mathcal{WS}_f$ does not depend on the choice of the vector in Wsupp_f , and is thus well defined.

Property 11. *Let $f \in \mathcal{B}_n$, then $\mathcal{WS}_f$ is independent of the choice of $b \in \text{Wsupp}_f$, *i.e.* $\mathcal{WS}_f$ is well defined.*

Proof. Let $b, c \in \text{Wsupp}_f$. We define V_b as $\text{Span}(a + b \mid a \in \text{Wsupp}_f)$ and $V_c = \text{Span}(a + c \mid a \in \text{Wsupp}_f)$. Let $a \in \text{Wsupp}_f$, then by definition $a + b$ is in V_b . Moreover, $c \in \text{Wsupp}_f$, thus $c + b$ is also in V_b . Therefore, $(a + b) + (c + b) \in V_b$, *i.e.* $a + c \in V_b$. We have $\{a + c \mid a \in \text{Wsupp}_f\} \subset V_b$. Since V_c is by definition of the Span the minimal vector space including $\{a + c \mid a \in \text{Wsupp}_f\}$, we deduce that $V_c \subset V_b$.

We apply the same reasoning starting from V_c instead of V_b and thus $V_b \subset V_c$. Hence, $V_b = V_c$. $\square$

The Walsh space is essential to extend our way of comparing Walsh supports geometrical properties.

Definition 8 (Equivalence of Supports). *Let $f \in \mathcal{B}_n$ and $g \in \mathcal{B}_m$. Wsupp_f and Wsupp_g are said to be equivalent if there exist $a \in \text{Wsupp}_f$, $b \in \text{Wsupp}_g$ and a linear isomorphism $L : \mathcal{WS}_f \rightarrow \mathcal{WS}_g$ such that $L(a + \text{Wsupp}_f) = b + \text{Wsupp}_g$.*

Remark 7. We can restate this definition using an affine transformation. L is a linear isomorphism from $\mathcal{WS}_f$ to $\mathcal{WS}_g$. Let $A : a + \mathcal{WS}_f \rightarrow b + \mathcal{WS}_g$ be defined by $A(u) = b + L(a + u)$. A is a well-defined isomorphism from $a + \mathcal{WS}_f$ to $b + \mathcal{WS}_g$ and $A(\text{Wsupp}_f) = \text{Wsupp}_g$.

Remark 8. A sufficient condition for two Walsh supports Wsupp_f and Wsupp_g to verify an equivalence as in Definition 8 is to have f and g EA equivalent. We recall from Definition 4 that it means that there exist $L : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^m$ a linear automorphism, $\alpha, \beta \in \mathbb{F}_2^n$ and $\gamma \in \mathbb{F}_2^m$ such that: $g(x) = f(\alpha + L(x)) \oplus \beta \cdot x \oplus \gamma$. From Property 1, we deduce that $\text{Wsupp}_g = \beta + L^T(\text{Wsupp}_f)$. Let $a \in \text{Wsupp}_f$ and $b = \beta + L^T(a) \in \text{Wsupp}_g$. We have:

$$\beta + L^T(\text{Wsupp}_f) = \beta + L^T(a + a + \text{Wsupp}_f) = \beta + L^T(a) + L^T(\text{Wsupp}_f).$$

The main motivation of this definition is to study the geometrical representation of a Walsh support as in [CM05]. In particular, if any subspace F of $\mathbb{F}_2^n$ is included in Wsupp_f then there will exist a subspace $G = L(F)$ of $\mathbb{F}_2^m$ such that $G \subset \text{Wsupp}_g$.

Proposition 1. *Let $f \in \mathcal{B}_n$ and $g \in \mathcal{B}_m$ such that Wsupp_f and Wsupp_g are equivalent with $L(a + \text{Wsupp}_f) = b + \text{Wsupp}_g$ as in Definition 8, then if $U_1, \dots, U_r$ are pairwise disjoint affine subspaces of $\mathbb{F}_2^n$ such that*

$$\text{Wsupp}_g = \bigsqcup_{i=1}^r U_i,$$

then the $V_i = a + L^{-1}(b + U_i)$ are pairwise disjoint affine subspaces of $\mathbb{F}_2^n$ such that

$$\text{Wsupp}_f = \bigsqcup_{i=1}^r V_i$$

where each V_i is isomorphic to U_i . Moreover, parallelism is preserved, i.e. if U_i is parallel to U_j , then V_i is parallel to V_j .

Proof. Since Wsupp_f and Wsupp_g are equivalent by assumption, there exist $a \in \text{Wsupp}_f$, $b \in \text{Wsupp}_g$ and a linear isomorphism $L : \mathcal{WS}_f \rightarrow \mathcal{WS}_g$ such that L induces a bijection between $a + \text{Wsupp}_f$ and $b + \text{Wsupp}_g$ (Definition 8).

Let $A : b + \mathcal{WS}_g \rightarrow a + \mathcal{WS}_f$ defined by $A(x) = a + L^{-1}(b + x)$. A is well defined and is an affine automorphism from $b + \mathcal{WS}_g$ to $a + \mathcal{WS}_f$. Since $\text{Wsupp}_f \subset a + \mathcal{WS}_f$ and $\text{Wsupp}_g \subset b + \mathcal{WS}_g$, we deduce that A is well defined on Wsupp_g .

If Wsupp_g admits a decomposition into pairwise distinct affine subspaces U_i , i.e.

$$\text{Wsupp}_g = \bigsqcup_{i=1}^r U_i,$$

then we can define for each $i \in [1, r]$, $V_i = A(U_i)$. We show that:

1. Each $V_i \subset \mathbb{F}_2^n$ is an affine subspace isomorphic to U_i ,
2. The V_i 's are pairwise disjoint,
3. $\text{Wsupp}_f = \bigsqcup_{i=1}^r V_i$,
4. Parallelism is preserved between the U_i 's and V_i 's.

1. V_i is obtained through an affine isomorphism of U_i , thus V_i is an affine subspace of $\mathbb{F}_2^n$ isomorphic to U_i .

2. We reason by contradiction. Let $i \neq j$ and assume that $V_i \cap V_j \neq \emptyset$. Let $v \in V_i \cap V_j$, then there exist $u \in U_i$ such that $A(u) = v$. We also have $v \in V_j$, then $A(u) \in V_j$. Since A is bijective from U_j to V_j , we deduce that $u \in U_j$, thus $u \in U_i \cap U_j$. We reach a contradiction because $U_i \cap U_j = \emptyset$. Thus, $V_i \cap V_j = \emptyset$.

3. We derive the following:

$$\text{Wsupp}_f = A(\text{Wsupp}_g) = A\left(\bigsqcup_{i=1}^r U_i\right) = \bigsqcup_{i=1}^r A(U_i) = \bigsqcup_{i=1}^r V_i.$$

4. Affine isomorphisms are known to preserve parallelism. However for the sake of completeness we give a proof that parallelism is preserved. If U_i and U_j are parallel, i.e. there exists $x \in \mathbb{F}_2^m$ such that $U_i = x + U_j$, then we derive the following:

$$\begin{aligned} V_i &= A(U_i) = a + L^{-1}(b + U_i) = a + L^{-1}(b + x + U_j) = a + L^{-1}(x + \underbrace{b + U_j}_{\subset \mathcal{WS}_g}) \\ &= a + L^{-1}(x) + L^{-1}(b + U_j) = L^{-1}(x) + A(U_j) = L^{-1}(x) + V_j. \end{aligned}$$

Thus, we deduce that V_i and V_j are parallel. □

As we will mainly consider functions constructed using Siegenthaler's construction to use the results from Section 3, we show a simple bound on the dimension of the Walsh space of such a construction based on the Walsh spaces of its subfunctions.

Proposition 2. Let $f, g \in \mathcal{B}_n$ such that there exists $a \in \text{Wsupp}_f \cap \text{Wsupp}_g$. We define h as $h = \text{Sieg}[f, g]$, then

$$\dim(\mathcal{W}_h) \leq \dim(\mathcal{W}_f) + \dim(\mathcal{W}_g) - \dim(\mathcal{WS}_f \cap \mathcal{WS}_g) + 1$$

Proof. Using Property 9, we know that if $(a, b) \in \text{Wsupp}_h$ then $a \in \text{Wsupp}_f \cup \text{Wsupp}_g$ and $b \in \mathbb{F}_2$, that is

$$\text{Wsupp}_h \subset (\text{Wsupp}_f \cup \text{Wsupp}_g) \times \mathbb{F}_2.$$

In particular, there exists $b \in \mathbb{F}_2$ such that $u = (a, b) \in \text{Wsupp}_h$. Thus, we have

$$u + \text{Wsupp}_h \subset [(a + \text{Wsupp}_f) \cup (a + \text{Wsupp}_g)] \times \mathbb{F}_2.$$

By Definition 7, $\mathcal{WS}_f = a + \text{Wsupp}_f$, $\mathcal{WS}_g = a + \text{Wsupp}_g$ and $\mathcal{WS}_h = \text{Span}(u + \text{Wsupp}_h)$. Hence, we have $\text{Span}(u + \text{Wsupp}_h) \subset \text{Span}((a + \text{Wsupp}_f) \cup (a + \text{Wsupp}_g)) \times \mathbb{F}_2$, i.e.

$$\mathcal{WS}_h \subset (\mathcal{WS}_f + \mathcal{WS}_g) \times \mathbb{F}_2.$$

Thus, we deduce the inequality on the dimension $\dim(\mathcal{WS}_h) \leq \dim(\mathcal{WS}_f + \mathcal{WS}_g) + 1$ which leads to our claim by the formula $\dim(\mathcal{WS}_f + \mathcal{WS}_g) = \dim(\mathcal{W}_f) + \dim(\mathcal{W}_g) - \dim(\mathcal{WS}_f \cap \mathcal{WS}_g)$. $\square$

4.2 Reduction of Boolean Functions

We introduce the notions of reduction of Boolean functions and equivalence of Walsh supports. The informal idea is to extend the relation of extended affine equivalence (Definition 4) in order to preserve somehow the Walsh spectrum (up-to a constant factor) and the Walsh supports geometrical properties, i.e. parallelism and linear structures. The main difference is to consider injective transformations and not automorphism and to ensure that the bijection between the two Walsh supports is preserved.

Definition 9 (Reduction). Let $f \in \mathcal{B}_n$ and $g \in \mathcal{B}_m$ such that $m \leq n$. g is said to be a reduction of f if there exist a linear transformation $L : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$, $\alpha \in \mathbb{F}_2^n$, $\beta \in \mathbb{F}_2^m$ and $\gamma \in \mathbb{F}_2$ such that

1. For all $x \in \mathbb{F}_2^m$, $g(x) = f(\alpha + L(x)) + \beta \cdot x + \gamma$,
2. L is injective and denote $L' : \mathbb{F}_2^m \rightarrow L(\mathbb{F}_2^m)$ the induced linear isomorphism of L ,
3. L'^T induces a bijection between $a + \text{Wsupp}_f$ and $b + \text{Wsupp}_g$ where $a \in \text{Wsupp}_f$ and $b \in \text{Wsupp}_g$.

To ease the following proofs, we begin by deriving properties on the reduction.

Proposition 3. Let $f \in \mathcal{B}_n$, $g \in \mathcal{B}_m$ and $h \in \mathcal{B}_\ell$.

- If g is a reduction of f and h is a reduction of g , then h is a reduction of f , i.e. the reduction is transitive.
- If f and g are EA-equivalent (Definition 4), then f is a reduction of g , and vice versa.

Proof. We start with the first statement. If g is a reduction of f and h is a reduction of g , by Definition 9 there exist: $L : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$, $\alpha \in \mathbb{F}_2^n$, $\beta \in \mathbb{F}_2^m$ and $\gamma \in \mathbb{F}_2$ such that

$$\forall x \in \mathbb{F}_2^m, g(x) = f(\alpha + L(x)) + \beta \cdot x + \gamma,$$

and similarly there exist: $M : \mathbb{F}_2^\ell \rightarrow \mathbb{F}_2^m$, $\alpha' \in \mathbb{F}_2^m$, $\beta' \in \mathbb{F}_2^\ell$ and $\gamma' \in \mathbb{F}_2$ such that

$$\forall x \in \mathbb{F}_2^\ell, h(x) = g(\alpha' + M(x)) + \beta' \cdot x + \gamma'.$$

We show that h is a reduction of f .

1. For all $x \in \mathbb{F}_2^\ell$, we compute:

$$\begin{aligned}
h(x) &= g(\alpha' + M(x)) + \beta' \cdot x + \gamma' \\
&= f(\alpha + L(\alpha' + M(x))) + \beta \cdot (\alpha' + M(x)) + \gamma + \beta' \cdot x + \gamma' \\
&= f(\alpha + L(\alpha') + L \circ M(x)) + \beta \cdot M(x) + \beta' \cdot x + \beta \cdot \alpha' + \gamma + \gamma' \\
&= f(\delta + L \circ M(x)) + (M^T(\beta) + \beta') \cdot x + \eta
\end{aligned}$$

where $\delta = \alpha + L(\alpha')$ and $\eta = \beta \cdot \alpha' + \gamma + \gamma'$.

2. Since L and M are both injective, their composition $L \circ M : \mathbb{F}_2^\ell \rightarrow \mathbb{F}_2^n$ is also injective. Let L' and M' denote the induced isomorphism $L' : \mathbb{F}_2^m \rightarrow L(\mathbb{F}_2^m)$ and $M' : \mathbb{F}_2^\ell \rightarrow M(\mathbb{F}_2^\ell)$ of L and M respectively. Then, the induced isomorphism of $L \circ M$ is equal to $(L \circ M)' = L' \circ M'$.
3. Since g is a reduction of f , by Definition 9 there exists $a \in \text{Wsupp}_f$ and $b \in \text{Wsupp}_g$ such that L'^T induces a bijection from $a + \text{Wsupp}_f$ to $b + \text{Wsupp}_g$. Similarly, since h is a reduction of g , there exist $b' \in \text{Wsupp}_g$ and $c \in \text{Wsupp}_h$ such that M'^T induces a bijection from $b' + \text{Wsupp}_g$ to $c + \text{Wsupp}_h$. Therefore:

$$\begin{aligned}
c + \text{Wsupp}_h &= M'^T(b' + \text{Wsupp}_g) \\
&= M'^T(\underbrace{b + b'}_{\in b' + \text{Wsupp}_g} + b + \text{Wsupp}_g) \\
&= M'^T(b + b') + M'^T(b + \text{Wsupp}_g) \\
&= M'^T(b + b') + M'^T(L'^T(a + \text{Wsupp}_f)) \\
&= c + c' + (L' \circ M')^T(a + \text{Wsupp}_f)
\end{aligned}$$

where $c' = c + M'^T(b + b') \in \text{Wsupp}_h$. Thus, $c' + \text{Wsupp}_h = (L' \circ M')^T(a + \text{Wsupp}_f)$ with $c' \in \text{Wsupp}_h$ and $a \in \text{Wsupp}_f$.

Hence, $(L \circ M)'$ satisfies the last condition of Definition 9, i.e. $(L \circ M)'^T$ induces a bijection from $c' + \text{Wsupp}_h$ to $a + \text{Wsupp}_f$ with $c' \in \text{Wsupp}_h$ and $a \in \text{Wsupp}_f$.

We conclude that h is a reduction of f .

Now, we handle the second statement of the proposition. If f and g are EA equivalent, then there exists a linear automorphism $L : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$, and vectors $\alpha, \beta \in \mathbb{F}_2^n$, and $\gamma \in \mathbb{F}_2$ such that $g(x) = f(L(\alpha + x)) + \beta \cdot x + \gamma$ for all $x \in \mathbb{F}_2^n$. Therefore:

1. $g(x) = f(L(\alpha + x)) + \beta \cdot x + \gamma$ for all $x \in \mathbb{F}_2^n$.
2. L is injective (and bijective) and its induced isomorphism is L .
3. We recall from Property 1 that $\text{Wsupp}_g = \beta + L^T(\text{Wsupp}_f)$. Let $a \in \text{Wsupp}_f$ and define $b = \beta + L^T(a)$ in Wsupp_g . Then:

$$L^T(a + \text{Wsupp}_f) = L^T(a) + \beta + \text{Wsupp}_g = b + \text{Wsupp}_g.$$

Since L^T is also a linear automorphism, it is injective and thus L^T induces a bijection from $a + \text{Wsupp}_f$ to $b + \text{Wsupp}_g$ with $a \in \text{Wsupp}_f$ and $b \in \text{Wsupp}_g$.

Thus, g is a reduction of f . Since EA equivalence is symmetric f and g can be switched. Therefore, the same reasoning can be used to conclude that f is a reduction of g . $\square$

We show that the reduction implies the equivalence of Walsh support and also allows us to study the spectrum on a smaller number of variables.

Proposition 4. Let $f \in \mathcal{B}_n$ and $g \in \mathcal{B}_m$ such that g is a reduction of f , then there exists an affine isomorphism $A : b + \mathcal{WS}_g \rightarrow a + \mathcal{WS}_f$ where $a \in \text{Wsupp}_f$ and $b \in \text{Wsupp}_g$ such that:

- $A(\text{Wsupp}_g) = \text{Wsupp}_f$, i.e. Wsupp_g and Wsupp_f are equivalent,
- For all $u \in \text{Wsupp}_g$, $|W_f(u)| = 2^{n-m} |W_g(A(u))|$,
- $\text{AWspec}_f = \{2^{n-m} \cdot w \times m(w) \mid (w, m(w)) \in \text{AWspec}_g\} \cup \{0 \times 2^n - 2^{n-m}\}$.

Proof. Let f and g such that g is a reduction of f . First, we assume that $0_n \in \text{Wsupp}_f$. Then, there exist $\alpha \in \mathbb{F}_2^n$, $\beta \in \mathbb{F}_2^m$, $\gamma \in \mathbb{F}_2$ and $L : \mathbb{F}_2^m \rightarrow \mathbb{F}_2^n$ an injective linear map such that:

$$g(x) = f(\alpha + L(x)) \oplus \beta \cdot x \oplus \gamma.$$

By Definition 9, L is injective and we denote by M the induced isomorphism $M : \mathbb{F}_2^m \rightarrow L(\mathbb{F}_2^m)$. In particular, $\dim(L(\mathbb{F}_2^m)) = m$. As long as we consider vectors $u \in L(\mathbb{F}_2^m)$, M^T and L^T are equal, i.e. for all $u \in L(\mathbb{F}_2^m)$, $M^T(u) = L^T(u)$. From the definition of reduction, M^T induces a bijection between $b + \text{Wsupp}_g$ and $a + \text{Wsupp}_f$, where $a \in \text{Wsupp}_f$ and $b \in \text{Wsupp}_g$, i.e.

$$b + \text{Wsupp}_g = M^T(a + \text{Wsupp}_f).$$

Therefore, $a + \text{Wsupp}_f \subset L(\mathbb{F}_2^m)$. Since $0_n \in \text{Wsupp}_f$, we deduce that $a + 0_n \in a + \text{Wsupp}_f \subset L(\mathbb{F}_2^m)$ and thus that $\text{Wsupp}_f = a + (a + \text{Wsupp}_f) \subset L(\mathbb{F}_2^m)$. Therefore, $M^T(u)$ is well defined for $u \in \text{Wsupp}_f$. Then, we compute $W_g(\beta + M^T(u))$:

$$\begin{aligned} W_g(\beta + M^T(u)) &= \sum_{x \in \mathbb{F}_2^m} (-1)^{f(\alpha + M(x)) \oplus \beta \cdot x \oplus \gamma \oplus (\beta + M^T(u)) \cdot x} = (-1)^\gamma \sum_{x \in \mathbb{F}_2^m} (-1)^{f(\alpha + M(x)) \oplus (M^T(u)) \cdot x} \\ &= (-1)^\gamma \sum_{x \in \mathbb{F}_2^m} (-1)^{f(\alpha + M(x)) \oplus u \cdot M(x)} = (-1)^\gamma \sum_{\alpha + y \in L(\mathbb{F}_2^m)} (-1)^{f(y) \oplus u \cdot (\alpha + y)} \\ &= (-1)^{u \cdot \alpha \oplus \gamma} \sum_{\alpha + y \in L(\mathbb{F}_2^m)} (-1)^{f(y) \oplus u \cdot y}. \end{aligned}$$

Using Poisson's formula (Property 4) with $E = L(\mathbb{F}_2^m)^\perp$, $d = u \in \text{Wsupp}_f$ and $e = \alpha$, the right-hand side becomes:

$$\begin{aligned} |E| (-1)^{d \cdot e} \sum_{y \in e + E^\perp} (-1)^{f(y) \oplus d \cdot y} &= |L(\mathbb{F}_2^m)^\perp| (-1)^{u \cdot \alpha} \sum_{y \in \alpha + L(\mathbb{F}_2^m)} (-1)^{f(y) \oplus u \cdot y} \\ &= |L(\mathbb{F}_2^m)^\perp| W_g(\beta + L^T(u)) \\ &= 2^{n-m} W_g(\beta + L^T(u)). \end{aligned}$$

Since $\dim(L(\mathbb{F}_2^m)) = m$, we have $|L(\mathbb{F}_2^m)^\perp| = 2^{n-m}$. The left-hand side of Poisson's formula becomes:

$$\sum_{v \in d + E} (-1)^{e \cdot v} W_f(v) = \sum_{\substack{v \in u + L(\mathbb{F}_2^m)^\perp \\ v \in \text{Wsupp}_f}} (-1)^{\alpha \cdot v} W_f(v).$$

We show that the RHS sum has a unique element for $v \in \text{Wsupp}_f$. If $v \in \text{Wsupp}_f \cap (u + L(\mathbb{F}_2^m)^\perp)$, then for any $x \in \mathbb{F}_2^m$: $(u + v) \cdot L(x) = 0$. Therefore, for any $x \in \mathbb{F}_2^m$, $L^T(u + v) \cdot x = 0$ and thus: $L^T(u + v) = 0_m$. Since $\text{Wsupp}_f \subset L(\mathbb{F}_2^m)$, we have the following: $L^T(u + v) = M^T(u + v) = 0_m$. By injectivity of M^T , $M^T(u + v) = 0_m$ implies that $v = u$. Reciprocally, if $v = u$, then $v \in \text{Wsupp}_f \cap (u + L(\mathbb{F}_2^m)^\perp)$. Therefore, we deduce that:

$$\sum_{v \in u + L(\mathbb{F}_2^m)^\perp} (-1)^{\alpha \cdot v} W_f(v) = (-1)^{\alpha \cdot u} W_f(u).$$

Hence, using Poisson's formula, we deduce that:

$$W_f(u) = (-1)^{\alpha \cdot u} 2^{n-m} W_g(\beta + M^T(u)).$$

Moreover, $\text{Wsupp}_g = \beta + M^T(\text{Wsupp}_f)$. In particular, we deduce that $\beta = \beta + M^T(0_n) \in \text{Wsupp}_g$.

So far we assumed that $0_n \in \text{Wsupp}_f$, now we remove this restriction and assume only that g is a reduction of f . Let f' be defined as $f'(x) = f(x) \oplus a \cdot x$ where $a \in \text{Wsupp}_f$. Then, $0_n \in \text{Wsupp}_{f'}$. Accordingly to Proposition 3, f' is EA equivalent to f and consequently f is a reduction of f' . Then, using the transitivity of the reduction, we deduce that g is a reduction of f' . We apply our result for the case $0_n \in \text{Wsupp}_{f'}$ and therefore: $|W_{f'}(u)| = 2^{n-m} |W_g(\beta + M^T(u))|$. Using Property 1 we can study the transformation of the Walsh transform under EA equivalence, it allows us to deduce that for $u \in \text{Wsupp}_f$:

$$|W_f(u)| = |W_{f'}(a + u)| = 2^{n-m} W_g(\beta + M^T(a + u)).$$

We denote by $A : a + \mathcal{WS}_f \rightarrow \beta + \mathcal{WS}_g$ the affine isomorphism defined by: $A(u) = \beta + M^T(a + u)$. Then, we observe that:

- $A(\text{Wsupp}_f) = \text{Wsupp}_g$, i.e. Wsupp_f and Wsupp_g are equivalent,
- For all $u \in \text{Wsupp}_f$, $|W_f(u)| = 2^{n-m} W_g(A(u))$,
- If $u \notin \text{Wsupp}_f$, then $W_f(u) = 0$, then:

$$\text{AWSpec}_f = \{2^{n-m} \cdot w \times m(w) \mid (w, m(w)) \in \text{AWSpec}_g\} \cup \{0 \times (2^n - 2^{n-m})\}$$

□

Proposition 4 allows us to study the Walsh spectrum and support of a function f by considering another function g reducing it and having a fewer number of variables. Two questions arise: does g always exist? If it exists, what is the minimal number of variables ℓ needed to defined a reduction g ? If a function f has a Walsh space of dimension m , it implies that its Walsh support can be map with an affine transformation to a set of $\mathbb{F}_2^m$. Therefore, a naive lower bound on the minimal ℓ such that there exists a reduction $g \in \mathcal{B}_\ell$ of f is m . We show in Lemma 2 and Theorem 2 that this naive bound is achieved, i.e. there exists a reduction of f in $\mathcal{B}_m$.

Lemma 2. *Let $f \in \mathcal{B}_{n,s}$ such that $\dim(\mathcal{WS}_f) < n$, then there exists $g \in \mathcal{B}_{n-1,s}$ such that g is a reduction of f .*

Proof. We begin by choosing an extended equivalent function having a Walsh support easier to manipulate. In our case, we want to define a function f' such that $\text{Wsupp}_{f'} \subset \mathbb{F}_2^{n-1} \times \{0\}$. Since $\dim(\mathcal{WS}_f) < n$, for any $a \in \text{Wsupp}_f$ there exists a linear automorphism $L : \mathbb{F}_2^n \rightarrow \mathbb{F}_2^n$ such that $L(a + \mathcal{WS}_f) \subset \mathbb{F}_2^{n-1} \times \{0\}$. We define $f' \in \mathcal{B}_n$ by $f'(x) = f \circ L^T(x) + L(a) \cdot x$ where L^T is the adjoint of L . Using Property 1, we observe that $\text{Wsupp}_{f'} = L(a) + L(\text{Wsupp}_f)$. Therefore, $\mathcal{WS}_{f'} = L(a + \mathcal{WS}_f) \subset \mathbb{F}_2^{n-1} \times \{0\}$. In particular, since $a + a \in a + \text{Wsupp}_f$, we have $L(a + a) \in \text{Wsupp}_{f'}$ and thus $0_n \in \text{Wsupp}_{f'}$.

We define $g \in \mathcal{B}_{n-1}$ such that for all $x \in \mathbb{F}_2^{n-1}$ we have $g(x) = f'(x, 0)$. Using Property 10, we have:

$$\forall u \in \mathbb{F}_2^{n-1}, W_g(u) = \frac{1}{2} \left(W_{f'}(u, 0) + \underbrace{W_{f'}(u, 1)}_{\notin \mathbb{F}_2^{n-1} \times \{0\}} \right) = \frac{W_{f'}(u)}{2}. \quad (3)$$

Then, we show that g is a reduction of f' . We define the linear transformation $M : \mathbb{F}_2^{n-1} \rightarrow \mathbb{F}_2^n$ as $M(x) = (x, 0)$ for all $x \in \mathbb{F}_2^{n-1}$. m satisfies the following properties:

1. We have $g = f' \circ M$.

2. M is linear and injective and we denote by $M' : \mathbb{F}_2^{n-1} \rightarrow \mathbb{F}_2^{n-1} \times \{0\}$ its induced isomorphism.
3. $M'^T : \mathbb{F}_2^{n-1} \times \{0\} \rightarrow \mathbb{F}_2^{n-1}$ is defined by $M'^T(x_1, \dots, x_n) = (x_1, \dots, x_{n-1})$ for any $x \in \mathbb{F}_2^{n-1} \times \{0\}$, i.e. the linear transformation obtained by removing the last variable. According to Equation 3, $W_{f'}(u, v) \neq 0$ if and only if $u \in \text{Wsupp}_g$ and $v = 0$, thus $M'^T(\text{Wsupp}_{f'}) = \text{Wsupp}_g$. Since $0_n \in \text{Wsupp}_{f'}$ and $0_{n-1} \in \text{Wsupp}_g$, we deduce that M'^T induces a bijection from $0_n + \text{Wsupp}_{f'}$ to $0_{n-1} + \text{Wsupp}_g$.

Therefore, g is a reduction of f' . Since f' is in the EA equivalent class of f (Definition 4), according to Proposition 3 f' is a reduction of f and we can apply the transitivity of the reduction to conclude that g is a reduction of f . □

Theorem 2. *Let $f \in \mathcal{B}_n$ such that $\dim \mathcal{WS}_f = m$, then for all $\ell \in [m, n]$ there exists $g^\ell \in \mathcal{B}_\ell$ such that g is a reduction of f . Reciprocally, if $g \in \mathcal{B}_m$ is a reduction of f , then $\dim(\mathcal{WS}_f) \leq m$.*

Proof. The first part of the proof is obtained by applying recursively Lemma 2 on f . We process by induction on ℓ . Let $g^n \in \mathcal{B}_n$ such that $g^n = f$, then g^n is a reduction of f , therefore the result is verified for $\ell = n$.

Next, let assume that for some $\ell \in [m+1, n]$, $g^\ell \in \mathcal{B}_\ell$ is a reduction of f . Since Wsupp_f and $\mathcal{WS}_g$ are equivalent, their Walsh spaces are isomorphic and in particular they have the same dimension, i.e. $\dim(\mathcal{WS}_{g^\ell}) = \dim(\mathcal{WS}_f) = m$. Since $\ell \geq m+1$, then $\ell > m$ and consequently we can apply Lemma 2: There exists $g^{\ell-1} \in \mathcal{B}_{\ell-1}$ such that $g^{\ell-1}$ is a reduction of g^ℓ . Then, we apply the transitivity of the reduction (Proposition 3) to conclude that $g^{\ell-1}$ is a reduction of f .

Therefore, it allows us to conclude that for all $\ell \in [m, n]$ g^ℓ is a reduction of f and is in $\mathcal{B}_\ell$.

Reciprocally, let assume that $g \in \mathcal{B}_m$ is a reduction of $f \in \mathcal{B}_n$. Since Wsupp_f and Wsupp_g are equivalent (Proposition 4), then $\dim(\mathcal{WS}_f) = \dim(\mathcal{WS}_g)$. By definition, $\mathcal{WS}_g$ is a vector subspace of $\mathbb{F}_2^m$, thus $\dim(\mathcal{WS}_f) \leq m$. □

Corollary 3. *Let $s \in \mathbb{N}$. If there exist $n \in \mathbb{N}$ and f such that $f \in \mathcal{B}_{n,s}$, then $\dim(\mathcal{WS}_f) \geq \min\{m \in \mathbb{N} \mid \mathcal{B}_{m,s} \neq \emptyset\}$.*

Proof. We assume that there exists $n \in \mathbb{N}$ such $f \in \mathcal{B}_{n,s}$. Then, we perform the proof by contradiction. We denote $\dim(\mathcal{WS}_f)$ and $\min\{m \in \mathbb{N} \mid \mathcal{B}_{m,s} \neq \emptyset\}$ by d and ℓ respectively. If $d < \ell$, then according to Theorem 2, there exists $g^d \in \mathcal{B}_d$ such that g^d is a reduction of f . By Definition 9, Wsupp_f is in bijection with Wsupp_{g^d} and therefore $|\text{Wsupp}_{g^d}| = |\text{Wsupp}_f| = s$. Then, we reach a contradiction: by definition of ℓ , $\mathcal{B}_{p,s} = \emptyset$ for $p < \ell$, and thus $\mathcal{B}_{d,\ell} = \emptyset$. Finally we can conclude $\dim(\mathcal{WS}_f) \geq \ell$. □

Theorem 2 can be used to reduce the number of variables whenever we study the structure of a Walsh support. In particular, the Walsh support can be represented using $\mathbb{F}_2^{\dim(\mathcal{WS}_f)}$.

4.3 A Lemma on Plateaued Functions

Plateaued functions (Definition 5) have been widely studied as a class of functions that generalizes the class of bent functions. We derive a lemma on one relation that must be fulfilled by the Walsh support of any plateaued function. In Section 6 this result will be used to discard possible Walsh supports' structures.

Lemma 3. *Let $f \in \mathcal{B}_{n,s}$ be a plateaued function with $s > 1$ (Definition 5) then*

$$\sum_{u \in \text{Wsupp}_f} u = 0.$$

Proof. Since f is plateaued, the integers s and λ are powers of two [Car21] (Section 6.2.2). In particular, there are an even number of terms in the sum $\sum_{u \in \text{Wsupp}_f} u$. If we denote by f^a the function defined by $f^a(x) = f(x) + a \cdot x$, we have:

$$\sum_{u \in \text{Wsupp}_{f^a}} u = \sum_{u \in a + \text{Wsupp}_f} u = \sum_{u \in \text{Wsupp}_f} a + u = \sum_{u \in \text{Wsupp}_f} u.$$

Therefore, we can prove the result for any f^a . In particular, we can choose $g = f^a$ such that $0_n \in \text{Wsupp}_g$. For $u \in \text{Wsupp}_g$, we can write $W_g(u) = (-1)^{v_u \lambda}$ with $v_u \in \mathbb{F}_2$. Using Property 2 we obtain:

$$\begin{aligned} (-1)^{g(x)} &= \frac{1}{2^n} \left[W_g(0_n) + \sum_{u \in \mathbb{F}_2^n \setminus \{0_n\}} W_g(u) (-1)^{u \cdot x} \right] \\ &= \frac{1}{2^n} \left[\pm 2^n - \sum_{u \in \mathbb{F}_2^n \setminus \{0_n\}} W_g(u) + \sum_{u \in \mathbb{F}_2^n \setminus \{0_n\}} W_g(u) (-1)^{u \cdot x} \right] \\ &= \pm 1 + \frac{1}{2^n} \left[\sum_{u \in \text{Wsupp}_g \setminus \{0_n\}} W_g(u) ((-1)^{u \cdot x} - 1) \right]. \end{aligned}$$

Since $(-1)^{g(x)} \in \{-1, 1\}$, we deduce that:

$$\frac{1}{2^n} \left[\sum_{u \in \text{Wsupp}_g \setminus \{0_n\}} W_g(u) ((-1)^{u \cdot x} - 1) \right] \in \{0, \mp 2\}.$$

We can rewrite the sum. We define as p the number of $u \in (\text{Wsupp}_f \setminus \{0_n\})$ such that $u \cdot x = 1$ and q as the number of $u \in (\text{Wsupp}_f \setminus \{0_n\})$ such that $u \cdot x = 1$ and $v_u = 1$, we can derive the following:

$$\begin{aligned} \frac{1}{2^n} \left[\sum_{u \in \text{Wsupp}_g \setminus \{0_n\}} W_g(u) ((-1)^{u \cdot x} - 1) \right] &= \frac{\lambda}{2^n} \left[\sum_{u \in \text{Wsupp}_g \setminus \{0_n\}} (-1)^{v_u} ((-1)^{u \cdot x} - 1) \right] \\ &= \frac{\lambda}{2^n} (2q - 2(p - q)) \\ &= \frac{\lambda}{2^{n-1}} (2q - p) \in \{0, \mp 2\}. \end{aligned}$$

Since $s > 1$, it implies $\lambda < 2^n$, and we deduce that $\frac{\lambda}{2^{n-1}} = 2^{-\ell}$ with $\ell \geq 0$. Hence, $(2q - p) \in \{0, \mp 2^{\ell+1}\}$. Thus, $p = 0 \pmod 2$. Thereafter,

$$\bigoplus_{u \in \text{Wsupp}_g \setminus \{0_n\}} u \cdot x = \bigoplus_{\substack{u \in \text{Wsupp}_g \setminus \{0_n\} \\ u \cdot x = 0}} u \cdot x \oplus \bigoplus_{\substack{u \in \text{Wsupp}_g \setminus \{0_n\} \\ u \cdot x = 1}} u \cdot x = p \pmod 2.$$

Therefore, we deduce that $(\sum_{u \in \text{Wsupp}_g} u) \cdot x = 0$ for any $x \in \mathbb{F}_2^n$. Hence, it allows us to conclude:

$$\sum_{u \in \text{Wsupp}_f} u = \sum_{u \in \text{Wsupp}_g} u = 0.$$

□

5 Experimental Results

In this section, we present the results obtained from an exhaustive search over Boolean functions in 6 variables. The goal is to characterize the number of inequivalent Walsh supports of cardinality at most 16, as well as the dimension of their associated Walsh spaces. These results will then serve as base cases for the inductive proofs in the next section, which aim to determine which cardinalities are impossible.

Using Theorem 2, Proposition 1 and Proposition 4 from Section 4, we can develop the main routine used in Section 6 to discard certain cardinalities. Whenever we establish a bound on the dimension of a Walsh space, we know that the geometric structure contained in the Walsh support of the function can be studied using a function having fewer variables. In particular, from classifications of Boolean functions in small dimensions we can deduce properties of Walsh supports of small cardinality.

The number of EA equivalent classes of $\mathcal{B}_n$ are known to quickly increase. For $n \geq 7$, they are already more than $6 \cdot 10^{19}$ EA classes. However, for $n \leq 6$, it is reasonable to perform an exhaustive search on every equivalent classes using classifications from [CM05] or [Lan09]. In Table 1, we display the classification of Walsh support sizes for functions in 6 variables, based on the EA-classification from [Lan09]. The main observation is that for $s \in \{10, 13\}$ and $n \leq 6$, there exists only one EA class for a Walsh support of size s .

In the following, we show that the functions with Walsh supports of size 10 and 13, respectively listed in Table 1, have Walsh spaces of dimension 4 and 5, respectively. Therefore, we can represent the Walsh support of each of these functions in the space $\mathbb{F}_2^{\dim(\mathcal{WS}_f)}$ using isomorphism.

Property 12. *Let $s \in \{10, 13\}$, then $\mathcal{B}_{6,s}$ contains only one EA class. Let $f \in \mathcal{B}_{6,10}$ and $g \in \mathcal{B}_{6,13}$ be the representatives of these classes listed in Table 1, then:*

- $\dim(\mathcal{WS}_f) = 4$,
- $\dim(\mathcal{WS}_g) = 5$.

Moreover, for $n \leq 6$, $\mathcal{B}_{n,s} = \emptyset$ for $s \in \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\}$.

Proof. We begin with preliminary remarks to ease the understanding of the proof. From Corollary 3, we deduce that $\dim(\mathcal{WS}_f) \geq 4$ and $\dim(\mathcal{WS}_g) \geq 5$ since experimentally $\mathcal{B}_{4,10} \neq \emptyset$ and $\mathcal{B}_{5,13} \neq \emptyset$. Accordingly to Property 9, for $f' \in \mathcal{B}_n$, $\text{Sieg}[f', f']$ has a Walsh support equal to $\text{Wsupp}_f \times \{0\}$.

Then, we upper bound the dimension. We denote by $\text{tt}_{f'}$ the truth table of $f' \in \mathcal{B}_n$, then:

- We have $\text{tt}_f = 2a802a802a802a80$. We denote by h the functions such that $\text{tt}_h = 2a80$. From Remark 3, we observe that $h' = \text{Sieg}[h, h]$ admits the following truth table: $\text{tt}_{h'} = 2a80|2a80$. Similarly, $\text{Sieg}[h', h']$ admits the following truth table: $2a802a80|2a802a80$. Thus, $f = \text{Sieg}[h', h']$. Therefore, we deduce from the second preliminary remark that

$$\text{Wsupp}_f = \text{Wsupp}_{h'} \times \{0\} = \text{Wsupp}_h \times \{(0, 0)\}.$$

Since $h \in \mathcal{B}_4$, we have $\text{Wsupp}_h \subset \mathbb{F}_2^4$ and thus $\text{Wsupp}_f \subset \mathbb{F}_2^4 \times \{(0, 0)\}$. Consequently, $\mathcal{WS}_f \subset \mathbb{F}_2^4 \times \{(0, 0)\}$ which implies that $\dim(\mathcal{WS}_f) \leq 4$. From the lower bound derived in the preliminary remark we conclude that $\dim(\mathcal{WS}_f) = 4$.

- We have $\text{tt}_g = d8885000d8885000$. We denote by $h \in \mathcal{B}_5$ the function with truth table $\text{tt}_h = d8885000$. From Remark 3, we observe that $\text{Sieg}[h, h]$ admits the following truth table: $d8885000|d8885000$. Thus, $g = \text{Sieg}[h, h]$. Thereafter, we deduce from the second preliminary remark that $\text{Wsupp}_g = \text{Wsupp}_h \times \{0\}$. Since $h \in \mathcal{B}_5$, we have $\text{Wsupp}_h \subset \mathbb{F}_2^5$ and thus $\text{Wsupp}_g \subset \mathbb{F}_2^5 \times \{0\}$. Consequently, $\mathcal{WS}_g \subset \mathbb{F}_2^4 \times \{0\}$ which implies that $\dim(\mathcal{WS}_g) \leq 5$. The lower bound derived in the preliminary remark allows us to conclude, $\dim(\mathcal{WS}_g) = 5$.

s	f (hexadecimal)	Number of EA classes	s	f (hexadecimal)	Number of EA classes
1	0000000000000000	1	39	6a361e82e8141428	21
4	8888888888888888	1	40	d52a2a2a7f808080	370
8	8080808080808080	1	41	98d0b070e8a0c000	73
10	2a802a802a802a80	1	42	6280ea2a48aac000	270
13	d8885000d8885000	1	43	622a264c2e666a00	266
16	6ac06ac06ac06ac0	8	44	9b4cec4cd700a000	765
18	2aaa80002aaa8000	1	45	7880f880f8007800	815
20	0888800008888000	4	46	9deaeaeab7c0c0c0	2321
21	f8788000f8788000	1	47	99804c801900cc00	3255
22	a82820a0a82820a0	6	48	9eedaaaa34447000	5620
23	eac08000eac08000	2	49	10092a00853640c0	6465
24	ea6a40c0ea6a40c0	6	50	4369260cdaf0eac0	9978
25	5222700052227000	6	51	92e2e2e2b0c0c0c0	10966
26	8660eca08660eca0	10	52	1666daaabccc7000	11079
27	62404040e2c0c0c0	2	53	68284888e0a0c000	7459
28	e6662aaa4ccc8000	30	54	972848881fa0c000	6114
29	57802080d700a000	8	55	ff4020e0f74828e8	3903
30	3780e020b70060a0	32	56	52227888daaaf000	2419
31	9720408017a0c000	22	57	9580ea80bf00c000	792
32	7f8080807f808080	85	58	95eabf40ea6ac0c0	520
33	9000d888b8887888	26	59	59808c80d9000c00	109
34	82220aaa2888a000	112	60	95eaeaeabfc0c0c0	91
35	d95980008c0c8000	87	61	14c4d08094445000	6
36	e4446ccc2888a000	95	62	f9202c2079a0aca0	4
37	e828488860a0c000	130	64	96665aaa3cccf000	75815
38	172848889fa0c000	183			

Table 1: Classification of 6-variable Boolean functions with respect to the extended affine equivalence, grouped by Walsh support size. We give the size of the Walsh support of one function from each group, its hexadecimal truth table and the number of EA classes having a Walsh support of that size.

Regarding the impossible values, from the experimental results displayed in Table 1 there are no 6-variable functions with a Walsh support of the 12 listed cardinalities. These cardinalities cannot exist for smaller values of n , otherwise they would exist for $n = 6$ due to the construction given in Remark 3 where $|\text{Wsupp}_f| = |\text{Wsupp}_g| = |\text{Wsupp}_h|$.

□

In the following we show how one parts of the functions with Walsh support of size 16 have a Walsh space of dimension only 4. In Table 2 we give a representative of each EA equivalent class of 6-variable function having a Walsh support of cardinality 16 and their associated absolute Walsh spectrum.

f (hexadecimal)	AWspec_f	$\dim(\mathcal{WS}_f)$
6ac06ac06ac06ac0	$\{0 \times 48, 16 \times 16\}$	4
e62a4c80e62a4c80	$\{0 \times 48, 16 \times 16\}$	5
e4789c00e4789c00	$\{0 \times 48, 16 \times 16\}$	5
7222faaad8885000	$\{0 \times 48, 16 \times 16\}$	6
24d87488ac50fc00	$\{0 \times 48, 16 \times 16\}$	6
8000800080008000	$\{0 \times 48, 8 \times 15, 56 \times 1\}$	4
0888088808880888	$\{0 \times 48, 40 \times 1, 24 \times 3, 8 \times 12\}$	4
eac0eac0eac0eac0	$\{0 \times 48, 8 \times 10, 24 \times 6\}$	4

Table 2: Table of the representatives of each EA equivalent classes having a Walsh support of cardinality 16 and their associated absolute Walsh spectrum.

Property 13. *Let f be the representative of one of the eight EA classes for $s = 16$ in Table 2. If f is not plateaued, then $\dim(\mathcal{WS}_f) = 4$.*

Proof. There are three non-plateaued EA classes represented in Table 2. Due to the small dimension of the functions ($n = 6$) we can experimentally compute the support and determine the dimension $\dim(\mathcal{WS}_f)$. We bring however a direct proof of the claim by observing the truth tables. Let h_1, h_2 and h_3 in $\mathcal{B}_6$ respectively defined with the following truth tables in hexadecimal:

$$\text{tt}_1 = 8000800080008000 \quad \text{tt}_2 = 0888088808880888 \quad \text{tt}_3 = \text{eac0eac0eac0eac0}.$$

They share one property; the string tt_i can be written as $\text{tt}_i = \text{xyzt|xyzt|xyzt|xyzt}$ where x, y, z, t are hexadecimal digits where $|$ denotes the concatenation. Let $f_i \in \mathcal{B}_4$ be defined with the truth table: xyzt .

By definition $h = \text{Sieg}[f, g]$ verifies that if tt_f and tt_g are the truth tables of f and g respectively, then the truth table of h is equal to $\text{tt}_f | \text{tt}_g$ (see Remark 3). Consequently, $g_i = \text{Sieg}[f_i, f_i]$ is the function defined by the hexadecimal truth table xyzt|xyzt . Similarly, $\text{Sieg}[g_i, g_i]$ is the function defined by the hexadecimal truth table $\text{xyzt|xyzt|xyzt|xyzt}$, i.e. $h_i = \text{Sieg}[g_i, g_i]$. Accordingly to Property 9, for $f \in \mathcal{B}_n$, $\text{Sieg}[f, f]$ has a Walsh support equal to $\text{Wsupp}_f \times \{0\}$. Therefore, $\text{Wsupp}_{h_i} = \text{Wsupp}_{f_i} \times \{(0, 0)\}$. Since $\text{Wsupp}_{f_i} \subset \mathbb{F}_2^4$, we deduce that $\dim(\mathcal{WS}_{h_i}) = 4$.

□

Remark 9. In particular, we have four classes such that f is plateaued and $\dim(\mathcal{WS}_f) > 4$, i.e. Wsupp_f is not an affine space. These supports have already been studied by Hodžić et al. [HPWZ19].

6 Walsh Support Structure for Small Cardinalities

In this section, we study the Walsh supports of cardinality at most 19. The main motivation is to prove which of these cardinalities are non-achievable, *i.e.* for which $r \in [1, 19]$ we have for all $n \in \mathbb{N}$, $\mathcal{B}_{n,r} = \emptyset$, and in the following section (Section 7) we will show all cardinalities greater than 19 are possible.

We proceed step by step by running a sieving method on the possible (s, t, k, ξ) -constructions (as defined in Theorem 1). Consequently, each subsection depends on the previous ones, as the proof unfolds in a sequential manner. To help the reader navigate the argument, we briefly outline the structure of each subsection below:

1. We use or find characterizations of the Walsh supports of cardinalities $s \leq r - 1$.
2. Using these characterizations, we deduce relations between (s, t, k, ξ) for $s, t \leq r - 1$.
3. Using Theorem 1, if (s, t, k, ξ) is not impossible, then we construct a set S such that $\mathcal{C}^{\leq r-1} \subset S$.
4. Finally, we use Corollary 2: if $r' \in [1, r]$ is not in S , it is not in $\mathcal{C}^{\leq r-1}$ and thus r' cannot be constructed.

For $s, t \in \mathbb{N}$, we define the Construction Table ($\text{CT}^{s,t}$) as a double-entry table where each cell indexed by (k, ξ) contains the value $2(s+t) - 2k - \xi$. A cell (k, ξ) in $\text{CT}^{s,t}$ is colored if one of the results shows that the (s, t, k, ξ) -construction is not possible. The tables $\text{CT}^{s,t}$ thus serve as a way to keep track of all possible constructions for $s, t \in \mathbb{N}$.

Propositions 8, 12, 6, and 11 are mostly technical and are used to color the construction tables $\text{CT}^{s,10}$ and $\text{CT}^{s,13}$. We recommend following the proofs of these propositions alongside Figure 1, which provides a geometric representation (up to isomorphism) of the Walsh supports of cardinalities 4, 8, 10, and 13.

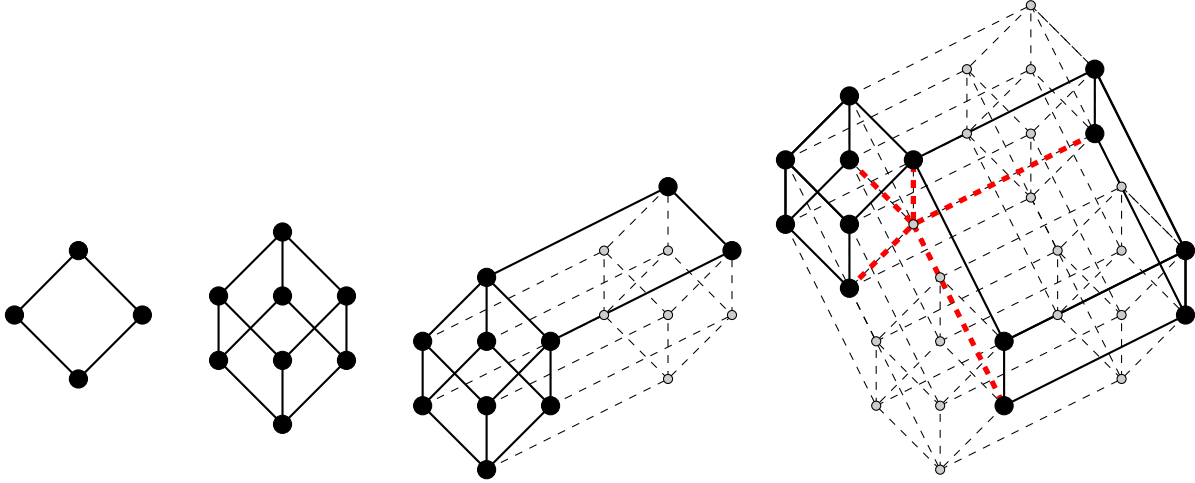


Fig. 1: Representation of each structure with cardinality $s \in \{4, 8, 10, 13\}$ from left to right.

6.1 Study of $\mathcal{C}^{\leq 8}$

We begin by studying the (s, t, k, ξ) -constructions with $s, t \leq 8$. Thanks to Property 6 and Property 7, if f is a function having of Walsh support of cardinality s less than 8, then $s \in \{1, 4, 8\}$ and Wsupp_f is an affine space. These two observations heavily simplify our sieving method for the first cardinalities, as studied in the following proposition.

Proposition 5. *Let $n \in \mathbb{N}^*$, $f \in \mathcal{B}_{n,s}$ and $g \in \mathcal{B}_{n,t}$, we denote $|K(f, g)|$ by k and $|\Xi(f, g)|$ by ξ . The following holds:*

1. If $s = 1$, then:
 - (a) If $t = 1$, $k = \xi$,
 - (b) If $t \neq 1$, $\xi = 0$.
2. If $s = 4$ and $t = 4$, then:
 - (a) $k \in \{0, 1, 2, 4\}$,
 - (b) $\xi = k$.
3. If $s = 4$ and $t = 8$, then:
 - (a) $k \in \{0, 1, 2, 4\}$,
 - (b) $\xi = 0$.
4. If $s = 8$ and $t = 8$, then:
 - (a) $k \in \{0, 1, 2, 4, 8\}$,
 - (b) $\xi \in \{k, k-1, k-2\}$.

Proof. Wsupp_f and Wsupp_g are affine subspaces of $\mathbb{F}_2^n$ for any $s, t \in \{1, 4, 8\}$. Therefore, in these cases $K(f, g)$ is either empty or an affine subspace of $\mathbb{F}_2^n$.

1. If $s = 1$, we recall that $\text{AWspec}_f = \{0 \times (2^n - 1), 2^n \times 1\}$ (Property 5).
 - (a) If $t = 1$, we know that $\text{AWspec}_f = \text{AWspec}_g = \{0 \times (2^n - 1), 2^n \times 1\}$. Therefore, $u \in K(f, g)$ if and only if $|W_f(u)| = |W_g(u)| = 2^n$, i.e. $u \in \Xi(f, g)$. Thus, $k = \xi$.
 - (b) If $t \neq 1$, then $2^n \notin \text{AWspec}_g$. We prove it by contradiction. If $2^n \in \text{AWspec}_g$, then according to Parseval relation (Property 3), we have $\sum_{u \in \text{Wsupp}_g} |W_g(u)|^2 = 2^{2n}$. However, there exists by hypothesis $u \in \text{Wsupp}_g$ such that $|W_g(u)| = 2^n$ and since $t > 1$, there exists $v \in \text{Wsupp}_g \setminus \{u\}$.

$$|W_g(v)|^2 + |W_g(u)|^2 \leq \sum_{u \in \text{Wsupp}_g} |W_g(u)|^2 = 2^{2n}$$

Therefore, $|W_g(v)|^2 \leq 0$, i.e. $W_g(v) = 0$ leading to a contradiction since $v \in \text{Wsupp}_g$, hence $2^n \notin \text{AWspec}_g$. Since AWspec_f and AWspec_g share no common value, we must have $\xi = 0$.

2. If $s = t = 4$, $K(f, g)$ is either empty or an affine subspace of dimension at most 2. Thus, $k \in \{0, 1, 2, 4\}$. Moreover, $\text{AWspec}_f = \text{AWspec}_g = \{0 \times (2^n - 4), 2^{n-1} \times 4\}$. Therefore, if $u \in K(f, g)$, we have $|W_f(u)| = |W_g(u)| = 2^{n-1}$. Thus, $\xi = k$.
3. If $s = 4$ and $t = 8$, $K(f, g)$ is either empty or an affine subspace of dimension at most 2. Thus, $k \in \{0, 1, 2, 4\}$. Moreover, the spectra of f and g have no value in common, thereafter $\xi = 0$.
4. If $s = t = 8$, $K(f, g)$ is either empty or an affine subspace of dimension at most 3. Thus, $k \in \{0, 1, 2, 4, 8\}$. We have $\text{AWspec}_f = \text{AWspec}_g = \{0 \times (2^n - 8), 2^{n-2} \times 7, (3 \cdot 2^{n-2}) \times 1\}$. Therefore, if $u \in K(f, g)$ is such that $|W_f(u)| \neq |W_g(u)|$, either $|W_f(u)| = 3 \cdot 2^{n-2}$ or $|W_g(u)| = 3 \cdot 2^{n-2}$. Thus, there are at most two vectors $u \in K(f, g)$ such that $|W_f(u)| \neq |W_g(u)|$, i.e. $\xi \in \{k, k-1, k-2\}$.

□

The possible values of $\mathcal{C}^{s,t}$ (based on the value of s and t) are given in general by Corollary 1 and Proposition 1. For s and t in $\{4, 8\}$, we use Proposition 5 to characterize the possible 4-tuples (s, t, k, ξ) as defined in Theorem 1. In Tables $\text{CT}^{4,4}$, $\text{CT}^{4,8}$ and $\text{CT}^{8,8}$, we represent for fixed (s, t) (respectively $(4, 4)$, $(4, 8)$, $(8, 8)$) the possible and impossible 4-tuples (s, t, k, ξ) . The corresponding results are stated in Proposition 6.

Proposition 6. Let $n \in \mathbb{N}^*$, $f \in \mathcal{B}_{n,s}$, $g \in \mathcal{B}_{n,t}$, denote $k = |K(f, g)|$ and $\xi = |\Xi(f, g)|$ (see Section 3). If $(s, t) = (4, 4)$ (resp. $(4, 8)$, $(8, 8)$), the couples (k, ξ) corresponding to the colored cells in Table $\text{CT}^{4,4}$ (resp. $\text{CT}^{4,8}$, $\text{CT}^{8,8}$) are impossible. Only the numbers in bold could be obtained from (s, t) .

Proof. First, we fill the content of each cell in any table as the value $r = 2(s+t) - 2k - \xi$. We have to show that if a cell having entries (k, ξ) is colored in the chosen table, then we cannot have $k = |K(f, g)|$ and $\xi = |\Xi(f, g)|$ at the same time. Every cell such that $\xi > k$ are colored in black, since it is not possible to have $\xi > k$. Then, we distinguish between the different cases using the items of Proposition 5:

- $(s, t) = (4, 4)$: The red colored cells of Table $CT^{4,4}$ are such that $k \neq \xi$ and according to Item 2a, $|K(f, g)| = |\Xi(f, g)|$. The blue colored cell of Table $CT^{4,4}$ have entries $(k, \xi) = (3, 3)$ and $|K(f, g)| \neq 3$ according to Item 2b. Accordingly, $\mathcal{C}^{4,4} \subset \{4, 10, 13, 16\}$.
- $(s, t) = (4, 8)$: The red colored cells of Table $CT^{4,8}$ are such that $\xi \neq 0$ and according to Item 3b, $|\Xi(f, g)| = 0$. The blue colored cell of Table $CT^{4,8}$ has entries $(k, \xi) = (3, 0)$ and $|K(f, g)| \neq 3$ according to Item 3b. This results in $\mathcal{C}^{4,8} \subset \{16, 20, 22, 24\}$.
- $(s, t) = (8, 8)$: The blue colored cells of Table $CT^{8,8}$ are such that k is not a power of two and $|K(f, g)| \in \{1, 2, 4, 8\}$ according to Item 4b. The red colored cells of Table $CT^{8,8}$ are such that $\xi < k-2$ since according to Item 4b, $|\Xi(f, g)| \geq |K(f, g)| - 2$. The green colored cell Table $CT^{8,8}$ has entries $(k, \xi) = (8, 7)$ and according to Lemma 1, $|\Xi(f, g)| \neq |K(f, g)| - 1$ if $s = t = k$.

□

Since there is no function having Walsh support of size 2, 3, 5, 6 or 7 according to Property 7, Tables $CT^{1,t}$, $CT^{4,4}$, $CT^{4,8}$ and $CT^{8,8}$ give the set of Walsh supports cardinalities that might be possible to construct from two supports of size at most 8.

Corollary 4. *We have $\mathcal{C}^{\leq 8} \subset \{1, 4, 8, 10, 13, 16, 18, 20, 21, 22, 24, 26, 27, 28, 29, 30, 32\}$ and $\mathcal{B}_{n,9} = \emptyset$ for all $n \in \mathbb{N}$.*

Proof. We list all the numbers appearing in bold in Tables $CT^{4,4}$, $CT^{4,8}$, $CT^{8,8}$ and we add the bold cells obtained using Table $CT^{1,t}$ with $t \in \{4, 8\}$, respectively $\{8, 10\}$ and $\{16, 18\}$. According to Proposition 6, we deduce that $\mathcal{C}^{\leq 8} \subset \{1, 4, 8, 10, 13, 16, 18, 20, 21, 22, 24, 26, 27, 28, 29, 30, 32\}$.

As a consequence of Corollary 2, since $9 \notin \mathcal{C}^{\leq 8}$, we deduce that for all $n \in \mathbb{N}$, $\mathcal{B}_{n,9} = \emptyset$. □

$\xi \backslash k$	0	1	2	3	4
0	16	14	12	10	8
1		13	11	9	7
2			10	8	6
3				7	5
4					4

$CT^{4,4}$: Every k, ξ pair possible for $s = 4$ and $t = 4$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 6).

$\xi \backslash k$	0	1	2	3	4
0	24	22	20	18	16
1		21	19	17	15
2			18	16	14
3				15	13
4					12

$CT^{4,8}$: Every k, ξ possible pair for $s = 4$ and $t = 8$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 6).

6.2 Walsh Supports of Size 10 and study of $\mathcal{C}^{\leq 10}$

Using the tables and results established in Section 6.1, we can characterize the (s, t, k, ξ) -constructions allowing us to obtain the functions of $\mathcal{B}_{n,10}$.

Proposition 7. *Let $h \in \mathcal{B}_{n,10}$, $f \in \mathcal{B}_{n-1,s}$ and $g \in \mathcal{B}_{n-1,t}$ such that $h = \text{Sieg}[f, g]$ and $s \leq t$, then one of the following assertions is true:*

- $s = 1$ and $t = 4$ with $|K(f, g)| = 0$ and $|\Xi(f, g)| = 0$,

$\xi \backslash k$	0	1	2	3	4	5	6	7	8
0	32	30	28	26	24	22	20	18	16
1		29	27	25	23	21	19	17	15
2			26	24	22	20	18	16	14
3				23	21	19	17	15	13
4					20	18	16	14	12
5						17	15	13	11
6							14	12	10
7								11	9
8									8

$CT^{8,8}$: Every k, ξ possible pair for $s = 8$ and $t = 8$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 6).

$\xi \backslash k$	0	1
0	4	2
1		1

$CT^{1,1}$: Every k, ξ possible pair for $s = 1$ and $t = 1$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the construction (s, t, k, ξ) may be possible, then the cell is in bold.

$\xi \backslash k$	0	1
0	$2t+2$	$2t$
1		$2t-1$

$CT^{1,t}$: Every k, ξ possible pair for $s = 1$ and $t > 1$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the construction (s, t, k, ξ) may be possible, then the cell is in bold.

- $s = 4$ and $t = 4$ with $|K(f, g)| = 2$ and $|\Xi(f, g)| = 2$,
- $s = 8$ and $t = 8$ with $|K(f, g)| = 8$ and $|\Xi(f, g)| = 6$,
- $s = 10$ and $t = 10$ with $|K(f, g)| = 10$ and $|\Xi(f, g)| = 10$.

Proof. According to Tables $CT^{1,t}$, $CT^{4,4}$, $CT^{4,8}$, $CT^{8,8}$ the only possible 4-tuples (s, t, k, ξ) to obtain 10 are $(1, 4, 0, 0)$, $(4, 4, 2, 2)$, $(8, 8, 8, 6)$ and $(10, 10, 10, 10)$. For each couple with s and t lower than 10, we give the values of k and ξ and an example of such construction, with the truth table of f and g in hexadecimal.

(s, t, k, ξ)	f	g
$(1, 4, 0, 0)$	cc	41
$(4, 4, 2, 2)$	5f	41
$(8, 8, 8, 6)$	10	70

□

Based on the restricted number of possibilities to obtain a Wash support of size 10, we can study its properties, first the dimension of the Walsh space, then the structure of such support. We note it is the first cardinality that is not a power of two, and therefore the smallest example of Walsh support that is not an affine subspace.

Lemma 4. *Let $n \geq 4$ and $h \in \mathcal{B}_{n,10}$, then $\dim(\mathcal{WS}_h) = 4$.*

Proof. According to Proposition 7, h can be obtained solely from a (s, t, k, ξ) -construction with (s, t, k, ξ) equals to one of the following 4-tuples: $(4, 1, 0, 0)$, $(4, 4, 2, 2)$, $(8, 8, 8, 6)$ and $(10, 10, 10, 10)$. We begin by proving that $\dim(\mathcal{WS}_h) = 4$ if $s, t < 10$ using Proposition 2 when $k > 0$:

$$\dim(\mathcal{WS}_h) \leq \dim(\mathcal{WS}_f) + \dim(\mathcal{WS}_g) - \dim(\mathcal{WS}_f \cap \mathcal{WS}_g)$$

Let $f \in \mathcal{B}_{n-1,s}$ and $g \in \mathcal{B}_{n-1,t}$ such that $h = \text{Sieg}[f, g]$. We distinguish between the different 4-tuples (s, t, k, ξ) with s and t lower than 10:

- If $(s, t, k, \xi) = (4, 1, 0, 0)$, then $\dim(\mathcal{WS}_f) = 0$ and $\dim(\mathcal{WS}_g) = 2$ (Property 6). From Property 9, we deduce that $\text{Wsupp}_h = (\text{Wsupp}_f \sqcup \text{Wsupp}_g) \times \mathbb{F}_2$ since Wsupp_f and Wsupp_g are disjoint. Let $a \in \text{Wsupp}_f$, then:

$$\mathcal{WS}_h \subset \text{Span}((a, 0) + (\text{Wsupp}_f \sqcup \text{Wsupp}_g) \times \mathbb{F}_2) = (\mathcal{WS}_f + \text{Span}(a + \text{Wsupp}_g)) \times \mathbb{F}_2.$$

Since $a \notin \text{Wsupp}_g$, we have $\mathcal{WS}_f \cap \text{Span}(a + \text{Wsupp}_g) = \{0_n\}$ and thus $\dim(\mathcal{WS}_h) = \dim(\mathcal{WS}_f) + \dim(\text{Span}(a + \text{Wsupp}_g)) + 1 = 2 + 1 + 1$, i.e. $\dim(\mathcal{WS}_h) = 4$.

- If $(s, t, k, \xi) = (4, 4, 2, 2)$. We recall that Wsupp_f and Wsupp_g are affine planes (Property 6). Therefore, $\dim(\mathcal{WS}_f) = \dim(\mathcal{WS}_g) = 2$. Since $k = 2$, $\dim(\mathcal{WS}_f \cap \mathcal{WS}_g) = 1$. Then, we deduce that $\dim(\mathcal{WS}_h) \leq 2 + 2 - 1 + 1 = 4$ (Proposition 2).
- If $(s, t, k, \xi) = (8, 8, 8, 6)$. We recall that Wsupp_f and Wsupp_g are affine cubes (Property 6). Therefore, $\dim(\mathcal{WS}_f) = \dim(\mathcal{WS}_g) = 3$. Since $k = 8$, then $\text{Wsupp}_f = \text{Wsupp}_g$ and thus $\mathcal{WS}_f = \mathcal{WS}_g$ (by Definition 7). Thus $\mathcal{WS}_f \cap \mathcal{WS}_g = \text{Wsupp}_g$ and it follows that $\dim(\mathcal{WS}_h) \leq 3 + 3 - 3 + 1 = 4$ (Proposition 2).

The minimal $m \in \mathbb{N}$ such that $\mathcal{B}_{m,10} \neq \emptyset$ is $m = 4$. Accordingly to Corollary 3, $\dim(\mathcal{WS}_h) \geq 4$ and thus $\dim(\mathcal{WS}_h) = 4$.

One case is remaining: the case $(s, t, k, \xi) = (10, 10, 10, 10)$. Let us assume that $\dim(\mathcal{WS}_f) = 4$. Since $\mathcal{WS}_f = \mathcal{WS}_g$ due to $\xi = k = 10$, and thus $\mathcal{WS}_f \cap \mathcal{WS}_g = \mathcal{WS}_f$, we have $\dim(\mathcal{WS}_h) \leq 4 + 4 - 4 + 1 = 5$

by Proposition 2. We prove by contradiction that $\dim(\mathcal{WS}_h) \leq 4$. Let us assume that $\dim(\mathcal{WS}_h) = 5$, thus there exists $h^5 \in \mathcal{B}_{5,10}$ such that h^5 is a reduction of h (Theorem 2). However, h^5 is also in the only EA class of $\mathcal{B}_{5,10}$ and thus $\dim(\mathcal{WS}_{h^5}) = 4$ (Property 12). Therefore, $\dim(\mathcal{WS}_h) = 4$. Finally, we can conclude by recursion that for all $n \geq 4$ and any $h \in \mathcal{B}_{n,10}$, we have $\dim(\mathcal{WS}_f) = 4$. $\square$

Theorem 3 (Characterization of Walsh Supports of Size 10). *Let $f \in \mathcal{B}_{n,10}$, then we have*

- $\text{Wsupp}_f = V \cup D$ where V is an affine subspace of dimension 3 and D is an affine subspace of dimension 1 such that $V \cap D = \emptyset$,
- D is parallel to one edge of V ,
- $V = \{u \in \text{Wsupp}_f \mid W_f(u) = \pm 2^{n-2}\}$ and $D = \{u \in \text{Wsupp}_f \mid W_f(u) = \pm 2^{n-1}\}$,
- $\text{AWspec}_f = \{0 \times (2^n - 10), 2^{n-1} \times 2, 2^{n-2} \times 8\}$.

Proof. Since $f \in \mathcal{B}_{n,10}$, then $\dim(\mathcal{WS}_f) = 4$ (Lemma 4). Using Theorem 2, there exists $g \in \mathcal{B}_{4,10}$ such that g is a reduction of f (Definition 9), and from Property 12 there is a unique EA class in $\mathcal{B}_{4,10}$. Therefore, using the transitivity of the reduction and that two EA equivalent functions are reductions of each other (Proposition 3), any function $h \in \mathcal{B}_{4,10}$ is a reduction of g , and thus a reduction of f . Then, we build a function $h \in \mathcal{B}_{4,10}$ with $(4, 1, 0, 0)$ -construction which is one of the possible constructions (Proposition 7) in order to characterize the structure of any function of $\mathcal{B}_{n,10}$.

We begin by defining the Walsh support the function g_1 of cardinality 4. Let $a_0 = 0_3$, $a_1 = (0, 1, 0)$, $a_2 = (0, 0, 1)$ and $a_3 = (0, 1, 1)$. Then, $H = \{a_0, a_1, a_2, a_3\}$ is a vector space of dimension 2. We define $g_1 \in \mathcal{B}_{3,4}$ such that $\text{Wsupp}_{g_1} = H$ (Property 6). Then, we consider a vector not in Wsupp_{g_1} , say $a_4 = (1, 0, 0)$, and $g_2 \in \mathcal{B}_{3,1}$ such that $\text{Wsupp}_{g_2} = \{a_4\}$.

We define the function h as $h = \text{Sieg}[g_1, g_2]$. By construction, $h \in \mathcal{B}_{4,10}$. Since $K(g_1, g_2) = \emptyset$, we deduce from Property 9 that:

$$\text{Wsupp}_h = (\text{Wsupp}_{g_1} \times \mathbb{F}_2) \sqcup (\text{Wsupp}_{g_2} \times \mathbb{F}_2) = (H \times \mathbb{F}_2) \sqcup (\{a_4\} \times \mathbb{F}_2).$$

Then, we observe that $V' = H \times \mathbb{F}_2$ is a vector space of dimension 3 and $D' = a_4 \times \mathbb{F}_2$ is an affine space of dimension 1 such that $V' \cap D' = \emptyset$. We illustrate this construction in Figure 2. We denote $(a_i, 0)$ and $(a_i, 1)$

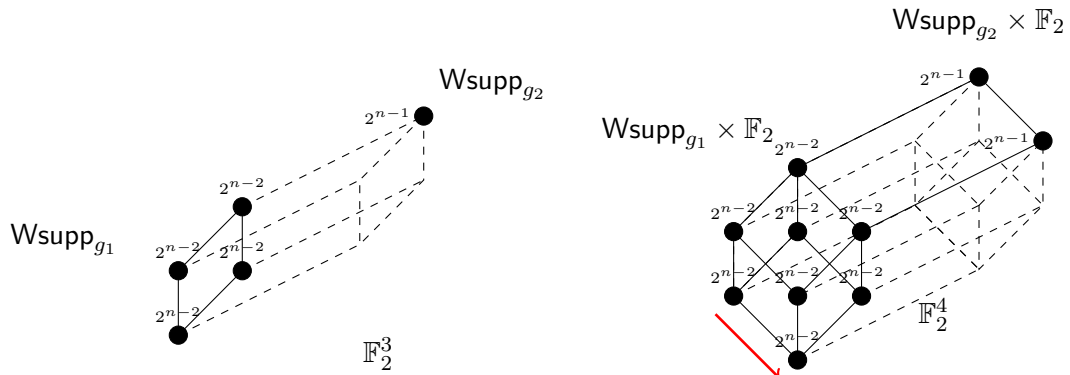


Fig. 2: Illustration of the construction in Theorem 3.

by u_i and v_i respectively. Then, we define $E'_i = \{u_i, v_i\}$ for $i \in [0, 3]$. We notice that by construction the E'_i 's are pairwise parallel and E'_1 is one edge of the cube V' . D' is also parallel to E'_0 .

Since h is a reduction of f , Wsupp_f and Wsupp_h are equivalent, *i.e.* there exists an affine isomorphism $A : a + \mathcal{WS}_h \rightarrow b + \mathcal{WS}_f$ for some $a \in \text{Wsupp}_h$ and $b \in \text{Wsupp}_f$. Moreover, accordingly to Proposition 4, $|W_f(u)| = 2^{n-4} |W_h(A(u))|$ for all $u \in \text{Wsupp}_f$. Therefore, the partition in affine subspaces are preserved using Proposition 1, that is there exists $V = A(V')$ an affine subspace of dimension 3 and $D = A(D')$ an affine subspace of dimension 1 such that $\text{Wsupp}_f = V \sqcup D$. We define E_0, E_1, E_2, E_3 such that $E_i = A(E'_i) \subset V$. Since parallelism is preserved, these four lines are pairwise distinct parallel lines. Thus, the E_i 's define edges of V . Moreover, D is parallel to E_1 since E'_1 and D' are. Thus, D is parallel to (at least) one edge of V , this proves the second item of the statement.

Using Property 9, we compute $\text{AWspec}_h = \{0 \times 6\} \cup \{2^3 \times 2\} \cup \{2^2 \times 8\}$. Moreover, $V' = \{u \in \mathbb{F}_2^4 \mid |W_h(u)| = 2^2\}$ and $D' = \{u \in \mathbb{F}_2^4 \mid |W_h(u)| = 2^3\}$. Thus, using Proposition 4, we deduce that $\text{AWspec}_f = \{0 \times (2^n - 10), 2^{n-1} \times 2, 2^{n-2} \times 8\}$. Since $|W_f(u)| = 2^{n-4} |W_h(A(u))|$, then

$$\begin{aligned} \{u \in \text{Wsupp}_f \mid |W_f(u)| = 2^{n-1}\} &= \underbrace{A(\{u \in \text{Wsupp}_g \mid |W_h(u)| = 2^3\})}_{=D'} = D, \\ \{u \in \text{Wsupp}_f \mid |W_f(u)| = 2^{n-2}\} &= \underbrace{A(\{u \in \text{Wsupp}_g \mid |W_h(u)| = 2^2\})}_{=V'} = V. \end{aligned}$$

□

Now that the structure of Walsh supports of size 10 are known, we consider the possible 4-tuples (s, t, k, ξ) for Siegenthaler's construction where $t = 10$ and $s \leq 10$.

Proposition 8. *Let $f \in \mathcal{B}_{n,s}$ and $g \in \mathcal{B}_{n,10}$, we denote $k = |K(f, g)|$ and $\xi = |\Xi(f, g)|$, then the following holds:*

1. *If $s = 4$, we have:*
 - (a) *If $k = 4$ then $\xi \neq 1$,*
 - (b) *$\xi \in \{0, 1, 2\}$.*
2. *If $s = 8$, we have:*
 - (a) *$\xi \in \{k, \dots, k-3\}$,*
 - (b) *If $\xi \geq 5$, then $k = 8$ and $\xi = 7$.*
3. *If $s = 10$, we have:*
 - (a) *$\xi \in \{k, \dots, k-4\}$,*
 - (b) *If $k \geq 8$ and $\xi \geq 5$ then $\xi = k$. Moreover, If $k = 7$, then $\xi \neq 7$.*

Proof. We use Theorem 3 to write $\text{Wsupp}_g = V \cup D$ where V is an affine subspace of dimension 3 and D is of dimension 1 such that $W_g(u) = \pm 2^{n-2}$ for $u \in V$ and $W_g(u) = \pm 2^{n-1}$ for $u \in D$.

1. If $s = 4$, since there are two values 2^{n-1} in AWspec_g and Wsupp_f is only composed of 2^{n-1} and 0, we deduce $\xi \leq 2$.

If $k = 4$ and $\xi = 1$, then the intersection is one affine subspace of dimension 2 included in Wsupp_g such that only one vector verifies $W_f(u) = \pm 2^{n-1}$. Let H be such an affine subspace.

According to Theorem 3, $D = \{a, b\}$ is parallel to one edge of the cube V , *i.e.* there exists $u, v \in V$ such that $u + v = a + b$. We have $H = \{a, x, y, z\}$ with x, y, z pairwise distinct and not equal to a or b . The cube V can be split in two parts $V = H_1 \cup ((a + b) + H_1)$ where $(a + b) + H_1$ and H_1 are two distinct planes.

If x, y, z are in the same plane, then a belongs to this plane included in V , which is contradictory since $a \in D$ and $V \cap D = \emptyset$.

Otherwise, there are two vectors among x, y and z , say x, y , belonging to the same plane. Thus, $(a + b) + x, (a + b) + y$ and z belong to the same plane included in V and $a + (a + b) + x + (a + b) + y + z =$

$a + x + y + z = 0$. Hence, we are in the same situation as before. Accordingly, we deduce that $a \in V$, which leads to a contradiction. Therefore, H does not exist, allowing us to conclude that the couple $(k = 4, \xi = 1)$ is impossible.

2. If $s = 8$, we recall that according to Property 6, $\text{AWspec}_f = \{0 \times (2^n - 8), 2^{n-2} \times 7, (3 \cdot 2^{n-3}) \times 1\}$ and Wsupp_f is an affine space of dimension 3. Let $a \in \text{Wsupp}_f$ such that $W_f(a) = \pm 3 \cdot 2^{n-2}$.
 - (a) $u \in K(f, g) \setminus (\Xi(f, g) \cup \{a\})$ if and only if $|W_f(u)| = 2^{n-2}$ and $|W_g(u)| = 2^{n-1}$. Since there are two $u \in \text{Wsupp}_g$ such that $|W_g(u)| = 2^{n-1}$, we deduce that $|K(f, g) \setminus (\Xi(f, g) \cup \{a\})| \leq 2$. Hence, $|K(f, g) \setminus \Xi(f, g)| \leq 3$, i.e. $\xi \in \{k, k-1, k-2, k-3\}$.
 - (b) Since the only nonzero values in common in AWspec_f and AWspec_g are the 2^{n-2} , we have $\Xi(f, g) = \{u \in K(f, g) \mid |W_f(u)| = |W_g(u)| = 2^{n-2}\}$. Since, V and Wsupp_f are both affine spaces of dimension 3, $V \cap \text{Wsupp}_f$ is an affine space of dimension at most 3. Thus, if $\xi \geq 5$, we have $|\text{Wsupp}_f \cap V| \geq 5$. Therefore, $V = \text{Wsupp}_f$. Hence, the only vector of the intersection not in $\Xi(f, g)$ is a since $|W_f(a)| = 3 \cdot 2^{n-2}$ and $|W_g(a)| = 2^{n-2}$. Otherwise, if $u \in K(f, g) \setminus \{a\}$, we have $|W_f(u)| = |W_g(u)| = 2^{n-2}$. Thus, $k = 8$ and $\xi = 7$.

3. If $s = 10$:

- (a) given the cardinality of the intersection k , we have $\xi \in \{k, \dots, k-4\}$. Indeed, let us assume that $\xi \leq k-5$. If $W_f(u) \neq \pm 2^{n-1}$, it is equal to $\pm 2^{n-2}$ and vice versa. Thus, we may be able to find four vectors such that either $|W_f(u)| = 2^{n-1}$ and $|W_g(u)| = 2^{n-2}$ or $|W_f(u)| = 2^{n-2}$ and $|W_g(u)| = 2^{n-1}$ is true. In that case, all the remaining values element are associated to the value 2^{n-2} in the absolute spectrum, hence we cannot find any other element not in $\Xi(f, g)$. Thus, $\xi \geq k-4$.
- (b) Let V^f and V^g be the affine spaces of dimension 3 in each support from Theorem 3. $V^f \cap V^g$ is an affine subspace of dimension up to 3 included in $\Xi(f, g)$, since all the absolute values for these vectors are 2^{n-2} .

Let $a, b \in \text{Wsupp}_f$ such that $|W_f(a)| = |W_f(b)| = 2^{n-1}$. We define $\ell = |\Xi(f, g) \cap \{a, b\}|$, accordingly $|V^f \cap V^g| = \xi - \ell$ and we deduce the following:

- If $\ell = 0$, then ξ must be a power of 2,
- If $\ell = 1$, then we have $\xi \in \{k, k-1, k-2\}$,
- If $\ell = 2$, then all the other elements of the intersection are 2^{n-2} valued, thus $\xi = k$.

We consider the case $k \geq 8$ and $\xi \geq 5$, doing a disjunction on the value of ℓ . If $\ell = 0$, then $\xi \in [5, 10]$ is a power of 2 and thus $\xi = 8$. We also deduce that $|V^f \cap V^g| = \xi - 0 = 8$ and consequently $V^f = V^g$ and $k = \xi$. Then, if $\ell = 1$ we obtain $\xi \geq k-2$. Then, $\xi \geq 6$. Therefore, $|V^f \cap V^g| = \xi - \ell \geq 5$. $V^f \cap V^g$ is an affine space of dimension at most 3, therefore $|V^f \cap V^g| = 8$. Accordingly, this case implies $k = \xi$. Finally, for $\ell = 2$ we have already shown that $\xi = k$ hence we can conclude that $k \geq 8$ and $\xi \geq 5$ implies $k = \xi$.

Finally, we show that $k = 7$ implies $\xi \neq 7$. The condition $k = \xi$ is satisfied only when D^f intersects with D^g and V^f with V^g . The size of D^f being 2, $\xi = 7$ implies V^f and V^g intersect in 5 to 7 elements. Since V^f and V^g are affine spaces of dimension 3, their intersection cannot be exactly 5, 6 nor 7 elements. The contradiction allows to conclude.

□

As for the cardinalities lower than 10 in the precedent part, we study $\mathcal{C}^{s,10}$. We represent the different 4-tuples in Tables $\text{CT}^{4,10}$, $\text{CT}^{8,10}$ and $\text{CT}^{10,10}$. Based on Proposition 8, a cell is colored if the construction is impossible. The corresponding results are stated in Proposition 9.

Proposition 9. *Let $n \in \mathbb{N}^*$, $f \in \mathcal{B}_{n,s}$ and $g \in \mathcal{B}_{n,10}$, we denote $|K(f, g)|$ by k and $|\Xi(f, g)|$ by ξ . For $s = 4$ (resp. 8, 10), the couples (k, ξ) corresponding to the colored cells in Table $\text{CT}^{4,4}$ (resp. $\text{CT}^{4,8}$, $\text{CT}^{8,8}$) are*

$\xi \backslash k$	0	1	2	3	4
0	28	26	24	22	20
1		25	23	21	19
2			22	20	18
3				19	17
4					16

$CT^{4,10}$: Every pair k, ξ possible for $s = 4$ and $t = 10$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 9).

$\xi \backslash k$	0	1	2	3	4	5	6	7	8
0	36	34	32	30	28	26	24	22	20
1		33	31	29	27	25	23	21	19
2			30	28	26	24	22	20	18
3				27	25	23	21	19	17
4					24	22	20	18	16
5						21	19	17	15
6							18	16	14
7								15	13
8									12

$CT^{8,10}$: Every pair k, ξ possible for $s = 8$ and $t = 10$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 9).

$\xi \backslash k$	0	1	2	3	4	5	6	7	8	9	10
0	40	38	36	34	32	30	28	26	24	22	20
1		37	35	33	31	29	27	25	23	21	19
2			34	32	30	28	26	24	22	20	18
3				31	29	27	25	23	21	19	17
4					28	26	24	22	20	18	16
5						25	23	21	19	17	15
6							22	20	18	16	14
7								19	17	15	13
8									16	14	12
9										13	11
10											10

$CT^{10,10}$: Every pair k, ξ possible for $s = 10$ and $t = 10$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 9).

impossible. Moreover, $\mathcal{C}^{s,10}$ (Definition 6) is included in the set of values corresponding to non-colored cells of the corresponding table.

Proof. We show that if a cell having entries (k, ξ) is colored in the chosen table, then the pair $k = |K(f, g)|$, $\xi = |\Xi(f, g)|$ is impossible. Every cell such that $\xi > k$ is colored in black, since by definition $\xi \leq k$. Then, we distinguish between the different cases using the items of Proposition 8:

- $(s, t) = (4, 10)$: The red colored cells of Table $\text{CT}^{4,10}$ are such that $\xi > 2$ and according to Item 1b, $|\Xi(f, g)| \leq 2$. The blue colored cell of Table $\text{CT}^{4,10}$ have entries $(k, \xi) = (4, 1)$ and $(|K(f, g)|, |\Xi(f, g)|) \neq (4, 1)$ according to Item 1a.
- $(s, t) = (8, 10)$: The red colored cells of Table $\text{CT}^{8,10}$ are such that $\xi \geq 5$ and $(k, \xi) \neq (8, 7)$, and according to Item 2b, if $|\Xi(f, g)| \geq 5$, then $(|K(f, g)|, |\Xi(f, g)|) = (8, 7)$. The blue colored cells of Table $\text{CT}^{8,10}$ are such that $\xi < k - 3$ and $|\Xi(f, g)| \geq |K(f, g)| - 3$ according to Item 2a.
- $(s, t) = (10, 10)$: The red colored cells of Table $\text{CT}^{10,10}$ are such that for $k \geq 8$ and $\xi \geq 5$ they satisfy $\xi \neq k$ and the cell for $k = 7, \xi = 7$, since according to Item 3b, if $|K(f, g)| \geq 8$ and $|\Xi(f, g)| \geq 5$ then $|\Xi(f, g)| = |K(f, g)|$, and $(|K(f, g)|, |\Xi(f, g)|) \neq (7, 7)$. The blue colored cells of Table $\text{CT}^{10,10}$ are such that $\xi < k - 4$ and $|\Xi(f, g)| \geq |K(f, g)| - 4$ according to Item 3a.

We conclude on $\mathcal{C}^{s,10}$, we recall that the content of each cell in each table is the value $r = 2(s + t) - 2k - \xi$. From Theorem 1, if the construction (s, t, k, ξ) is possible then the Walsh support cardinality of the obtained function is $r = 2(s + t) - 2k - \xi$. Since each possible pair k, ξ such that $k \leq s$ is considered, the set $\mathcal{C}^{s,10}$ is contained in the set of numbers in bold in the corresponding table. □

Corollary 5. We have $\mathcal{C}^{\leq 10} \subset \{1, 4, 8, 10, 13, 16, 18, 20, 21, \dots, 38\} \cup \{40\}$. Moreover, $\mathcal{B}_{n,11} = \emptyset$ and $\mathcal{B}_{n,12} = \emptyset$ for all $n \in \mathbb{N}$.

Proof. Since for all $n \in \mathbb{N}^*$, $\mathcal{B}_{n,9} = \emptyset$ (Corollary 4), then $\mathcal{C}^{\leq 9} = \mathcal{C}^{\leq 8}$. We list the numbers appearing in bold in Tables $\text{CT}^{4,10}$, $\text{CT}^{8,10}$, $\text{CT}^{10,10}$ and we add the numbers corresponding to the bold cells obtained using Table $\text{CT}^{1,t}$ with $t = 10$ (20 and 22). These give a superset $(\{10, 13, 16, 18, 20, 21, \dots, 40\})$ of $\bigcup_{s=1}^{10} \mathcal{C}^{s,10}$. Since we already know a superset of $\mathcal{C}^{\leq 9}$ (Corollary 4), we deduce that $\mathcal{C}^{\leq 10} \subset \{1, 4, 8, 10, 13, 16, 18, 20, 21, \dots, 38\} \cup \{40\}$.

As a consequence of Corollary 2, since $11 \notin \mathcal{C}^{\leq 10}$, we deduce that for all $n \in \mathbb{N}^*$, $\mathcal{B}_{n,11} = \emptyset$. Thus, $\mathcal{C}^{\leq 10} = \mathcal{C}^{\leq 11}$ and similarly, since $12 \notin \mathcal{C}^{\leq 11}$, we conclude that for all $n \in \mathbb{N}^*$, $\mathcal{B}_{n,12} = \emptyset$. □

6.3 Walsh Supports of Size 13 and Study of $\mathcal{C}^{\leq 13}$

As for the Walsh supports of cardinality 10 in the previous part, we use the tables and results established in Section 6.1 and Section 6.2 to characterize the (s, t, k, ξ) -constructions allowing us to obtain the functions of $\mathcal{B}_{n,13}$.

Proposition 10. Let $h \in \mathcal{B}_{n,13}$, $f \in \mathcal{B}_{n-1,s}$ and $g \in \mathcal{B}_{n-1,t}$ such that $h = \text{Sieg}[f, g]$ and $s \leq t$, then one of the following is true:

- $s = 4$ and $t = 4$ with $|K(f, g)| = 1$ and $|\Xi(f, g)| = 1$,
- $s = 8$ and $t = 10$ with $|K(f, g)| = 8$ and $|\Xi(f, g)| = 7$,
- $s = 10$ and $t = 10$ with $|K(f, g)| = 9$ and $|\Xi(f, g)| = 9$,
- $s = 13$ and $t = 13$ with $|K(f, g)| = 13$ and $|\Xi(f, g)| = 13$.

Proof. According to Tables $CT^{1,t}$, $CT^{4,4}$, $CT^{4,8}$, $CT^{8,8}$, $CT^{4,10}$, $CT^{8,10}$, $CT^{10,10}$ and the fact that $\mathcal{B}_{n,r} = \emptyset$ for $r \in \{2, 3, 5, 6, 7, 9, 11, 12\}$ (Property 7, Corollary 4 and 5), the only possible 4-tuples (s, t, k, ξ) to obtain 13 are $(4, 4, 1, 1)$, $(8, 10, 8, 7)$, $(10, 10, 9, 9)$ and $(13, 13, 13, 13)$. For each possible 4-tuple with s and t lower than 13, we give an example of such construction, giving the truth table of f and g in hexadecimal:

(s, t, k, ξ)	f	g
$(4, 4, 1, 1)$	9565	d42b
$(8, 10, 8, 7)$	e1f0	4b00
$(10, 10, 9, 9)$	f92d	b103

□

Despite 13 is small than 16, no Walsh support of size 13 is contained in a 4-dimensional subspace, which also explains why there exist no Walsh support of cardinality 13 for functions in $\mathcal{B}_4$. In the following lemma we establish the dimension of the Walsh space of the elements of $\mathcal{B}_{n,13}$.

Lemma 5. *Let $n \geq 5$ and $h \in \mathcal{B}_{n,13}$, then $\dim(\mathcal{WS}_h) = 5$.*

Proof. According to Proposition 10, h is obtained from a (s, t, k, ξ) -construction with (s, t, k, ξ) equals to one of the following 4-tuples: $(4, 4, 1, 1)$, $(8, 10, 8, 7)$, $(10, 10, 9, 9)$ and $(13, 13, 13, 13)$. We begin by proving that $\dim(\mathcal{WS}_h) = 5$ if $s, t < 13$ using Proposition 2 since $k > 0$ for all constructions:

$$\dim(\mathcal{WS}_h) \leq \dim(\mathcal{WS}_f) + \dim(\mathcal{WS}_g) - \dim(\mathcal{WS}_f \cap \mathcal{WS}_g) + 1$$

Let $f \in \mathcal{B}_{n-1,s}$ and $g \in \mathcal{B}_{n-1,t}$ such that $h = \text{Sieg}[f, g]$. We consider the possible different 4-tuples (s, t, k, ξ) :

- If $(s, t, k, \xi) = (4, 4, 1, 1)$. Wsupp_f and Wsupp_g are affine planes (Property 6), then $\dim(\mathcal{WS}_f) = \dim(\mathcal{WS}_g) = 2$. Moreover, $\mathcal{WS}_f \cap \mathcal{WS}_g = \{0_{n-1}\}$. It follows that $\dim(\mathcal{WS}_h) \leq 2 + 2 - 0 + 1 = 5$ (Proposition 2).
- If $(s, t, k, \xi) = (8, 10, 8, 7)$. Wsupp_f is an affine cube, then $\dim(\mathcal{WS}_f) = 3$ (Property 6), and $\dim(\mathcal{WS}_g) = 4$ (Lemma 4). Since $k = 8$, then $\text{Wsupp}_f \subset \text{Wsupp}_g$ and thus $\mathcal{WS}_f \subset \mathcal{WS}_g$. It follows that $\mathcal{WS}_f \subset \mathcal{WS}_g = \mathcal{WS}_g$. Therefore, $\dim(\mathcal{WS}_h) \leq 3 + 4 - 3 + 1 = 5$ (Proposition 2).
- If $(s, t, k, \xi) = (10, 10, 9, 9)$, then $\dim(\mathcal{WS}_f) = \dim(\mathcal{WS}_g) = 4$ (Lemma 4). Since $k = 9$, $\mathcal{WS}_f$ and $\mathcal{WS}_g$ share at least nine vectors. Thus, $\mathcal{WS}_f \cap \mathcal{WS}_g$ is a vector space of dimension at least 4. We deduce then that $\mathcal{WS}_f = \mathcal{WS}_g$. It follows that $\mathcal{WS}_f \cap \mathcal{WS}_g = \mathcal{WS}_f$ and thus $\dim(\mathcal{WS}_h) \leq 4 + 4 - 4 + 1 = 5$ (Proposition 2).

The minimal $m \in \mathbb{N}$ such that $\mathcal{B}_{m,13} \neq \emptyset$ is $m = 5$. Accordingly to Corollary 3, $\dim(\mathcal{WS}_h) \geq 5$ and thus $\dim(\mathcal{WS}_h) = 5$.

One case is remaining: if $(s, t, k, \xi) = (13, 13, 13, 13)$. We assume that $\dim(\mathcal{WS}_f) = 5$. Since $\mathcal{WS}_f = \mathcal{WS}_g$, we have $\dim(\mathcal{WS}_h) \leq 5 + 1 = 6$. We prove by contradiction that $\dim(\mathcal{WS}_h) \leq 5$. We assume that $\dim(\mathcal{WS}_h) = 6$, thus there exist $h^6 \in \mathcal{B}_{6,13}$ such that h^6 is a reduction of h (Theorem 2). However, h^6 is also in the only EA class of $\mathcal{B}_{6,13}$ and thus $\dim(\mathcal{WS}_{h^6}) = 5$ (Property 12). Therefore, $\dim(\mathcal{WS}_h) = 5$.

We conclude by induction that for all $n \geq 5$ and any $h \in \mathcal{B}_{n,13}$, we have $\dim(\mathcal{WS}_f) = 5$.

□

We give more properties on the Walsh spectrum and the structure of the Walsh support of cardinality 13 in the following theorem, corollary and proposition.

Theorem 4. *Let $n \in \mathbb{N}^*$ and $f \in \mathcal{B}_{n,13}$, the following holds:*

- $\text{AWspec}_f = \{0 \times (2^n - 13), 2^{n-2} \times 12, 2^{n-1} \times 1\}$,
- there exist V_1, V_2 two affine subspaces of dimension 3 such that $V_1 \cap V_2 = \{a, b\}$ and $\text{Wsupp}_f = V_1 \cup V_2 \setminus \{b\}$,
- $|W_f(a)| = 2^{n-1}$,
- $\mathcal{WS}_f = (a + V_1) + (a + V_2)$.

Proof. Since $f \in \mathcal{B}_{n,13}$, then $\dim(\mathcal{WS}_f) = 5$ (Lemma 5). Accordingly to Theorem 2, there exists $g \in \mathcal{B}_{5,13}$ such that g is a reduction of f (Definition 9). A consequence of Property 12 is that there is a unique EA class in $\mathcal{B}_{5,13}$. Therefore, using the transitivity of the reduction and that two EA equivalent functions are reductions of each other (Proposition 3), any function $h \in \mathcal{B}_{5,13}$ is a reduction of g , and thus a reduction of f .

Then, we construct a function $h \in \mathcal{B}_{5,13}$ with the $(4, 4, 1, 1)$ -construction which is one of the possible constructions for $h \in \mathcal{B}_{5,13}$ (Proposition 7). We study the geometrical properties of Wsupp_h to deduce those of Wsupp_f by reduction. We begin by constructing the Walsh supports of the two subfunctions g_1 and g_2 . We consider the following vectors of $\mathbb{F}_2^4$:

$$\begin{aligned} a_1 &= (1, 0, 0, 0) & a_2 &= (0, 1, 0, 0) \\ a_3 &= (1, 1, 0, 0) & a_4 &= (0, 0, 0, 0) \\ a_5 &= (0, 0, 1, 0) & a_6 &= (0, 0, 0, 1) \\ a_7 &= (0, 0, 1, 1) \end{aligned}$$

Thus $H_1 = \{a_1, a_2, a_3, a_4\}$ and $H_2 = \{a_4, a_5, a_6, a_7\}$ define two affine subspaces of $\mathbb{F}_2^4$ such that $H_1 \cap H_2 = \{a_4\}$. Then, we define $g_1, g_2 \in \mathcal{B}_{4,4}$ such that $\text{Wsupp}_{g_1} = H_1$, $\text{Wsupp}_{g_2} = H_2$ and $W_{g_1}(a_4) = W_{g_2}(a_4) = 2^{n-1}$. Now we define $h = \text{Sieg}[g_1, g_2]$, thereafter according to Property 9, $W_h(a_4, 0) = W_{g_1}(a_4) + W_{g_2}(a_4) = 2^n$. Moreover, by construction $h \in \mathcal{B}_{5,13}$, we illustrate its construction in Figure 3.

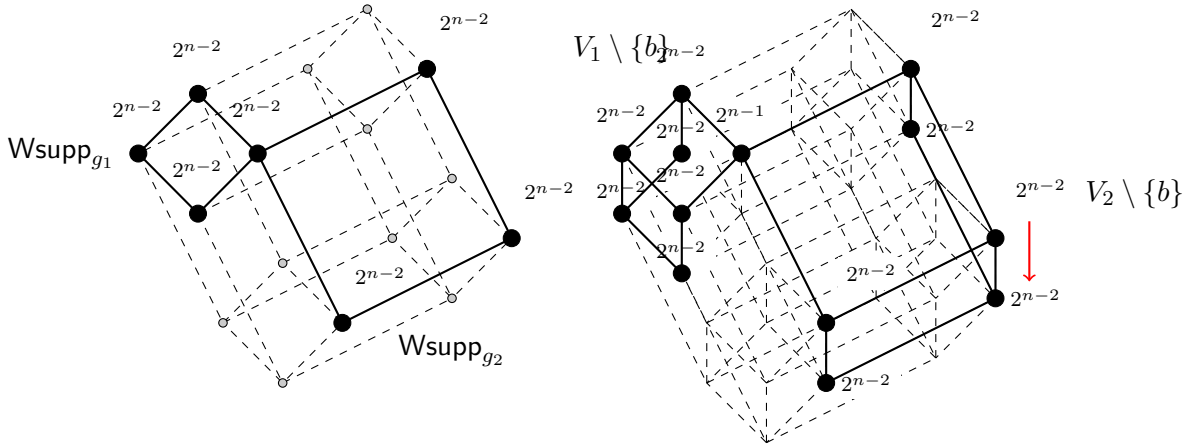


Fig. 3: Illustration of the construction of the Walsh support of cardinality 13.

We derive the absolute Walsh spectrum of h . From Property 9, we deduce that $\forall c \in \mathbb{F}_2^4$ and $\forall d \in \mathbb{F}_2$:

$$W_h(c, d) = \begin{cases} W_{g_1}(c) = \pm 2^{4-1} & \text{if } c \in H_1 \setminus \{a_4\}, \\ W_{g_2}(c) = \pm 2^{4-1} & \text{if } c \in H_2 \setminus \{a_4\}, \\ 2^{4-1} + 2^{4-1} & \text{if } (c, d) = (a_4, 0), \\ 2^{4-1} - 2^{4-1} & \text{if } (c, d) = (a_4, 1), \\ 0 & \text{otherwise.} \end{cases} \quad (4)$$

We denote by a' and b' the vectors $(a_4, 0)$ and $(a_4, 1)$ respectively. Therefore, $\text{Wsupp}_h = (H_1 \times \mathbb{F}_2 \cup H_2 \times \mathbb{F}_2) \setminus \{b'\}$. $V'_i = H_i \times \mathbb{F}_2$ is an affine space of dimension 3 and $(H_1 \times \mathbb{F}_2) \cap (H_2 \times \mathbb{F}_2) = \{a', b'\}$. We denote $(a_i, 0)$ and $(a_i, 1)$ by u_i and v_i respectively for $i \neq 4$ and define $D'_i = \{u_i, v_i\}$.

$$\text{Wsupp}_h = \underbrace{D'_1 \sqcup D'_2 \sqcup D'_3}_{=V'_1 \setminus \{a', b'\}} \sqcup \underbrace{D'_5 \sqcup D'_6 \sqcup D'_7}_{=V'_2 \setminus \{a', b'\}} \sqcup \{a'\}.$$

In particular, we deduce that $\text{Span}(D'_1 \sqcup D'_2 \sqcup D'_3) = V'_1$ and $\text{Span}(D'_5 \sqcup D'_6 \sqcup D'_7) = V'_2$.

Since h is a reduction of f , from Proposition 4 we deduce that Wsupp_h and Wsupp_f are equivalent, that is there exists an affine isomorphism $A : x + \mathcal{WS}_h \rightarrow y + \mathcal{WS}_f$ with $x \in \text{Wsupp}_h$ and $y \in \text{Wsupp}_f$, and $|W_f(A(u))| = 2^{n-5} |W_h(u)|$ for all $u \in \text{Wsupp}_h$. Therefore:

$$\text{Wsupp}_f = \underbrace{D_1 \sqcup D_2 \sqcup D_3}_{=V_1 \setminus \{a, b\}} \sqcup \underbrace{D_5 \sqcup D_6 \sqcup D_7}_{=V_2 \setminus \{a, b\}} \sqcup \{a\}.$$

where $D_i = A(D'_i)$ for $i \neq 4$, $V_i = A(V'_i)$ for $i \in \{1, 2\}$, $a = A(a')$ and $b = A(b')$. The latter vector is well defined since $b' \in V'_1$, and thus $b' \in x + \mathcal{WS}_h$. Since A is an affine isomorphism, V_1 and V_2 are both affine spaces of dimension 3. Moreover, they verify that $V_1 \cap V_2 = \{a, b\}$ and $\text{Wsupp}_f = (V_1 \cup V_2) \setminus \{b\}$.

From Relation 4, we deduce that $\text{AWspec}_h = \{0 \times (2^5 - 13), 2^3 \times 12, 2^4 \times 1\}$. Moreover, $|W_h(u, v)| = 2^4$ if and only if $(u, v) = a'$. Since $|W_f(A(u))| = 2^{n-5} |W_h(u)|$ (Proposition 4), we deduce that $\text{AWspec}_f = \{0 \times (2^n - 13), 2^{n-2} \times 12, 2^{n-1} \times 1\}$. We recall that A induces a bijection from Wsupp_h to Wsupp_f (Definition 8), then $|W_f(u)| = 2^{n-1}$ if and only if $u = A(a')$, i.e. $u = a$.

Finally, since $a \in V_1 \cap V_2$, $(a + V_1) + (a + V_2)$ is a vector space of dimension five (as they are two cubes sharing a unique line). Additionnaly, we have $(a + V_1) + (a + V_2) \subset \text{Span}(a + \text{Wsupp}_f) = \mathcal{WS}_f$ and $\dim(\mathcal{WS}_f) = 5$ (Lemma 5). Therefore, $\mathcal{WS}_f = (a + V_1) + (a + V_2)$. □

Corollary 6. *Let $f \in \mathcal{B}_{n,13}$, then there is no affine space V of dimension 3 such that $V \subset \text{Wsupp}_f$.*

Proof. Let V_1 and V_2 be the affine subspaces of dimension 3 as defined in Theorem 4, we perform a proof by contradiction. We assume that there exists V an affine space of dimension 3 such that $V \subset \text{Wsupp}_f$. We observe that we cannot have neither $V_1 = V$ nor $V_2 = V$ since neither $V_1 \subset \text{Wsupp}_f$ nor $V_2 \subset \text{Wsupp}_f$.

Let $a \in \text{Wsupp}_f$ and $b \in \mathbb{F}_2^n$ such that $V_1 \cap V_2 = \{a, b\}$. Without loss of generality, we can assume that $a = 0_n$. Indeed, if $a \neq 0_n$, we can consider f' defined as $f'(x) = f(x) \oplus a \cdot x$ and thus $W_{f'}(u) = W_f(u + a)$ (Property 1). Therefore, V is an affine cube included in Wsupp_f if and only if $a + V$ is an affine cube included in $\text{Wsupp}_{f'}$.

First, we notice that $V_1 \setminus \{a, b\}, V_2 \setminus \{a, b\}$ is a partition of $\text{Wsupp}_f \setminus \{a\}$. Thereafter, $V \setminus \{a\}$ admits $V \cap (V_1 \setminus \{a, b\}), V \cap (V_2 \setminus \{a, b\})$ as a partition. Since $b \notin V \cap (V_i \setminus \{a, b\})$, we deduce that:

$$V \setminus \{a\} = ([V \cap V_1] \setminus \{a\}) \sqcup ([V \cap V_2] \setminus \{a\}). \quad (5)$$

If $a \in V$, then we deduce that $7 = |V \cap V_1| - 1 + |V \cap V_2| - 1$, and thus $9 = |V \cap V_1| + |V \cap V_2|$. This implies that either $V \cap V_1$ or $V \cap V_2$ has a cardinality of 1 and the other one a cardinality of 8, which would contradict our preliminary observation (i.e. $V_i \neq V$). Therefore, $a \notin V$. Thus, we deduce that:

$$8 = |V \cap V_1| + |V \cap V_2|.$$

Since $V_i \neq V$, we add a constraint to this equation, that is $|V \cap V_i| \leq 4$ for $i \in \{1, 2\}$. Therefore, we conclude that $|V \cap V_1| = |V \cap V_2| = 4$ necessarily.

We have shown so far that $V = (V \cap V_1) \sqcup (V \cap V_2)$ where both $V \cap V_i$ that we denote by H_i are planes. We conclude by showing that this implies that the dimension of $\mathcal{WS}_f$ is at most 4. We observe that $a, b \notin H_i$ for $i \in \{1, 2\}$ since $0_n, b \notin V$. Therefore, $\text{Span}(H_i \cup \{0_n\}) = V_i$ for $i \in \{1, 2\}$ since they share at least five vectors. Finally, observe that:

$$\mathcal{WS}_f = \text{Span}(0_n + \text{Wsupp}_f) = \text{Span}(V_1 \cup V_2) = \text{Span}(H_1 \cup H_2 \cup \{0_n\}).$$

We notice that $\text{Span}(H_1 \cup H_2 \cup \{0_n\}) = \text{Span}(H_1 \cup H_2)$. Hence, $\mathcal{WS}_f = \text{Span}(V)$. V is an affine cube not containing 0_n , therefore the smallest vector space containing it is of dimension 4. We reach a contradiction: $\dim(\mathcal{WS}_f) = 4$ and $\dim(\mathcal{WS}_f) = 5$ according to Lemma 5. $\square$

Proposition 11. *Let $f \in \mathcal{B}_{n,13}$, we denote by E the set $\{u \in \text{Wsupp}_f \mid |W_f(u)| = 2^{n-2}\}$. Then, we have $\sum_{u \in E} u = 0_n$.*

Proof. Let V_1 and V_2 be the affine subspaces of dimension 3 as defined in Theorem 4. We have $V_1 \cap V_2 = \{a, b\}$. $\{a, b\}$ is a line of V_1 and V_2 , consequently if $u \in V_i$ then $u + a + b \in V_i$ (since it is an affine combination). If we consider $u \in E$, then $u \notin \{a, b\}$, since $b \notin \text{Wsupp}_f$ and $W_f(a) = \pm 2^{n-1}$, and thus $u + a + b \in \text{Wsupp}_f$. Therefore, we can define six pairs $u_i, v_i \in E$ such that $u_i + v_i = a + b$. Since there are exactly 12 elements in E , we deduce that

$$\sum_{u \in E} u = \sum_{i=1}^6 u_i + v_i = \sum_{i=1}^6 (a + b) = 0_n.$$

$\square$

From the results on the structure of the Walsh supports of cardinality 13 and the values of the corresponding Walsh spectra, we can study the possible 4-tuples for Siegenthaler's construction with $s \leq 13$ and $t = 13$. We study the possible constructions in Proposition 12, in a similar way as the Proposition 8 in Section 6.2 for the case $t = 10$.

Proposition 12. *Let $f \in \mathcal{B}_{n,s}$ and $g \in \mathcal{B}_{n,13}$, we denote $|K(f, g)|$ by k and $|\Xi(f, g)|$ by ξ , then the following holds:*

1. If $s = 4$, we have $\xi \in \{0, 1\}$.
2. If $s = 8$, we have:
 - (a) $\xi \in \{k, \dots, k - 2\}$,
 - (b) $k \neq 8$.
3. If $s = 10$, we have:
 - (a) $\xi \in \{k, \dots, k - 3\}$,
 - (b) $k \neq 10$ and if $k = 9$ then $\xi \neq 9$.
4. If $s = 13$,
 - (a) We have $\xi \in \{k, k - 1, k - 2\}$,
 - (b) $\xi \neq 11$.

Proof. We recall that according to Theorem 4, $\text{AWspec}_g = \{0 \times (2^n - 13), 2^{n-2} \times 12, 2^{n-1} \times 1\}$.

1. $s = 4$: According to Property 6, $\text{AWspec}_f = \{0 \times (2^n - 4), 2^{n-1} \times 4\}$. Thus, $\Xi(f, g) = \{u \in K(f, g) \mid |W_f(u)| = |W_g(u)| = 2^{n-1}\}$. Since there is only one 2^{n-1} valued vector in Wsupp_g , then $|\Xi(f, g)| \leq 1$, i.e. $\xi \in \{0, 1\}$.
2. $s = 8$: According to Property 6, $\text{AWspec}_f = \{0 \times (2^n - 8), 2^{n-2} \times 7, (3 \cdot 2^{n-2}) \times 1\}$ and Wsupp_f is an affine space of dimension 3.
 - (a) Let $a \in \text{Wsupp}_f$ such that $W_f = \pm 3 \cdot 2^{n-2}$.
 - If $a \notin K(f, g)$ then $u \in K(f, g) \setminus \Xi(f, g)$ if and only if $W_f(u) = \pm 2^{n-2}$ and $W_g(u) = \pm 2^{n-1}$, thus there is at most one $u \in K(f, g) \setminus \Xi(f, g)$ since there is only 2^{n-1} valued vector in Wsupp_g . Thus, $\xi \geq k - 1$.
 - Otherwise, $a \in K(f, g) \setminus \Xi(f, g)$ since there is no $\pm 3 \cdot 2^{n-2}$ valued vector in Wsupp_f . Then, $u \in K(f, g) \setminus (\Xi(f, g) \cup \{a\})$ if and only if $W_f(u) = \pm 2^{n-2}$ and $W_g(u) = \pm 2^{n-1}$ and thus there is at most one $u \in K(f, g) \setminus (\Xi(f, g) \cup \{a\})$ since there is only one 2^{n-1} valued vector in Wsupp_g . Hence in this case we obtain $\xi \geq k - 2$.

In both cases, we have $\xi \in \{k, k - 1, k - 2\}$.
 - (b) If $k = 8$, then $\text{Wsupp}_f \subset \text{Wsupp}_g$. However, Wsupp_f is an affine space of dimension 3 and according to Corollary 6, Wsupp_f cannot be included in Wsupp_g .
3. $s = 10$: According to Theorem 3, $\text{AWspec}_f = \{0 \times (2^n - 10), 2^{n-2} \times 8, 2^{n-1} \times 2\}$ and $V = \{u \in \text{Wsupp}_f \mid |W_f(u)| = 2^{n-2}\}$ is an affine space of dimension 3.
 - (a) $u \in K(f, g) \setminus \Xi(f, g)$ if either $W_f(u) = \pm 2^{n-2}$ and $W_g(u) = \pm 2^{n-1}$ or $W_f(u) = \pm 2^{n-1}$ and $W_g(u) = \pm 2^{n-2}$. There is at most one u verifying the first condition since there is only one 2^{n-1} valued vector in Wsupp_g and there are at most two u 's verifying the second condition since there is only two 2^{n-1} valued vector in Wsupp_f . Thus, $|K(f, g) \setminus \Xi(f, g)| \leq 3$, i.e. $\xi \in \{k, k - 1, k - 2, k - 3\}$.
 - (b) – If $k = 10$, then $\text{Wsupp}_f \subset \text{Wsupp}_g$ and thus $V \subset \text{Wsupp}_g$.
 - Let assume that $k = 9$ and $\xi = 9$. Since there are only 8 2^{n-2} valued vectors in Wsupp_f , at least one 2^{n-1} valued vector, say a , of Wsupp_f is in $\Xi(f, g)$. All the vectors of $\Xi(f, g) \setminus \{a\}$ are 2^{n-2} valued since Wsupp_g has only one 2^{n-1} valued vector for g . Thus, we deduce that eight 2^{n-2} valued vectors of Wsupp_f are in $\Xi(f, g)$, i.e. $V \subset K(f, g) \subset \text{Wsupp}_g$.

In both cases, we deduce that $V \subset \text{Wsupp}_g$. However, according to Corollary 6, as an affine space of dimension 3, V cannot be included in Wsupp_g . Thus, $k \neq 10$ and if $k = 9$ then $\xi \neq 9$.
4. $s = 13$: Let $a \in \text{Wsupp}_f$ be the unique vector such that $|W_f(a)| = 2^{n-1}$ (From the characterization of Theorem 4).
 - (a) $u \in K(f, g) \setminus \Xi(f, g)$ if either $W_f(u) = \pm 2^{n-2}$ and $W_g(u) = \pm 2^{n-1}$ or $W_f(u) = \pm 2^{n-1}$ and $W_g(u) = \pm 2^{n-2}$. There is at most one u verifying the first condition since there is only one 2^{n-1} valued vector in Wsupp_g and there is at most one u verifying the second condition since there is only one 2^{n-1} valued vector in Wsupp_f . Thus, $|K(f, g) \setminus \Xi(f, g)| \leq 2$, i.e. $\xi \in \{k, k - 1, k - 2\}$.
 - (b) We prove by contradiction that ξ cannot be equal to 11. We assume that $\xi = 11$, then $k \in \{13, 12, 11\}$. If $a \in \Xi(f, g)$, then all the other vectors of the intersection are 2^{n-2} valued, and thus $k = \xi$. Thereafter, if $k \neq \xi$ then $a \notin \Xi(f, g)$. We denote by E^f and E^g the set of the 2^{n-2} valued vectors of f and g , respectively. Since $a \notin \Xi(f, g)$, $\Xi(f, g) = E^f \cap E^g$ is a set of exactly 11 vectors. Accordingly to Proposition 11,

$$\sum_{u \in E^f} u = 0_n \quad \text{and} \quad \sum_{u \in E^g} u = 0_n.$$

Therefore, $\sum_{u \in \Xi(f, g)} u$ is a vector in $E^f \cap E^g$, which contradicts the fact that $\xi = 11$ (conditioned on $k \neq \xi$).

Then, we deduce that $k = \xi = 11$. If $a \notin \Xi(f, g)$, then using the same reasoning, we show that $\sum_{u \in \Xi(f, g)} u \in \Xi(f, g)$, which again contradicts that $\xi = 11$. Hence, the only remaining possibility is $a \in \Xi(f, g)$. We denote by V_1, V_2 the cubes in Theorem 4 such that $\text{Wsupp}_f = (V_1 \cup V_2) \setminus \{b\}$. For $i \in \{1, 2\}$ we define $K_i = K(f, g) \cap V_i$. We know that $K(f, g) = K_1 \cup K_2$ and that $K_1 \cap K_2 = \{a\}$. Thus, if we denote $|K_i|$ by k_i , we have $k_1 + k_2 - 1 = 11$, i.e. $k_1 + k_2 = 12$. Note that for $i \in \{1, 2\}$, $K_i = (\text{Wsupp}_f \cap V_i) \cap \text{Wsupp}_g$. Since, $\text{Wsupp}_f \cap V_i$ have exactly seven elements, we have that $k_i \leq 7$. Thus, we deduce that $k_i \in \{5, 6, 7\}$. Therefore, we deduce that $k_1 \geq 5$ and $k_2 \geq 5$. Since $a \in K(f, g) = \Xi(f, g)$ we have

$$\Xi(f, g) \subset a + \text{Span}(a + \text{Wsupp}_f \cap \text{Wsupp}_g) \subset a + \mathcal{WS}_f \cap \mathcal{WS}_g$$

Consequently, $(a + \mathcal{WS}_f \cap \mathcal{WS}_g) \cap V_i$ has at least five vectors for $i \in \{1, 2\}$. Thereafter, $a + V_i \subset \mathcal{WS}_f \cap \mathcal{WS}_g$. Thus, $(a + V_1) + (a + V_2) \subset \mathcal{WS}_f \cap \mathcal{WS}_g$. Using Theorem 4, $(a + V_1) + (a + V_2) = \mathcal{WS}_f$, hence, we deduce the following:

$$\mathcal{WS}_f \subset \mathcal{WS}_f \cap \mathcal{WS}_g \subset \mathcal{WS}_f.$$

Hence, $\mathcal{WS}_f = \mathcal{WS}_g$. It follows that $\mathcal{WS}_f \cap \mathcal{WS}_g = \mathcal{WS}_f$.

We define $h = \text{Sieg}[f, g]$ and accordingly to Theorem 1, Wsupp_h has cardinality $2(s+t) - 2k - \xi = 19$. Using Proposition 2 and the fact that $\dim(\mathcal{WS}_f) = 5$ (Lemma 5), we show that:

$$\dim(\mathcal{WS}_h) \leq \dim(\mathcal{WS}_f) + \dim(\mathcal{WS}_g) - \dim(\mathcal{WS}_f \cap \mathcal{WS}_g) + 1 = 5 + 5 - 5 + 1 = 6.$$

Therefore, if $k = \xi = 11$, there would exist a function $h \in \mathcal{B}_{n+1, 19}$ such that $\dim(\mathcal{WS}_h) \leq 6$. However, according to Property 12 (and Table 1), $\mathcal{B}_{6, 19} = \emptyset$ and to Corollary 3, $\dim(\mathcal{WS}_h) \geq \min\{m \in \mathbb{N} \mid \mathcal{B}_{m, 19} \neq \emptyset\} \geq 7$. We reach a contradiction, allowing us to conclude $\xi \neq 11$. $\square$

As for the cardinalities at least 10 in the precedent part, we study $\mathcal{C}^{s, 13}$. We represent the different 4-tuples in Tables $\text{CT}^{4, 13}$, $\text{CT}^{8, 13}$, $\text{CT}^{10, 13}$ and $\text{CT}^{13, 13}$. Based on Proposition 12, a cell is colored if the construction is impossible. The corresponding results are stated in Proposition 13.

Proposition 13. *Let $f \in \mathcal{B}_{n, s}$, $g \in \mathcal{B}_{n, 13}$, we denote $|K(f, g)|$ by k and $|\Xi(f, g)|$ by ξ (notations from Section 3). Then, the couples (k, ξ) corresponding to the colored cells in Table $\text{CT}^{4, 13}$ (resp. $\text{CT}^{8, 13}$, $\text{CT}^{10, 13}$ and $\text{CT}^{13, 13}$) are impossible. Moreover, $\mathcal{C}^{s, 13}$ (Definition 6) is included in the set of values corresponding to non-colored cells of the corresponding table.*

Proof. We show that if a cell having entries (k, ξ) is colored in the chosen table, then the pair $k = |K(f, g)|$ and $\xi = |\Xi(f, g)|$ is impossible. Every cell such that $\xi > k$ is colored in black, since is impossible to have $\xi > k$ by definition. Then, we distinguish between the different cases using the items of Proposition 12:

- $s = 4$: The blue colored cells of Table $\text{CT}^{4, 13}$ are such that $\xi > 1$ and according to Item 1, $|\Xi(f, g)| \leq 1$.
- $s = 8$: The red colored cells of Table $\text{CT}^{8, 13}$ are such that $k = 8$ and according to Item 2b $|K(f, g)| \neq 8$. The blue colored cells of Table $\text{CT}^{8, 13}$ are such that $\xi < k - 3$ and $|\Xi(f, g)| \geq |K(f, g)| - 2$ according to Item 2a.
- $s = 10$: The red colored cells of Table $\text{CT}^{10, 13}$ are such that $k = 10$ or $(k, \xi) = (9, 9)$, since according to Item 3b, $|K(f, g)| \neq 10$ and $(|K(f, g)|, |\Xi(f, g)|) \neq (9, 9)$. The blue colored cells of Table $\text{CT}^{10, 13}$ are such that $\xi < k - 3$ and $|\Xi(f, g)| \geq |K(f, g)| - 3$ according to Item 3a.
- $s = 13$: The green colored cell of Table $\text{CT}^{13, 13}$ is such that $(k, \xi) = (13, 12)$ and according to Lemma 1, $(|K(f, g)|, |\Xi(f, g)|) \neq (13, 12)$. The blue colored cells of Table $\text{CT}^{13, 13}$ are such that $\xi < k - 2$ and $|\Xi(f, g)| \geq |K(f, g)| - 2$ according to Item 4a. The red colored cells of Table $\text{CT}^{13, 13}$ are such that $\xi = 11$ and $|\Xi(f, g)| \neq 11$ according to Item 4b.

$\xi \backslash k$	0	1	2	3	4
0	34	32	30	28	26
1		31	29	27	25
2			28	26	24
3				25	23
4					22

CT^{4,13}: Every pair k, ξ possible for $s = 4$ and $t = 13$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 13).

$\xi \backslash k$	0	1	2	3	4	5	6	7	8
0	42	40	38	36	34	32	30	28	26
1		39	37	35	33	31	29	27	25
2			36	34	32	30	28	26	24
3				33	31	29	27	25	23
4					30	28	26	24	22
5						27	25	23	21
6							24	22	20
7								21	19
8									18

CT^{8,13}: Every pair k, ξ possible for $s = 8$ and $t = 13$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 13).

$\xi \backslash k$	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	52	50	48	46	44	42	40	38	36	34	32	30	28	26
1		49	47	45	43	41	39	37	35	33	31	29	27	25
2			46	44	42	40	38	36	34	32	29	28	26	24
3				43	41	39	37	35	33	31	28	27	25	23
4					40	38	36	34	32	30	27	26	24	22
5						37	35	33	31	29	26	25	23	21
6							34	32	30	28	25	24	22	20
7								31	29	27	24	23	21	19
8									28	26	23	22	20	18
9										25	22	21	19	17
10											21	20	18	16
11												19	17	15
12													16	14
13														13

CT^{10,13}: Every pair k, ξ possible for $s = 10$ and $t = 13$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 13).

CT^{13,13}: Every pair k, ξ possible for $s = 13$ and $t = 13$. For $\xi \leq k$, we write in the cell (ξ, k) the value of $2(s+t) - 2k - \xi$. If the cell is colored then the (s, t, k, ξ) -construction is impossible (Proposition 13).

The content of each cell in any table is the value $r = 2(s + t) - 2k - \xi$. We recall that from Theorem 1, if the construction (s, t, k, ξ) is possible then the Walsh support cardinality of the obtained function would be $r = 2(s + t) - 2k - \xi$. Thus, if $r \in \mathcal{C}^{s,13}$ with $s = 4$ (resp. $s = 8, s = 10, s = 13$) then there would exist a construction (s, t, k, ξ) and r would not be in a colored cell in Table $\text{CT}^{4,13}$ (resp. $\text{CT}^{8,13}, \text{CT}^{10,13}, \text{CT}^{13,13}$). Hence, $\mathcal{C}^{s,13}$ is included in the set of values corresponding to non-colored cells of the corresponding tables. $\square$

Corollary 7. *We have $\mathcal{C}^{\leq 13} \subset \{1, 4, 8, 10, 13, 16, 18\} \cup [20, 50] \cup \{52\}$. Moreover, $\mathcal{B}_{n,14} = \emptyset$ and $\mathcal{B}_{n,15} = \emptyset$ for all $n \in \mathbb{N}$.*

Proof. Since for all $n \in \mathbb{N}$, $\mathcal{B}_{n,11} = \emptyset$ and $\mathcal{B}_{n,12} = \emptyset$ (Corollary 5), then $\mathcal{C}^{\leq 10} = \mathcal{C}^{\leq 11} = \mathcal{C}^{\leq 12}$. We list the bold numbers in Tables $\text{CT}^{4,13}, \text{CT}^{8,13}, \text{CT}^{10,13}, \text{CT}^{13,13}$ and those of the bold cells from Table $\text{CT}^{1,t}$ with $t = 13$ (26 and 28). These give a superset of $\bigcup_{s=1}^{13} \mathcal{C}^{s,13}$, in particular we know that $14, 15, 17, 19 \notin \bigcup_{s=1}^{13} \mathcal{C}^{s,13}$. Since we already know a superset of $\mathcal{C}^{\leq 12}$ (Corollary 5), we deduce a superset of $\mathcal{C}^{\leq 13}$. Therefore, $\mathcal{C}^{\leq 13} \subset \{1, 4, 8, 10, 13, 16, 18\} \cup [20, 50] \cup \{52\}$.

As a consequence of Corollary 2, since $14 \notin \mathcal{C}^{\leq 13}$, we deduce that for all $n \in \mathbb{N}$, $\mathcal{B}_{n,14} = \emptyset$. Thus $\mathcal{C}^{\leq 13} = \mathcal{C}^{\leq 14}$ and similarly, since $15 \notin \mathcal{C}^{\leq 14}$, we conclude that for all $n \in \mathbb{N}$, $\mathcal{B}_{n,15} = \emptyset$. $\square$

6.4 Walsh Supports of Size 16, and Proving the Biggest Impossible Cardinalities

We finish this section with a study of the Walsh supports of size 16. It is not as extensive as the other parts of this section, the main goal is to derive properties sufficient to show the nonexistence of Walsh supports of cardinality 17 and 19.

The study of the Walsh supports of size 10 and 13 were heavily simplified by the observation that there exists only one type of structure for them. The Walsh supports of size 16 are the first to exhibit multiple kinds of geometrical structures and spectra. In [CM05] (Table 2), we can already observe that some function have a Walsh support of size 16 but with completely different geometrical arrangement, *e.g.* the dimension of the Walsh space can be equal to 4, 5 or 6. First, we study the possible constructions leading to a Walsh support of cardinality 16.

Proposition 14. *Let $h \in \mathcal{B}_{n,16}$, $f \in \mathcal{B}_{n-1,s}$ and $g \in \mathcal{B}_{n-1,t}$ such that $h = \text{Sieg}[f, g]$ and $s \leq t$, then one of the following is true:*

- $s = 1$ and $t = 8$ with $|K(f, g)| = 1$ and $|\Xi(f, g)| = 0$,
- $s = 4$ and $t = 4$ with $|K(f, g)| = 0$ and $|\Xi(f, g)| = 0$,
- $s = 4$ and $t = 8$ with $|K(f, g)| = 4$ and $|\Xi(f, g)| = 0$,
- $s = 10$ and $t = 10$ with $|K(f, g)| = 8$ and $|\Xi(f, g)| = 8$,
- $s = 13$ and $t = 13$ with $|K(f, g)| = 12$ and $|\Xi(f, g)| = 12$,
- $s = 16$ and $t = 16$ with $|K(f, g)| = 16$ and $|\Xi(f, g)| = 16$.

Proof. Since there is no function having a Walsh support of size 14 or 15 (Corollary 7), we deduce that $\mathcal{C}^{\leq 15} = \mathcal{C}^{\leq 13}$. In particular, all the possible constructions of $h \in \mathcal{B}_{n,16}$ using (s, t, k, ξ) constructions (Theorem 1) are in the non-colored cells of Tables $\text{CT}^{4,4}, \text{CT}^{4,8}, \text{CT}^{8,8}, \text{CT}^{4,10}, \text{CT}^{8,10}, \text{CT}^{10,10}, \text{CT}^{4,13}, \text{CT}^{8,13}, \text{CT}^{10,13}, \text{CT}^{13,13}$ and in $\text{CT}^{1,t}$ with $t \in \{4, 8, 10, 13\}$. We look for each table for which entries (k, ξ) correspond to a non-colored cell with the value 16, and the tuple (s, s, s, s) is always possible.

We deduce that (s, t, k, ξ) may be equal to $(1, 8, 1, 0), (4, 4, 0, 0), (4, 8, 4, 0), (10, 10, 8, 8), (13, 13, 12, 12)$ and $(16, 16, 16, 16)$. We give an example of each constructions with $s < 16$ and $t < 16$, providing the truth table of f and g in hexadecimal:

(s, t, k, ξ)	f	g
$(1, 8, 1, 0)$	00	b7
$(4, 4, 0, 0)$	47	ed
$(4, 8, 4, 0)$	b0	b7
$(10, 10, 8, 8)$	2177	7ddb
$(13, 13, 12, 12)$	e7e788bb	c3ff5c53

□

The higher number of possible constructions for a Walsh support of cardinality 16 and the different geometrical structures of these supports harden the study of Walsh support of size 16 (and higher cardinalities). In particular, 16 is the first cardinality for which different Walsh spectra are possible, we study them in the following proposition.

Proposition 15. *Let $f \in \mathcal{B}_{n,16}$, then we have one of the following true:*

- $\text{AWspec}_f = \{0 \times (2^n - 16), 2^{n-2} \times 16\}$,
- $\text{AWspec}_f = \{0 \times (2^n - 16), 2^{n-3} \times 10, (3 \cdot 2^{n-3}) \times 6\}$,
- $\text{AWspec}_f = \{0 \times (2^n - 16), 2^{n-3} \times 12, (3 \cdot 2^{n-3}) \times 3, (5 \cdot 2^{n-3}) \times 1\}$,
- $\text{AWspec}_f = \{0 \times (2^n - 16), 2^{n-3} \times 15, (7 \cdot 2^{n-3}) \times 1\}$.

Moreover, if f is not plateaued, then $\dim(\mathcal{WS}_f) = 4$.

Proof. We define h as $h = \text{Sieg}[f, g]$. We begin by the enumeration of all possible constructions (s, t, k, ξ) such that $h \in \mathcal{B}_{n,16}$ and we use the relation from Proposition 2 whenever the intersection of the Walsh supports is non-empty:

$$\dim(\mathcal{WS}_h) \leq \dim(\mathcal{WS}_f) + \dim(\mathcal{WS}_g) - \dim(\mathcal{WS}_g \cap \mathcal{WS}_f) + 1.$$

We deduce the spectra from Property 9 stating that:

$$W_h(u, v) = W_f(u) + (-1)^v W_g(u) = \begin{cases} W_f(u) + (-1)^v W_g(u) & \text{if } u \in K(f, g), \\ W_f(u) & \text{if } u \in \text{Wsupp}_f \setminus \text{Wsupp}_g, \\ (-1)^v W_g(u) & \text{if } u \in \text{Wsupp}_g \setminus \text{Wsupp}_f, \\ 0 & \text{otherwise.} \end{cases}$$

We perform a disjunction based on the possible 4-tuple constructions.

- $(s, t) = (1, 8)$. In that case, we have $\dim(\mathcal{WS}_f) = 0$ and $\dim(\mathcal{WS}_g) = 3$. Therefore, $\dim(\mathcal{WS}_h) \leq 4$ and $\dim(\mathcal{WS}_h) \geq 4$ (since $\mathcal{WS}_h$ is a vector space containing 16 vectors at least). Moreover, let $a \in \text{Wsupp}_g$ such that $|W_g(a)| = 3 \cdot 2^{n-3}$ (Property 6). Let $c \in K(f, g)$, we shall distinguish the obtained spectrum whether $a \neq c$ or $a = c$:

- If $a \neq c$, then for some $d \in \mathbb{F}_2$ and for all $u \in \text{Wsupp}_g \setminus \{a, c\}$, $v \in \mathbb{F}_2$:

$$|W_h(u, v)| = 2^{n-3}, \quad |W_h(a, v)| = 3 \cdot 2^{n-3}, \quad |W_h(c, d)| = \underbrace{3 \cdot 2^{n-3}}_{2^{n-1}-2^{n-3}}, \quad |W_h(c, d+1)| = \underbrace{5 \cdot 2^{n-3}}_{2^{n-1}+2^{n-3}}.$$

Therefore, $\text{Wsupp}_h = \{0 \times (2^n - 16), 2^{n-3} \times 12, 3 \cdot (2^{n-3}) \times 3, (5 \cdot 2^{n-3}) \times 1\}$.

- If $a = c$, then for some $b \in \mathbb{F}_2$ and for all $u \in \text{Wsupp}_g \setminus \{a\}$, $v \in \mathbb{F}_2$:

$$|W_h(u, v)| = 2^{n-3}, \quad |W_h(a, b)| = \underbrace{2^{n-3}}_{2^{n-1}-3 \cdot 2^{n-3}}, \quad |W_h(a, b+1)| = \underbrace{7 \cdot 2^{n-3}}_{2^{n-1}+3 \cdot 2^{n-3}}.$$

Therefore, $\text{Wsupp}_h = \{0 \times (2^n - 16), 2^{n-3} \times 15, (7 \cdot 2^{n-3}) \times 1\}$.

- $(s, t) = (4, 4)$. In that case, $k = 0$ and $\xi = 0$. Therefore, we deduce that $\text{AWSpec}_h = \text{AWSpec}_f \cup \text{AWSpec}_g = \{0 \times (2^n - 16), 2^{n-2} \times 16\}$.
- $(s, t) = (4, 8)$. In that case, we have $k = 4$ and $\xi = 0$. Therefore, $\dim(\mathcal{WS}_h) \leq 3 + 2 - 2 + 1 = 4$ and $\dim(\mathcal{WS}_h) \geq 4$. There are two spectra possible using that construction. We denote by $a \in \text{Wsupp}_g$ the element such that $W_g(a) = \pm 3 \cdot 2^{n-3}$ (Property 6).
 - If $a \in K(f, g)$, then we have for all $u \in \text{Wsupp}_g \setminus K(f, g)$ and $v \in \mathbb{F}_2$, $|W_h(u, v)| = 2^{n-3}$. If $u \in K(f, g)$ but $u \neq a$, then $\{|W_h(u, 0)|, |W_h(u, 1)|\} = \{2^{n-3}, 3 \cdot 2^{n-3}\}$ and if $u = a$, then $\{|W_h(a, 0)|, |W_h(a, 1)|\} = \{2^{n-3}, 5 \cdot 2^{n-3}\}$. Therefore, we deduce that $\text{AWSpec}_h = \{0 \times (2^n - 16), 2^{n-3} \times 12, (3 \cdot 2^{n-3}) \times 3, (5 \cdot 2^{n-3}) \times 1\}$.
 - If $a \notin K(f, g)$, then we have for all $u \in \text{Wsupp}_g \setminus K(f, g)$ and $v \in \mathbb{F}_2$ with $u \neq a$, $|W_h(u, v)| = 2^{n-3}$. If $u = a$, then $|W_h(a, v)| = 3 \cdot 2^{n-3}$. If $u \in K(f, g)$, then $\{|W_h(u, 0)|, |W_h(u, 1)|\} = \{2^{n-3}, (3 \cdot 2^{n-3})\}$. Therefore, we deduce that $\text{AWSpec}_h = \{0 \times (2^n - 16), 2^{n-3} \times 10, (3 \cdot 2^{n-3}) \times 6\}$.
- $(s, t) = (10, 10)$. In that case, we have $k = 8$ and $\xi = 8$. We recall that $\text{Wsupp}_f = V^f \cup D^f$ verifying the properties of Theorem 3. In particular, $V^f = \{a \in \text{Wsupp}_f \mid W_f(a) = \pm 2^{n-3}\}$ is an affine subspace of dimension 3. The same result holds for g and therefore $\text{Wsupp}_g = V^g \cup D^g$. We deduce then that $V^f \cap V^g = \{u \in \Xi(f, g) \mid |W_f(u)| = 2^{n-3}\}$. There are at most two u 's in $\Xi(f, g)$ such that $W_f(u) = \pm 2^{n-2}$ (since there are at most two such u 's in Wsupp_f). Therefore, there are at least six 2^{n-3} valued vectors in $\Xi(f, g)$, i.e. $|V^f \cap V^g| \geq 6$. Thus, $V^f \cap V^g$ is an affine subspace as the intersection of two affine subspaces. $V^f \cap V^g$ is therefore of dimension at least 3. Since V^f is of dimension 3, we deduce that $V^f \cap V^g$ is of dimension 3 and, in particular, has exactly eight elements. Since $V^f \cap V^g \subset K(f, g)$ and $|V^f \cap V^g| = |K(f, g)|$, we conclude that $V_f = V_g = K(f, g)$. Therefore, we can compute AWSpec_h . If $u \in K(f, g)$, then $u \in V_f \cap V_g$ and $|W_f(u)| = |W_g(u)|$, thus there exists $v_u \in \mathbb{F}_2$ such that $|W_h(u, v_u)| = 2^{n-3} + 2^{n-3} = 2^{n-2}$ and $|W_h(u, v_u + 1)| = 0$. If $u \notin K(f, g)$, then $u \in D^f \cup D^g$, thus for $v \in \mathbb{F}_2$, $|W_h(u, v)| = 2^{n-2}$. Therefore, $\text{AWSpec}_h = \{0 \times (2^n - 16), 2^{n-2} \times 16\}$.
- $(s, t) = (13, 13)$. In that case, we have $k = 12$ and $\xi = 12$, thus $K(f, g) = \Xi(f, g)$. According to Theorem 4, the spectrum of a function in $\mathcal{B}_{n-1,13}$ is

$$\{0 \times (2^{n-1} - 13), 2^{n-3} \times 12, 2^{n-2} \times 1\}.$$

Denote by $a^f \in \text{Wsupp}_f$ (resp. $a^g \in \text{Wsupp}_g$) the unique vector 2^{n-2} valued in Wsupp_f (resp. Wsupp_g). We begin by showing that $a^f \notin \Xi(f, g)$. If $a^f \in \Xi(f, g)$, we must have $a^f = a^g$ since a^g is the unique 2^{n-2} valued vectors of Wsupp_g . Let b^f and b^g be the unique vectors in Wsupp_f and Wsupp_g respectively and not in $K(f, g)$. Since, b^f and b^g are not 2^{n-2} valued, they are 2^{n-3} valued. According to Proposition 11:

$$\sum_{\substack{u \in \text{Wsupp}_f \\ W_f(u) = \pm 2^{n-3}}} u = b^f + \sum_{\substack{u \in \Xi(f, g) \\ u \neq a^f}} u = 0_{n-1}.$$

The same relation holds for g since $a^f = a^g$, thus

$$b^g + \sum_{\substack{u \in \Xi(f,g) \\ u \neq a^f}} u = 0_{n-1} = b^f + \sum_{\substack{u \in \Xi(f,g) \\ u \neq a^f}} u.$$

We deduce that $b^g = b^f$ which leads to a contradiction since $b^f \notin K(f, g)$. Thus, $a^f \neq a^g$.

Consequently, we can compute the spectra of h . Let $b \in K(f, g)$, then there exists for each $v_b \in \mathbb{F}_2$, such that $|W_h(b, v_b)| = |W_f(b)| + |W_g(b)|$, and thus $|W_h(b, v_b)| = 2^{n-2}$. And for $v \in \mathbb{F}_2$, we have $|W_h(a^f, v)| = 2^{n-2}$ and $|W_h(a^g, v)| = 2^{n-2}$. Therefore, we deduce that $\text{AWspec}_h = \{0 \times (2^n - 16), 2^{n-2} \times 16\}$.

One case is still remaining: if $s = t = 16$. We begin by showing that if $f, g \in \mathcal{B}_{n-1,16}$ and $h \in \mathcal{B}_{n,16}$, then we must have $s = t = k = \xi$. Therefore, from Theorem 1, we have $t = 2(s + t) - 2k - \xi$. By Corollary 1, we deduce that $2t - t \leq s \leq t$, and thus $s = t$. Therefore, $2k + \xi = 3t$. Since $\xi \leq k$, we observe that $3t \leq 3k$ and thus $t \leq k \leq t$. We have proven that $s = t = k$, hence $\xi = 3t - 2k = t$.

We denote by S_n^{16} the spectra listed in the proposition for n . We then prove by induction that: the possible spectra for $h \in \mathcal{B}_{n,16}$ are in S_n^{16} and that if h is not plateaued, then $\dim(\mathcal{WS}_h) = 4$. If we consider $n = 4$, then h is constructed with one of the previous constructions with $s, t \neq 16$. Moreover, all the supports are affine spaces in that case since $\text{Wsupp}_h = \mathbb{F}_2^4$.

We assume that for all $n' \in [4, n]$ the possible spectra for $h \in \mathcal{B}_{n',16}$ are in $S_{n'}^{16}$ and that if h is not plateaued, then $\dim(\mathcal{WS}_h) = 4$. Let $h \in \mathcal{B}_{n+1,16}$ and f, g such that $h = \text{Sieg}[f, g]$. If $s, t \neq 16$, then $\text{AWspec}_h \in S_4^{16}$. Moreover, if h is not plateaued, we also have that $\dim(\mathcal{WS}_h) = 4$.

Otherwise ($s = t = 16$), $f \in \mathcal{B}_{n,16}$ and $g \in \mathcal{B}_{n,16}$ with $\xi = k = 16$. By hypothesis, we know that $\text{AWspec}_f \in S_n^{16}$. Since $\Xi(f, g) = \text{Wsupp}_f = \text{Wsupp}_g$, we deduce that for all $u \in \text{Wsupp}_f$, $|W_f(u)| = |W_g(u)|$. Therefore, for all $u \in \text{Wsupp}_f$ there exists v_u such that $|W_h(u, v_u)| = 2|W_f(u)|$. Hence, $\text{AWspec}_h \in S_{n+1}^{16}$. If h is not plateaued, then f is not plateaued either (because $|W_h(u, v_u)| = 2|W_f(u)|$). Hence, by hypothesis $\dim(\mathcal{WS}_f) = 4$. Since $\mathcal{WS}_f = \mathcal{WS}_g$ (because $\text{Wsupp}_f = \text{Wsupp}_g$), we have $\dim(\mathcal{WS}_f \cap \mathcal{WS}_g) = 4$. It follows that $\dim(\mathcal{WS}_h) \leq 4 + 4 - 4 + 1 = 5$ (Proposition 2). Then, we show that $\dim(\mathcal{WS}_h) \neq 5$. If $\dim(\mathcal{WS}_h) = 5$, then there would exist a reduction $h^5 \in \mathcal{B}_{5,16}$ of h (Theorem 2). We deduce from Proposition 4 that: $\dim(\mathcal{WS}_h) = \dim(\mathcal{WS}_{h^5})$ (by support equivalence) and h^5 is not plateaued since $\text{AWspec}_h = \{2^{n+1-5} |W_{h^5}(u)| \mid u \in \text{Wsupp}_{h^5}\} \cup \{0 \times (2^{n+1} - 13)\}$. However, this contradicts Property 13. Therefore, $\dim(\mathcal{WS}_h) \leq 4$, allowing us to conclude $\dim(\mathcal{WS}_h) = 4$. $\square$

The last part of Proposition 15 can be restated: if $f \in \mathcal{B}_{n,16}$ is not plateaued, then Wsupp_f must be an affine space of dimension 4. Using this observation with Lemma 3, we deduce that the sum of all the elements of the Walsh support is equal to zero.

Proposition 16. *Let $f \in \mathcal{B}_{n,16}$, then we have*

$$\sum_{u \in \text{Wsupp}_f} u = 0_n.$$

Proof. We distinguish between two cases:

- If $\text{AWspec}_f = \{0 \times (2^n - 16), 2^{n-2} \times 16\}$, then f is plateaued. Therefore, if we apply Lemma 3, we have $\sum_{u \in \text{Wsupp}_f} u = 0_n$.
- Otherwise (i.e. if f is not plateaued), according to Proposition 15, $\dim(\mathcal{WS}_f) = 4$, i.e. Wsupp_f is an affine space of dimension 4. In particular, it verifies $\sum_{u \in \text{Wsupp}_f} u = 0_n$.

□

From the characterization of the absolute Walsh spectrum and Proposition 16, we can discard the two last impossible cardinalities, 17 and 19.

Proposition 17. *For all $n \in \mathbb{N}$, $\mathcal{B}_{n,17} = \emptyset$.*

Proof. The proof is by contradiction. We observe that $17 \notin \mathcal{C}^{\leq 15}$ since $\mathcal{C}^{\leq 15} = \mathcal{C}^{\leq 13}$ (Corollary 7). Accordingly to Corollary 2, if $17 \in \mathcal{C}^{s,t}$ (with $s \leq t$) then $t \leq 16$ and consequently we must have $t = 16$.

We use Corollary 1 to give a lower bound on s given t : $s \geq 2t - 17$. Then, if $t = 16$, we have $s = 16$. We assume that there exists $h \in \mathcal{B}_{n,17}$ constructed with a $(16, 16, k, \xi)$ construction as in Theorem 1. Thus, $17 = 2(16 + 16) - 2k - \xi$. Since $\xi \leq k$, the only solution is $k = 16$ and $\xi = 15$, which contradicts Lemma 1. Hence we conclude that for any $n \in \mathbb{N}$, $\mathcal{B}_{n,17} = \emptyset$. □

Proposition 18. *For all $n \in \mathbb{N}$, $\mathcal{B}_{n,19} = \emptyset$.*

Proof. We observe that $19 \notin \mathcal{C}^{\leq 15}$ since $\mathcal{C}^{\leq 15} = \mathcal{C}^{\leq 13}$ (Corollary 7). Accordingly to Corollary 2, if $19 \in \mathcal{C}^{s,t}$ then $t \leq 19$ and consequently we must have $t = 16$ or $t = 18$ because $\mathcal{B}_{n,17} = \emptyset$ for all $n \in \mathbb{N}$ (Proposition 17). We use Corollary 1 to give a lower bound on s given t : $s \geq 2t - 19$. Then, if $t = 16$, we have $s \in \{13, 16\}$ and if $t = 18$, we have $s = 18$.

– If $t = 16$, then either $s = 13$ or $s = 16$:

1. $s = 13$. Therefore, $19 = 2(13 + 16) - 2k - \xi$ and thus $2k + \xi = 39$. Since $\xi \leq k$, we have $k = 13$ and $\xi = 13$. However, according to Lemma 1, if $s \neq t$ and $k = s$, then $\xi \neq k$. Thus, this construction is not possible.
2. $s = 16$. Therefore, $19 = 2(16 + 16) - 2k - \xi$ and thus $2k + \xi = 45$. Since $\xi \leq k$, we have either $k = 16$ and $\xi = 13$ or $k = 15$ and $\xi = 15$.

(a) If $k = 16$ and $\xi = 13$: Assume that f is plateaued. AWspec_g is equal to one of the possible spectra for 16 in Proposition 15. Since $\xi < k$, there exists one vector $u \in K(f, g)$ such that $|W_g(u)| \neq |W_f(u)| = 2^{n-2}$ and thus g cannot be plateaued. However, if g is not plateaued, there is no $2^{n-2} \in \text{AWspec}_g$ according to Proposition 15 and therefore for all $u \in K(f, g)$ $|W_g(u)| \neq |W_f(u)| = 2^{n-2}$, and thus $\xi = 0$. We deduce that f cannot be plateaued.

If f is not plateaued, then according to Proposition 15 $\dim(\mathcal{WS}_f) = 4$. Since $|K(f, g)| = 16$, we have $\text{Wsupp}_f = \text{Wsupp}_g$ and particularly $\mathcal{WS}_f = \mathcal{WS}_g$ and thus $\mathcal{WS}_f \cap \mathcal{WS}_g = \mathcal{WS}_f$.

$$\dim(\mathcal{WS}_h) \leq \dim(\mathcal{WS}_f) + \dim(\mathcal{WS}_g) - \dim(\mathcal{WS}_f \cap \mathcal{WS}_g) + 1 \leq 4 + 4 - 4 + 1 = 5.$$

Thus, if h is constructed in such way, it would verify $\dim(\mathcal{WS}_h) \leq 5$ and thus according to Theorem 2, there would exist a function $h' \in \mathcal{B}_{6,19}$. However, we already know that there is no function of six variables having a Walsh support of size 19 according to Property 12 (and Table 1). Hence, this construction is not valid.

(b) If $k = 15$ and $\xi = 15$. According to Proposition 16, we have

$$\sum_{u \in \text{Wsupp}_f} u = 0 \quad \sum_{u \in \text{Wsupp}_g} u = 0.$$

Let a^f (resp. a^g) be the vector in Wsupp_f (resp. Wsupp_g) and not in $K(f, g)$. We have then

$$a^f = \sum_{\substack{u \in \text{Wsupp}_f \\ u \neq a^f}} u = \sum_{u \in K(f, g)} u = \sum_{\substack{u \in \text{Wsupp}_g \\ u \neq a^g}} u = a^g.$$

However, by Definition $a^f \neq a^g$ otherwise $a^f \in K(f, g)$. Thus, this construction is not valid.

- If $t = 18$, then we must have $s = 18$ and $19 = 2(18 + 18) - 2k - \xi$. Since $\xi \leq k$, the only solution is $k = 18$ and $\xi = 17$. However, according to Lemma 1, if $s = t = k = 18$, then $\xi \neq 17$.

□

Summing up the cardinalities discarded in this section, we list in the following corollary the 12 values proven impossible. We study in the following section all the proven possible cardinalities.

Corollary 8. *Let $s \in \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\}$, then for all $n \in \mathbb{N}$, $\mathcal{B}_{n,s} = \emptyset$.*

7 Construction of All Possible Cardinalities

We have proven that no Walsh support of size $s \in \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\}$ exists. However, it remains unclear whether this list is exhaustive—that is, whether these are the only impossible sizes for Walsh supports. The objective of this section is to demonstrate that, starting from $n = 7$ variables, all sizes between 1 and 2^n , except for these few exceptions, are indeed possible cardinalities of a Walsh support.

We present generic constructions which, given a function $f \in \mathcal{B}_{n,s}$, allow us to build functions whose Walsh support sizes are of the form $\ell \cdot s + p$. Such constructions are already known to yield supports of size $2s$ and $2s + 2$ under mild conditions.

Construction 2 ($s \rightarrow 2s$). *Let $n \in \mathbb{N}^*$, $f \in \mathcal{B}_{n,s}$ such that $s > 1$. Let $a \in \text{Wsupp}_f$ and $g \in \mathcal{B}_n$ be defined as $g(x) = a \cdot x$. Let $h \in \mathcal{B}_{n+1}$ be defined as $h = \text{Sieg}[f, g]$, then we have: $|\text{Wsupp}_h| = 2s$ and $\text{Wsupp}_h = \text{Wsupp}_f \times \mathbb{F}_2$.*

Proof. h is a $(1, s, 1, 0)$ –construction as defined in Theorem 1, consequently $|\text{Wsupp}_h| = 2(s + 1) - 2 \cdot 1 - 0 = 2s$. As an affine function, g has the following Walsh support: $\text{Wsupp}_g = \{a\}$. Since $a \in \text{Wsupp}_f$, using Property 9 we have:

$$\forall (u, v) \in \mathbb{F}_2^n \times \mathbb{F}_2, W_h(u, v) = \begin{cases} W_f(u) & \text{if } u \in \text{Wsupp}_f \setminus \{a\}, \\ W_f(u) + (-1)^{v2^n} & \text{if } u = a, \\ 0 & \text{otherwise.} \end{cases}$$

Since $f \in \mathcal{B}_n$ and $|\text{Wsupp}_f| = s > 1$, $2^n \notin \text{AWspec}_f$. Therefore, for all $v \in \mathbb{F}_2$, $W_f(u) + (-1)^{v2^n} \neq 0$. We conclude that $\text{Wsupp}_h = \text{Wsupp}_f \times \mathbb{F}_2$. □

Construction 3 ($s \rightarrow 2s + 2$). *Let $n \in \mathbb{N}^*$, $f \in \mathcal{B}_{n,s}$ such that $s < 2^n$. Let $a \notin \text{Wsupp}_f$ and $g \in \mathcal{B}_n$ defined as $g(x) = a \cdot x$. Let $h \in \mathcal{B}_{n+1}$ defined as $h = \text{Sieg}[f, g]$, then we have: $|\text{Wsupp}_h| = 2s + 2$ and $\text{Wsupp}_h = (\text{Wsupp}_f \sqcup \{a\}) \times \mathbb{F}_2$.*

Proof. h is a $(1, s, 0, 0)$ –construction as defined in Theorem 1, consequently $|\text{Wsupp}_h| = 2(s + 1) - 2 \cdot 0 - 0 = 2s + 2$. As an affine function, g has the following Walsh support: $\text{Wsupp}_g = \{a\}$. Since $a \notin \text{Wsupp}_f$, using Property 9 we get:

$$\forall (u, v) \in \mathbb{F}_2^n \times \mathbb{F}_2, W_h(u, v) = \begin{cases} W_f(u) & \text{if } u \in \text{Wsupp}_f, \\ (-1)^{v2^n} & \text{if } u = a, \\ 0 & \text{otherwise.} \end{cases}$$

Since $f \in \mathcal{B}_n$ and $|\text{Wsupp}_f| = s > 1$, $2^n \notin \text{AWspec}_f$. Therefore, for all $v \in \mathbb{F}_2$, $W_f(u) + (-1)^{v2^n} \neq 0$. We conclude that $\text{Wsupp}_h = (\text{Wsupp}_f \sqcup \{a\}) \times \mathbb{F}_2$. □

These constructions rely on the characterization of Walsh supports of size 1, as well as on Siegenthaler's construction, here combining an arbitrary function with a function in $\mathcal{B}_{n,1}$. However, such constructions only yield Walsh supports of even sizes. To go beyond this limitation, we extend the method by leveraging the characterization of Walsh supports of size 4 (Property 6). In particular, to construct Walsh supports of odd sizes, we examine Siegenthaler's construction applied to an arbitrary function and a function in $\mathcal{B}_{n,4}$.

Lemma 6. *Let $n \geq 4$ and $f \in \mathcal{B}_{n,s}$ such that $1 < s < 2^n$. Then,*

- *If Wsupp_f is not an affine space, then there exists $h_1 \in \mathcal{B}_{n+2}$ such that $|\text{Wsupp}_{h_1}| = 4s + 3$,*
- *If $s < 2^n - 1$, then there exists $h_2 \in \mathcal{B}_{n+2}$ such that $|\text{Wsupp}_{h_2}| = 4s + 5$.*

Proof. We begin with the following preliminary result: Let $S \subset \mathbb{F}_2^\ell$. If there exists $x \in S$ such that for all distinct $y, z \in S \setminus \{x\}$, we have $y + z \in x + S$, then S is an affine subspace of $\mathbb{F}_2^\ell$. In the following we prove that $x + S$ is a vector subspace of $\mathbb{F}_2^\ell$. Since $x \in S$, we have $x + x = 0 \in x + S$. Let $x + y, x + z \in x + S$, we show that $(x + y) + (x + z) \in x + S$, i.e. $x + S$ is closed under addition, by distinguishing three cases:

- If $y = z$, then $(x + y) + (x + z) = 0 \in x + S$,
- If $y = x$ or $z = x$, say $y = x$, then $(x + y) + (x + z) = x + z \in x + S$,
- Otherwise, $y \neq z$ and $y, z \in S \setminus \{x\}$, then $(x + y) + (x + z) = y + z \in x + S$ by hypothesis.

Therefore, $x + S$ is a vector subspace of $\mathbb{F}_2^\ell$. Hence S is an affine subspace of $\mathbb{F}_2^\ell$. By contraposition, if S is not an affine space, then for all $x \in S$, there exist two distinct $y, z \in S \setminus \{x\}$ such that $y + z \notin x + S$, i.e. $x + y + z \notin S$.

Next, we show the existence of h_1 , that is the function with Walsh support of cardinality $4s + 3$. We assume that Wsupp_f is not an affine subspace of $\mathbb{F}_2^n$. We give a sketch of the construction:

1. We define a plane H_1 such that $|\text{Wsupp}_f \cap H_1| = 3$. Thus, denote by a the only vector in $H_1 \setminus \text{Wsupp}_f$.
2. Using Construction 3, we define f_1 such that Wsupp_{f_1} has cardinality $2s + 2$ and such that $\text{Wsupp}_{f_1} = (\text{Wsupp}_f \sqcup \{a\}) \times \mathbb{F}_2$.
3. Thus $H_1 \times \{0\} \subset \text{Wsupp}_{f_1}$ and there is a unique vector $u \in H_1 \times \{0\}$ such that $|W_{f_1}(u)| = 2^n$.
4. Then, we define g_1 such that $\text{Wsupp}_{g_1} = H_1 \times \{0\}$.
5. Finally, we prove that $|K(f_1, g_1)| = 4$ and $|\Xi(f_1, g_1)| = 1$, we conclude with Theorem 1 that $h_1 = \text{Sieg}[f_1, g_1]$ is such that Wsupp_{h_1} has cardinality $2(2s + 2) - 2 \cdot 4 - 1 = 4s + 3$.

We begin by showing the existence of a plane H_1 intersecting Wsupp_f in 3 points. Let $d \in \text{Wsupp}_f$. Since Wsupp_f is not an affine space, we can apply the preliminary result. There exist two distinct $b, c \in \text{Wsupp}_f \setminus \{d\}$ such that $b + c + d \notin \text{Wsupp}_f$. We define then $a = b + c + d$ and $H_1 = \{a, b, c, d\}$. Thus $|H_1 \cap \text{Wsupp}_f| = 3$.

We construct $f_1 \in \mathcal{B}_{n+1,2s+2}$ using Construction 3 with f and a , the element not in Wsupp_f . Therefore, f_1 verifies that $\text{Wsupp}_{f_1} = \text{Wsupp}_f \times \mathbb{F}_2 \cup \{(a, 0), (a, 1)\}$, and according to Property 9:

$$\forall (u, v) \in \mathbb{F}_2^n \times \mathbb{F}_2, |W_{f_1}(u, v)| = \begin{cases} |W_f(u)| & \text{if } u \in \text{Wsupp}_f, \\ 2^n & \text{if } u = a, \\ 0 & \text{otherwise.} \end{cases}$$

Since $f \in \mathcal{B}_{n,s}$ with $s > 1$, then $2^n \notin \text{AWspec}_f$. Therefore, $|W_{f_1}(u, v)| = 2^n$ if and only if $(u, v) \in \{(a, 0), (a, 1)\}$. In particular, we deduce that $(a, 0), (b, 0), (c, 0), (d, 0)$ are in Wsupp_{f_1} with $|W_{f_1}(a, 0)| = 2^n$ and $|W_{f_1}(u, 0)| \neq 2^n$ for $u \in \{b, c, d\}$.

Accordingly, $H_1 \times \{0\} = \{(a, 0), (b, 0), (c, 0), (d, 0)\}$ is an affine subspace of dimension 2 included in Wsupp_{f_1} . We define $g_1 \in \mathcal{B}_{n+1,4}$ such that $\text{Wsupp}_{g_1} = H_1 \times \{0\}$ using Property 6. According to Property 6, $\text{AWspec}_{g_1} = \{0 \times 2^{n+1} - 4, 2^n \times 4\}$. Therefore:

$$\Xi(f_1, g_1) = \{(u, v) \in K(f_1, g_1) \mid |W_{f_1}(u, v)| = |W_{g_1}(u, v)| = 2^n\}.$$

Then, we deduce that $|K(f_1, g_1)| = 4$ and $|\Xi(f_1, g_1)| = 1$ since $K(f_1, g_1) = H_1 \times \{0\}$ and $(a, 0)$ is the unique vector in $K(f_1, g_1)$ such that $|W_{f_1}(a, 0)| = |W_{g_1}(a, 0)| = 2^n$.

We finish this part by defining the function h_1 as $h_1 = \text{Sieg}[f_1, g_1]$. Using Theorem 1 with $k = 4$ and $\xi = 1$, we deduce that $|\text{Wsupp}_{h_1}| = 2(2s + 2 + 4) - 2 \cdot 4 - 1 = 4s + 3$.

Then, we show the existence of h_2 , that is the function with Walsh cardinality $4s + 5$. We give a sketch of the construction:

1. We construct f_2 such that Wsupp_{f_2} has cardinality $2s + 2$ using Construction 3 with d and f such that $\text{Wsupp}_f \cup \{d\}$ is not an affine space.
2. We define a plane H_2 such that $|H \cap f_2| = 3$.
3. Then, we define g_2 such that $\text{Wsupp}_{g_2} = H_2$.
4. Finally we prove that $|\Xi(f_2, g_2)| = 1$ and thus we conclude with Theorem 1 that $h_2 = \text{Sieg}[f_2, g_2]$ is such that Wsupp_{h_2} has cardinality $2(2s + 2) - 2 \cdot 3 - 1 = 4s + 5$.

To construct f_2 , we distinguish between two cases given the structure of Wsupp_f : if $\text{Wsupp}_f = V \setminus \{x\}$ where V is an affine space and $u \in \mathbb{F}_2^n$, then $|\text{Wsupp}_f| = 2^p - 1$ for some $p \in \mathbb{N}$. Since $s < 2^n - 1$, we deduce that $|\text{Wsupp}_f| \leq 2^{n-1} - 1$. In particular, there exists $d \neq u$ such that $d \notin \text{Wsupp}_f$. Otherwise, we can consider any $d \notin \text{Wsupp}_f$. In both cases, we have defined $d \notin \text{Wsupp}_f$, thus we construct $f_2 \in \mathcal{B}_{n+1, 2s+2}$ with d and f using Construction 3.

Then, we obtain $\text{Wsupp}_{f_2} = (\text{Wsupp}_f \times \mathbb{F}_2) \cup \{(d, 0), (d, 1)\}$. Thus, for $u \in \mathbb{F}_2^n$ and $v \in \mathbb{F}_2$, if $(u, v) \in \text{Wsupp}_{f_2}$, then $(u, v \oplus 1) \in \text{Wsupp}_{f_2}$. Since f_2 has been obtained from using Construction 3, we can use the same arguments as for f_1 above in the proof to show that it verifies $|W_{f_2}(u, v)| = 2^n$ if and only if $(u, v) \in \{(d, 0), (d, 1)\}$.

Then, we need to show that Wsupp_{f_2} is not an affine space. We prove it by contradiction. First, let us assume that Wsupp_{f_2} is an affine space. Since $\text{Wsupp}_{f_2} = (\text{Wsupp}_f \sqcup \{d\}) \times \mathbb{F}_2$, then Wsupp_{f_2} is an affine space if and only if $\text{Wsupp}_f \sqcup \{d\} = V'$ where V' is an affine space. If Wsupp_f is not of the form $V \setminus \{x\}$, we already reach a contradiction. Otherwise, the remaining cases implies $\text{Wsupp}_f \sqcup \{x\} = V$ and since we have defined d such that $d \neq x$, we have $V \cap V' = \text{Wsupp}_f$. $V \cap V'$ is an affine subspace, therefore for some $q \in \mathbb{N}$ we have $|V \cap V'| = 2^q = |\text{Wsupp}_f| = 2^p - 1$. Since $2^p - 1 = s > 1$, it is impossible. It allows us to conclude that Wsupp_{f_2} is not an affine space.

Then, we proceed by defining a plane H_2 such that $|H_2 \cap \text{Wsupp}_{f_2}| = 3$. Since Wsupp_{f_2} is not an affine space, we can apply the preliminary result of the proof. There exist two distinct elements of $\mathbb{F}_2^n \times \mathbb{F}_2$, $(b, v_b), (c, v_c) \in \text{Wsupp}_{f_2} \setminus \{(d, 0)\}$ such that $(d, 0) + (b, v_b) + (c, v_c) \notin \text{Wsupp}_{f_2}$. Then, we define $(a, v_a) = (d, 0) + (b, v_b) + (c, v_c)$ and $H_2 = \{(b, v_b), (c, v_c), (d, 0), (a, v_a)\}$, consequently $|H_2 \cap \text{Wsupp}_{f_2}| = 3$.

Neither (b, v_b) nor (c, v_c) can be equal to $(d, 1)$. Indeed, if $(b, v_b) = (d, 1)$, then $(a, v_a) = (c, v_c \oplus 1)$, but $(c, v_c \oplus 1) \in \text{Wsupp}_{f_2}$, which would contradict the definition of (a, v_a) . Thereafter, we deduce that H_2 is an affine subspace of dimension 2 having exactly three elements in common with Wsupp_{f_2} and exactly one common element with absolute Walsh value 2^n .

Then, we define $g_2 \in \mathcal{B}_{n+1, 4}$ such that $\text{Wsupp}_{g_2} = H_2$ using Property 6. We obtain:

$$\Xi(f_2, g_2) = \{(u, v) \in K(f_2, g_2) \mid |W_{f_2}(u, v)| = |W_{g_2}(u, v)| = 2^n\}.$$

We deduce that $|K(f_2, g_2)| = 3$ and $|\Xi(f_2, g_2)| = 1$ since $K(f_2, g_2) = H_2 \setminus \{(a, v_a)\}$ and $(d, 0)$ is the unique vector in $K(f_2, g_2)$ such that $|W_{f_2}(d, 0)| = |W_{g_2}(d, 0)| = 2^n$.

Finally, we construct the function h_2 as $h_2 = \text{Sieg}[f_2, g_2]$. We observe that h is a $(4, 2s + 2, 3, 1)$ -construction as defined in Theorem 1, consequently we conclude that $|\text{Wsupp}_{h_2}| = 2(2s + 2 + 4) - 2 \cdot 3 - 1 = 4s + 5$. $\square$

Remark 10. Thanks to the $4s + 3$ construction, given a function $f \in \mathcal{B}_{n, 2^n-1}$, we can construct at least one function $g \in \mathcal{B}_{n+2, 2^{n+2}-1}$. Functions whose spectrum has exactly one zero are quite rare and of particular

interest (see e.g. [CM05, LW24]). In Appendix A.2, we provide more details on the construction of such functions.

In the following we show that Lemma 6 can be used with $s = 2^{n-1}$ by exhibiting functions $f \in \mathcal{B}_{n,2^{n-1}}$ such that Wsupp_f is not an affine space. In particular, this will be used to build the function having Walsh support of size $2^{n+1} + 3$.

Lemma 7. *For $n \geq 4$, for all $k \geq n + 1$ there exists a function $f \in \mathcal{B}_{k,2^n}$ such that Wsupp_f is not an affine space.*

Proof. We begin the proof by noticing that if the result holds for $k \geq n + 1$, then the result also holds for $k + 1$. Let us assume that there exists a function $f \in \mathcal{B}_{k,2^n}$ such that Wsupp_f is not an affine space. We define $f' \in \mathcal{B}_{k+1}$ as $f'(x, y) = f(x)$ for $(x, y) \in \mathbb{F}_2^k \times \mathbb{F}_2$, i.e. $f' = \text{Sieg}[f, f]$. Therefore, according to Property 9:

$$W_{f'}(u, v) = \begin{cases} W_f(u) + (-1)^v W_f(u) & \text{if } u \in \text{Wsupp}_f \\ 0 & \text{Otherwise.} \end{cases}$$

Hence, $\text{Wsupp}_{f'} = \text{Wsupp}_f \times \{0\}$. Since Wsupp_f is not an affine space, then $\text{Wsupp}_{f'}$ is not an affine space neither.

Thanks to that preliminary remark, showing that for all $n \geq 4$ there exists $f \in \mathcal{B}_{n+1,2^n}$ such that Wsupp_f is not an affine space, is sufficient to prove this lemma. We prove it by induction. For $n = 4$, we are looking for a function $h \in \mathcal{B}_{5,16}$ such that Wsupp_h is not an affine space. To do so, we consider the construction $(4, 4, 0, 0)$ in Table CT^{4,4}, i.e. we define $h \in \mathcal{B}_{5,16}$ as $h = \text{Sieg}[f, g]$ with $f, g \in \mathcal{B}_{4,4}$ such that $\text{Wsupp}_f \cap \text{Wsupp}_g = \emptyset$.

Let $v_1 = (1, 0, 0, 0)$, $v_2 = (1, 1, 0, 0)$, $v_3 = (0, 1, 0, 0)$, $v_4 = (0, 0, 1, 0)$, $v_5 = (0, 1, 1, 0)$. We define $V_2 = \{0_4, v_1, v_2, v_3\}$ and $V_1 = \{0_4, v_3, v_4, v_5\}$. We translate V_2 such that it is disjoint with V_1 , thus we define $H_2 = V_2 + e_4$ where $e_4 = (0, 0, 0, 1)$. We denote $e_4 + v_i$ by u_i , thus $H_2 = \{e_4, u_1, u_2, u_3\}$. We represent in Figure 4 V_1, V_2 and H_2 in $\mathbb{F}_2^4$.

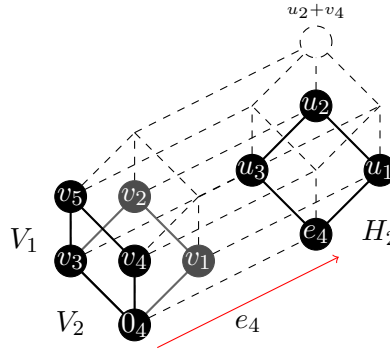


Fig. 4: Representation of V_1, V_2 and H_2 in $\mathbb{F}_2^4$ to illustrate the proof of Lemma 7. The nodes represent the vectors associated to their labels. A black node is in the set $V_1 \sqcup H_2$. The dashed node represent the vector $u_2 + v_4$.

We define $f \in \mathcal{B}_{4,4}$ and $g \in \mathcal{B}_{4,4}$ such that $\text{Wsupp}_f = V_1$ and $\text{Wsupp}_g = H_2$ using Property 6. Then, we define $h = \text{Sieg}[f, g]$, we have $\Xi(f, g) = K(f, g) = \emptyset$. Therefore, according to Theorem 1 $|\text{Wsupp}_h| = 16$. Moreover, $\text{Wsupp}_h = (V_1 \sqcup H_2) \times \mathbb{F}_2$ using Property 9.

Then, we show that Wsupp_h is not affine space. Since $0_5 \in \text{Wsupp}_h$, Wsupp_h is an affine subspace of $\mathbb{F}_2^5$ if and only if Wsupp_h is a vector subspace of $\mathbb{F}_2^5$. We recall that $\text{Wsupp}_h = (V_1 \sqcup H_2) \times \mathbb{F}_2$ and thus

Wsupp_h is a vector space if and only if $V_1 \sqcup H_2$ is a vector space. We show that $V_1 \sqcup H_2$ is not closed under addition by exhibiting $x, y \in \text{Wsupp}_h$ such that $x + y \notin \text{Wsupp}_h$. $u_2 = (1, 1, 0, 1) \in \text{Wsupp}_h$ and $v_4 = (0, 0, 1, 0) \in \text{Wsupp}_h$. However, $u_2 + v_4 = (1, 1, 1, 1) \notin V_1 \sqcup H_2$ (see Figure 4). Hence, $V_1 \sqcup H_2$ is not an affine subspace of $\mathbb{F}_2^4$ and thus Wsupp_h is not an affine subspace of $\mathbb{F}_2^5$.

Now, let us assume that for $n \geq 4$, there exists $f \in \mathcal{B}_{n+1, 2^n}$ such that Wsupp_f is not an affine space. Let $a \in \text{Wsupp}_f$, then, we use Construction 2 with a and f to build $f' \in \mathcal{B}_{n+2, 2^{n+1}}$. Moreover, since $2^n > 1$, f' satisfies $\text{Wsupp}_{f'} = \text{Wsupp}_f \times \mathbb{F}_2$. Since Wsupp_f is not an affine space, then $\text{Wsupp}_{f'}$ is not an affine space neither. □

Remark 11. The proof can be explained with classic affine geometry. If two planes V_1 and $a + V_2$ are such that their associated vector spaces, respectively V_1 and V_2 , share one common line, then the smallest space including V_1 and $a + V_2$ will be of dimension $\dim(V_1 + V_2) + 1 = 4$. It is the setting of the planes in the proof as illustrated in Figure 4.

Lemma 8. *Let $n \geq 4$, if $[s, t] \subset \mathcal{C}_n$ with $s > 1$ then $[4s + 2, 4t] \subset \mathcal{C}_n$.*

Proof. First, we handle the case of odd cardinalities. Since $t \in \mathcal{C}_n$, we have $t \leq 2^n$. Let $r \in [s, t - 1]$. We preform a disjunction of cases, considering when r is a power of two. Let assume that $r = 2^p$. We know that $p < n$ since $r < t \leq 2^n$. Since $n \geq p + 1$, there exists $f \in \mathcal{B}_{n, 2^p}$ such that Wsupp_f is not an affine space (by Lemma 7). Therefore, we can apply Lemma 6, to show that $4r + 3$ and $4r + 5$ are in $\mathcal{C}_{n+2}$. Otherwise, we assume that r is not a power of two and consequently Wsupp_f is not an affine space. If $r \leq t - 2$, then $r < 2^n - 1$. Therefore, we can also apply Lemma 6, to show that $4r + 3$ and $4r + 5$ is in $\mathcal{C}_{n+2}$. However, if $r = t - 1$, we cannot ensure that $r < 2^n - 1$ (because t may be equal to 2^n). Thereafter, by applying Lemma 6 we can only prove that $4r + 3 \in \mathcal{C}_{n+2}$, i.e. $4t - 1 \in \mathcal{C}_{n+2}$. Summing up these different cases, we get that all odd numbers in the range $[4s + 3, 4t - 1]$ belong to $\mathcal{C}_{n+2}$.

Then, we handle the case of even cardinalities. Since $1 < r$, for any $r \in [s, t]$, we can apply Construction 2 to show that $2r \in \mathcal{C}_{n+1}$ and applying Construction 2 again $4r \in \mathcal{C}_{n+2}$. For any $r < t$, we have $2r \leq 2^{n+1} - 2$ and $2r \in \mathcal{C}_{n+1}$, therefore we can apply Construction 3 and obtain $4r + 2 \in \mathcal{C}_{n+2}$. Summing up these different cases, we get that all even numbers in the range $[4s, 4t]$ belong to $\mathcal{C}_{n+2}$.

Finally, we have shown that $\{4s + 3, 4s + 5, 4s + 7, \dots, 4t - 1\} \subset \mathcal{C}_{n+2}$ and that $\{4s + 2, 4s + 4, \dots, 4t - 2, 4t\} \subset \mathcal{C}_{n+2}$, which allows us to conclude $[4s + 2, 4t] \subset \mathcal{C}_{n+2}$. □

Lemma 8 provides a key result to build whole ranges of $\mathcal{C}_{n+2}$ upon those of $\mathcal{C}_n$. Combining this result with experimental observations allows us to retrieve for any $n \in \mathbb{N}$ all the achievable cardinalities of Walsh supports.

Theorem 5 (Possible cardinalities of Walsh's supports). *We have the following:*

- $\mathcal{C}_1 = \{1\}$, $\mathcal{C}_2 = \{1, 4\}$, $\mathcal{C}_3 = \{1, 4, 8\}$, $\mathcal{C}_4 = \{1, 4, 8, 10, 16\}$,
- $\mathcal{C}_5 = [1, 2^5] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19, 27, 29, 30, 31\}$,
- $\mathcal{C}_6 = [1, 2^6] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19, 63\}$,
- For all $n \geq 7$:

$$\mathcal{C}_n = [1, 2^n] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\}.$$

Proof. For $n \leq 6$, an exhaustive search gives us that $\mathcal{C}_1 = \{1\}$, $\mathcal{C}_2 = \{1, 4\}$, $\mathcal{C}_3 = \{1, 4, 8\}$ and $\mathcal{C}_4 = \{1, 4, 8, 10, 16\}$. Using Carlet and Mesnager's classification [CM05] (Table 2) of $\mathcal{B}_5$, we deduce that $\mathcal{C}_5 = [1, 2^5] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19, 27, 29, 30, 31\}$ (see Property 8). Using Langevin's classification given in Table 1, we deduce that $\mathcal{C}_6 = [1, 2^6] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19, 63\}$.

Then, to handle the cases $n \geq 7$ we consider the following property P_n :

$$[2^{n-1} - 1, 2^n] \subset \mathcal{C}_n$$

We prove by induction that P_n is true for all $n \geq 7$. First, let us assume that P_k is true for one $k \geq 7$. Then, according to Lemma 8, $[4 \cdot (2^{k-1} - 1) + 2, 4 \cdot 2^n] \subset \mathcal{C}_{k+2}$, i.e. $[2^{k+1} - 2, 2^{k+2}] \subset \mathcal{C}_{k+2}$. In particular, $[2^{k+1} - 1, 2^{k+2}] \subset \mathcal{C}_{k+2}$ and therefore P_{k+2} is true. Accordingly, if we prove for some m that P_m and P_{m+1} are true, then P_n would be true for any $n \geq m$. In the following we show that P_7 and P_8 are true (allowing us to initiate the induction).

- We build a 7-variable function with Walsh support of cardinal r for all $r \in [2^6 - 1, 2^7]$, hence P_7 is true. An example of each function with $r \in \{2^6 - 1\} \cup [2^6 + 1, 2^7]$ is provided in Table 3, where the truth table is represented in hexadecimal. Since, $2^6 \in \mathcal{C}_6$, we conclude that $2^6 \in \mathcal{C}_7$ and thus $[2^6 - 1, 2^7] \subset \mathcal{C}_7$.
- As we have shown at the beginning of the proof, $[2^5 - 1, 2^6 - 2] \subset \mathcal{C}_6$, therefore by applying Lemma 8 we obtain $[2^7 - 2, 2^8 - 8] \subset \mathcal{C}_8$. Thereafter, only the range $[249, 256]$ is missing to show that P_8 is true. First, we retrieve the odd numbers of this range. Since 61 and 62 are not powers of two we can apply Lemma 6 to obtain Walsh support of size $61 \cdot 4 + 5 = 249$, $62 \cdot 4 + 3 = 251$ and $62 \cdot 4 + 5 = 253$ from functions with Walsh support of size 61 and 62 in $\mathcal{B}_6$. We also exhibit the function $f \in \mathcal{B}_{8,255}$ constructed by Lou and Wang [LW24] (Section IV) defined by the following hexadecimal truth table:

6880d555b33366668f0f5a5a3c3c696980ff55aa33cc66990ff05aa53cc36996.

Then, only the even numbers $\{250, 252, 254, 256\}$ are left. We have already seen that 125 and 127 are in $\mathcal{C}_7$. Therefore, applying Construction 2 since $125 > 1$ and Construction 3 since $127 < 2^7$ from function with Walsh support of size 125 and 127 in $\mathcal{B}_7$ we obtain 8-variable functions of Walsh support size $2 \cdot 125 = 250$, $2 \cdot 125 + 2 = 252$, $2 \cdot 127 = 254$ and $2 \cdot 127 + 2 = 256$. It allows us to conclude $[2^7 - 1, 2^8] \subset \mathcal{C}_8$, hence P_8 is true.

Since P_7 and P_8 are both true, we conclude that P_m is true for all $m \geq 7$, i.e. $[2^m - 1, 2^{m+1}] \subset \mathcal{C}_m$.

We recall that for any $m \in \mathbb{N}^*$ we have $\mathcal{C}_m \subset \mathcal{C}_{m+1}$ (By Remark 5). Thus, we deduce that for any $n \geq 7$:

$$\mathcal{C}_n = \mathcal{C}_6 \cup \bigcup_{m=7}^n \mathcal{C}_m \supset \mathcal{C}_6 \cup \bigcup_{m=7}^n [2^{m-1} - 1, 2^m].$$

Then,

$$\begin{aligned} \mathcal{C}_6 \cup \bigcup_{m=7}^n [2^{m-1} - 1, 2^m] &= ([1, 2^6] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19, 63\}) \cup [63, 2^n] \\ &= [1, 2^n] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\}. \end{aligned}$$

Hence $[1, 2^n] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\} \subset \mathcal{C}_n$.

Finally, since for any $r \in \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\}$ $r \notin \mathcal{C}_n$ (Corollary 8), we conclude that $\mathcal{C}_n = [1, 2^n] \setminus \{2, 3, 5, 6, 7, 9, 11, 12, 14, 15, 17, 19\}$. $\square$

8 Conclusion and Open Questions

In this article, we provided a complete characterization of the possible cardinalities of Walsh supports. We first examined these cardinalities through Siegenthaler's construction and, by leveraging its universality, established connections between the admissible support sizes in n variables and those in $n + 1$ variables.

We then introduced new tools and notions for studying Walsh supports, including the Walsh space, a reduction method, and an equivalence relation on supports. Our study also included experimental results: for $n = 6$, we enumerated the number of affine-equivalence classes for each support size. In addition, we fully characterized the geometric structure of Walsh supports in 10 and 13 variables, and partially in 16 variables. To rule out unattainable cases, we developed a sieving method that incrementally eliminates impossible values, which enabled us to determine all 12 impossible cardinalities. Finally, we proposed several constructions that map a support of size s to one of size $4s+r$, and we showed that these constructions suffice to generate all possible cardinalities for every $n \geq 7$, thereby resolving our initial problem.

By determining the sets $\mathcal{C}_n$, we resolved the open problem C3 posed by Preneel and Logachev [PL08], and we also settled the question raised by Lou and Wang [LW24] regarding the values of $n \in \mathbb{N}$ for which $2^n - 2 \in \mathcal{C}_n$. From Theorem 5, we conclude that $2^n - 2 \in \mathcal{C}_n$ if and only if $n \geq 6$. Moreover, the results established in this paper are sufficient to fully answer the broader question: for which $n \in \mathbb{N}$ does there exist a Walsh support of cardinality $2^n - p$ in $\mathbb{F}_2^n$, for any p .

Open Questions

We have shown that Walsh supports of cardinalities 10 and 13 have a unique structure. Although Walsh supports of cardinality 16 do not satisfy this uniqueness property, we proved in Section A.4 that there is a unique Walsh support of cardinality 18 (see Figure 5). The proofs of uniqueness strongly rely on the existence of a single EA class for $\mathcal{B}_{n,s}$, provided that n is large enough to contain all (s, t, k, ξ) constructions—a property seemingly satisfied for $s = 21$ and $n = 6$ (Table 1). While it appears reasonable to expect that uniqueness becomes increasingly rare as s grows, can we formally prove that $s = 18$ or $s = 21$ are the largest cardinalities exhibiting this property?

Our work is also limited by the fact that it constitutes a sign-blind study of the Walsh transform. In contrast, Hodžić, Horak and Pasalic [HPWZ19] focused solely on the signs of the Walsh spectrum of plateaued functions, which enabled them to obtain classification and construction results for these functions. It would be interesting to investigate intermediate approaches that combine support and sign information to extend our characterizations. A notable application would be the systematic construction of non-trivial plateaued functions. In Lemma 7, we provided several examples of such functions (in the sense of [HPWZ19]), *i.e.* plateaued functions whose Walsh support is not an affine space. Exploring their construction within the (s, t, k, ξ) -framework could lead to the identification of new families of plateaued functions and, potentially, to generic construction methods.

Finally, while this paper has focused exclusively on Boolean functions, a natural extension is to investigate Walsh supports in the p -ary setting. Determining the possible cardinalities of Walsh supports for p -ary functions, and understanding how these relate to the Boolean case, remains an open and challenging problem.

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A Appendix

A.1 List of Functions in Seven Variables

We have constructed an example of function for each Walsh support cardinality in [63, 128]. These functions are listed in Table 3. They have been constructed with three main techniques: generating random truth table corresponding to 7-variable functions; concatenating truth table of random function having Walsh supports of cardinality easy to construct (e.g. 4, 8, 10, 13) with random truth table of a 6-variable function, using Construction 2 and Construction 3 to retrieve all even cardinalities.

Notably, we have constructed a function whose Walsh support has a cardinality of 127:

$$f0ad2af77fdd5a876530653065306530,$$

which turns out to have a slightly better nonlinearity than the function constructed in [LW24] and the same algebraic immunity. In particular, this implies the existence of more than one EA class in seven variables for Walsh supports having a cardinality of $2^7 - 1$ since the nonlinearity is EA invariant.

n	s	$f(\text{hexadecimal})$
7	63	722793c97e7ebdbd74e2d1b8e274b8d1
7	65	0002000000002000105a290104022101
7	66	9000d888b88878880000000000000000
7	67	00000000040200000209af5b80000221
7	68	82220aaa2888a0000000000000000000
7	69	000001000000000200100d400c340c03
7	70	d95980008c0c80000000000000000000
7	71	00000000080020008214435124011aa0
7	72	e4446ccc2888a0000000000000000000
7	73	0000000004000020a0041c06032605e
7	74	e828488860a0c0000000000000000000
7	75	000000910000008064c4408226401acc
7	76	172848889fa0c0000000000000000000
7	77	0808000008000001356b4d0222de4265
7	78	6a361e82e81414280000000000000000
7	79	10000020000000000001f6182a0810682
7	80	d52a2a2a7f8080800000000000000000
7	81	0208000000000000bb909198d04f5b49
7	82	98d0b070e8a0c0000000000000000000
7	83	0000000008400000318181d9385fec8e
7	84	6280ea2a48aac0000000000000000000
7	85	00800000000000040ca1dd039ef7456cc
7	86	622a264c2e666a000000000000000000
7	87	000040002020100054c152472c5dcac1
7	88	9b4cec4cd700a0000000000000000000
7	89	00000800000090001004415000520425
7	90	7880f880f80078000000000000000000
7	91	020800020000000813830800a6609750
7	92	9deaeaeab7c0c0c00000000000000000
7	93	00000000000022000800043000080010
7	94	99804c801900cc000000000000000000
7	95	00000000200008002bf3c1a2787e1182
7	96	9eedaaa3444700000000000000000000
7	97	40000000000000000008140e10e0082020
7	98	10092a00853640c00000000000000000
7	99	0000000200000000a4f7180512ca104b
7	100	4369260cdaf0eac000000000000000000
7	101	0400000000400000bdc21f74a901c20c
7	102	92e2e2e2b0c0c0c00000000000000000
7	103	2804000400008001cb932246d614e401
7	104	1666daaabccc700000000000000000000
7	105	008000800800000084c75f857003ec8c
7	106	68284888e0a0c0000000000000000000
7	107	08082000100004009d7940fbbc35a0c3
7	108	972848881fa0c0000000000000000000
7	109	0000080004000000845f70ecbd1fa9c2
7	110	ff4020e0f74828e80000000000000000
7	111	00142000200a00802023c924709dd1e0
7	112	52227888daaaf0000000000000000000
7	113	0000c41000022004854a3e1b60e04102
7	114	9580ea80bf00c0000000000000000000
7	115	faea67927c300a3a9676f154deef21bb
7	116	95eabf40ea6ac0c00000000000000000
7	117	ca44c46bb643c06ab01cb360ad471d47
7	118	59808c80d9000c000000000000000000
7	119	22f9737486d0200b4d2b024b5b695a07
7	120	95eaeaeabfc0c0c00000000000000000
7	121	0700612040400000dd7b24f9eb85f363
7	122	14c4d08094445000000000000000000
7	123	6780f6452b8604c9981612240445a0c1
7	124	f9202c2079a0aca00000000000000000
7	125	1d1b47a31bbd9cd7eb2f95ef029d64fb
7	126	f9202c2079a0aca0cccccccccccccccc
7	127	f0ad2af77fdd5a876530653065306530
7	128	96665aaa3cccf00000000000000000000

Table 3: List of functions in $\mathcal{B}_7$ such that $|\text{Wsupp}_f| = s$ with $s \in [63, 128]$.

A.2 More Constructions for $2^n - 1$

In Section 6 and Section 7, we have provided constructions that are either impossible or always possible under specific assumptions. In practice, one can target a given cardinality by considering the integer relation $2(s + t) - 2k - \xi = r$ and testing if the (s, t, k, ξ) -construction is possible through random sampling on the cardinalities s and t . Our method for randomly sampling functions having a precise cardinality is, both for s and t , to start from a representative of one EA-class and apply a random EA transformation to that representative. We focused on building functions in $\mathcal{B}_{7,2^7-1}$ and $\mathcal{B}_{8,2^8-1}$. The existence of such functions has been proven in [LW24]. However, their work only provided a single EA-equivalence class for these cardinalities. Using our framework and Siegenthaler's construction, we were able to generate numerous functions in $\mathcal{B}_{n,2^n-1}$. To classify these functions into different EA-equivalence classes, we used EA-invariants: the absolute Walsh spectrum and the algebraic degree. As a result, we identified 97 EA-inequivalent classes for $2^7 - 1$ and 775 EA-inequivalent classes for $2^8 - 1$. Moreover, by Lemma 6, there exists a generic construction that takes as input a function in $\mathcal{B}_{n,2^n-1}$ and outputs a function in $\mathcal{B}_{n+2,2^{n+2}-1}$. Therefore, all these EA-inequivalent classes for $n = 7$ and $n = 8$ can be used to generate several inequivalent classes for larger n .

We compared our functions in $\mathcal{B}_{7,2^7-1}$ and $\mathcal{B}_{8,2^8-1}$ with those in [LW24], selecting as reference a function with the best nonlinearity and algebraic immunity. Our functions exhibit better nonlinearity for $2^7 - 1$ cardinality and equal algebraic immunity. Although our constructions for cardinality $2^8 - 1$ yield functions with lower nonlinearity, they achieve better overall algebraic immunity, which is even optimal in most cases. All our functions are listed in our [github repository](#).

$s = 2^7 - 1$	$f(\text{hexadecimal})$	NL	AI	$s = 2^8 - 1$	$f(\text{hexadecimal})$	NL	AI
[LW24]	f0ad2af77fdd5a87 6530653065306530	44	3	[LW24]	6880d555b33366668f0f5a5a3c3c6969 80ff55aa33cc66990ff05aa53cc36996	112	2
This work	48b7847b8b74748b 96e6da2a43ccf000	48	3	This work	8e4db271e8d42b17cf0c033f3f030ccf 6f6f6f6f66cccc9918f232a8fed87e28	96	4

Table 4: Comparison of the best nonlinearity and algebraic immunity for functions in $\mathcal{B}_{7,2^7-1}$ and $\mathcal{B}_{8,2^8-1}$ between our constructions and [LW24] constructions.

A.3 Walsh Supports for $s \leq 8$

It has already been shown in [PQ00] that there is no Walsh support of cardinality $s \in \{2, 3, 5, 6, 7\}$ and that the Walsh supports of cardinality 4 or 8 are affine spaces. For the sake of completeness, we provide different proofs only based on our results. Therefore, in this section we prove Property 6 and Property 7.

2 and 3 are impossible We observe in Table CT^{1,1} that 2 and 3 are not in bold font. Therefore, $2, 3 \notin \mathcal{C}^{\leq 1}$. According to Corollary 2, for all $n \in \mathbb{N}$ $\mathcal{B}_{n,2} = \emptyset$. Therefore, $\mathcal{C}^{\leq 1} = \mathcal{C}^{\leq 2}$ and thus $3 \notin \mathcal{C}^{\leq 2}$. Similarly, $\mathcal{B}_{n,3} = \emptyset$ for all $n \in \mathbb{N}$.

Walsh Support of Cardinality 4 We deduce from Corollary 1, that 4 can only be constructed with (s, t, k, ξ) -constructions with $s, t \leq 4$. We observe in Table CT^{1,1} that if $s, t \neq 4$ then $s = 1, t = 1, k = 0$

and $\xi = 1$. This construction is possible. Therefore, $h \in \mathcal{B}_{n,4}$ can be obtained by a $(1, 1, 0, 0)$ or $(4, 4, 4, 4)$ construction. If $s = 1$ and $t = 1$ with $h = \text{Sieg}[f, g]$, then $\text{Wsupp}_f = \{a\}$ and $\text{Wsupp}_g = \{b\}$, and using Property 9 we deduce that $\text{AWspec}_h = \{0 \times (2^n - 4), 2^{n-1} \times 4\}$ and $\text{Wsupp}_h = \{a, b\} \times \mathbb{F}_2$. Wsupp_h is an affine plane, and thus $\dim(\mathcal{WS}_h) = 2$.

There is a unique EA class for $\mathcal{B}_{6,4}$ (Table 1). Let $f, g \in \mathcal{B}_{n,4}$ such that $\dim(\mathcal{WS}_f) = \dim(\mathcal{WS}_g) = 2$ and $K(f, g) = \Xi(f, g) = \text{Wsupp}_f = \text{Wsupp}_g$, we deduce that $\mathcal{WS}_f \cap \mathcal{WS}_g = \mathcal{WS}_f$ (because $\mathcal{WS}_f = \mathcal{WS}_g$). Then, $h = \text{Sieg}[f, g]$ is such that $\dim(\mathcal{WS}_h) \leq 2 + 2 - 2 + 1 = 4$ (Proposition 2). However, if $\dim(\mathcal{WS}_h) = 3$, there would exist $h' \in \mathcal{B}_{6,4}$ such that $\dim(\mathcal{WS}_{h'}) = 3$ which is not the case (Table 1). Therefore, $\dim(\mathcal{WS}_h) = 2$ and thus Wsupp_h is an affine plane. Then, by recursion we show that $\dim(\text{Wsupp}_h) = 2$.

Conversely, we show that any affine plane of $\mathbb{F}_2^n$ is a Walsh support of cardinality 4. There exists $f' \in \mathcal{B}_{n,4}$ and thus $H' = \text{Wsupp}_{f'}$ an affine plane. Let $H \subset \mathbb{F}_2^n$ be an affine plane. There exists an affine automorphism $a + L^T$ such that $H = a + L^T(H')$. We define $f(x) = f'(L(x)) + a \cdot x$, then accordingly to Property 1, $\text{Wsupp}_f = a + L^T(H') = H$.

5, 6 and 7 are impossible cardinalities We observe in Tables $\text{CT}^{1,1}$, $\text{CT}^{4,4}$ and $\text{CT}^{1,t}$ with $t = 4$ that 5, 6, 7 are not in bold font. Therefore, $5, 6, 7 \notin \mathcal{C}^{\leq 4}$. According to Corollary 2, for all $n \in \mathbb{N}$ $\mathcal{B}_{n,5} = \emptyset$. Therefore, $\mathcal{C}^{\leq 4} = \mathcal{C}^{\leq 5}$ and thus $6 \notin \mathcal{C}^{\leq 5}$. Similarly, $\mathcal{B}_{n,6} = \emptyset$ for all $n \in \mathbb{N}$. Finally, we deduce that $\mathcal{C}^{\leq 5} = \mathcal{C}^{\leq 6}$ and thus $7 \notin \mathcal{C}^{\leq 6}$. Similarly, $\mathcal{B}_{n,7} = \emptyset$ for all $n \in \mathbb{N}$.

Walsh Support of Cardinality 8 We deduce from Corollary 1, that 8 can only be constructed with (s, t, k, ξ) -constructions with $s, t \leq 8$. We observe in Tables $\text{CT}^{1,1}$, $\text{CT}^{4,4}$, $\text{CT}^{4,8}$, $\text{CT}^{8,8}$ and $\text{CT}^{1,t}$ with $t \in \{4, 8\}$ that if $s, t \neq 8$ then $s = 1, t = 4, k = 1$ and $\xi = 0$. This construction is possible. Therefore, $h \in \mathcal{B}_{n,4}$ can be obtained by a $(1, 4, 1, 0)$ or $(8, 8, 8, 8)$ construction. If $s = 1$ and $t = 4$ with $h = \text{Sieg}[f, g]$, then $\text{Wsupp}_f = \{a\}$ and $\text{Wsupp}_g = \{a, b, c, d\}$ and Wsupp_g is an affine plane, and using Property 9 we deduce that $\text{AWspec}_h = \{0 \times (2^n - 8), 2^{n-2} \times 7, 3 \cdot 2^{n-2} \times 1\}$ and $\text{Wsupp}_h = \text{Wsupp}_g \times \mathbb{F}_2$. Wsupp_h is an affine cube as Wsupp_g is an affine plane, and thus $\dim(\mathcal{WS}_h) = 3$.

There is a unique EA class for $\mathcal{B}_{6,8}$ (Table 1). Let $f, g \in \mathcal{B}_{n,8}$ such that $\dim(\mathcal{WS}_f) = \dim(\mathcal{WS}_g) = 3$ and $K(f, g) = \Xi(f, g) = \text{Wsupp}_f = \text{Wsupp}_g$. We deduce that $\mathcal{WS}_f = \mathcal{WS}_g$ and thus $\mathcal{WS}_f \cap \mathcal{WS}_g = \mathcal{WS}_f$. Then, $h = \text{Sieg}[f, g]$ is such that $\dim(\mathcal{WS}_h) \leq 3 + 3 - 3 + 1$ (Proposition 2). However, if $\dim(\mathcal{WS}_h) = 4$, there would exist $h' \in \mathcal{B}_{6,8}$ such that $\dim(\mathcal{WS}_{h'}) = 4$ which is not the case (Table 1). Thereafter, $\dim(\mathcal{WS}_h) = 3$ and thus Wsupp_h is an affine cube. Then, by recursion we show that $\dim(\text{Wsupp}_h) = 3$.

Conversely, we show that any affine cube of $\mathbb{F}_2^n$ is a Walsh support of cardinality 8. There exists $f' \in \mathcal{B}_{n,8}$ and thus $H' = \text{Wsupp}_{f'}$ an affine cube. Let $H \subset \mathbb{F}_2^n$ be an affine cube. There exists an affine automorphism $a + L^T$ such that $H = a + L^T(H')$. We define $f(x) = f'(L(x)) + a \cdot x$, then accordingly to Property 1, $\text{Wsupp}_f = a + L^T(H') = H$.

A.4 Characterization of the Walsh Support of Size 18

In Table 1, we observe that there is a unique EA class for $\mathcal{B}_{6,18}$. Using Tables $\text{CT}^{s,t}$ for $s, t < 16$, we can deduce that the only (s, t, k, ξ) allowed to construct a function h of $\mathcal{B}_{n,18}$ are $(1, 8, 0, 0)$ and $(4, 10, 4, 2)$. Let $h \in \mathcal{B}_{n,18}$ and denote by f and g its subfunctions. For $t = 16$, we must have by Lemma 1, that $s \geq 14$ and thus we must have $s = t = 16$. If we try to solve the equation $18 = 2(16 + 16) - 2k - \xi$ with $\xi \leq k \leq 16$, we must have $2k + \xi = 46$. We deduce with the constraints that $k = 16$ and thus $\xi = 14$. This cannot be verified by f and g if they are plateaued. If either f or g is not plateaued, then $\mathcal{WS}_f$ is of dimension 4, which implies by Proposition 2, that $\dim(\mathcal{WS}_h) \leq 5$. Therefore, for all constructions with $s, t < 18$, we

deduce that $\dim(\mathcal{WS}_h) = 5$. Then, we can use the fact that there is a unique EA equivalent class in $\mathcal{B}_{6,18}$ which contains function having a Walsh support whose Walsh space is of dimension 5 to conclude similarly as in Theorem 3 or Theorem 4.

Consequently, we can represent the Walsh support of cardinality 18 up-to isomorphism in the vector space $\mathbb{F}_2^5$. Wsupp_h is for example equivalent to $T \sqcup D$ where T is an affine subspace of dimension 4 and D a line parallel to one edge of T . This representation is deduced from the construction $(1, 8, 0, 0)$, which is similar to the construction $(1, 4, 0, 0)$ of the Walsh support of cardinality 10.

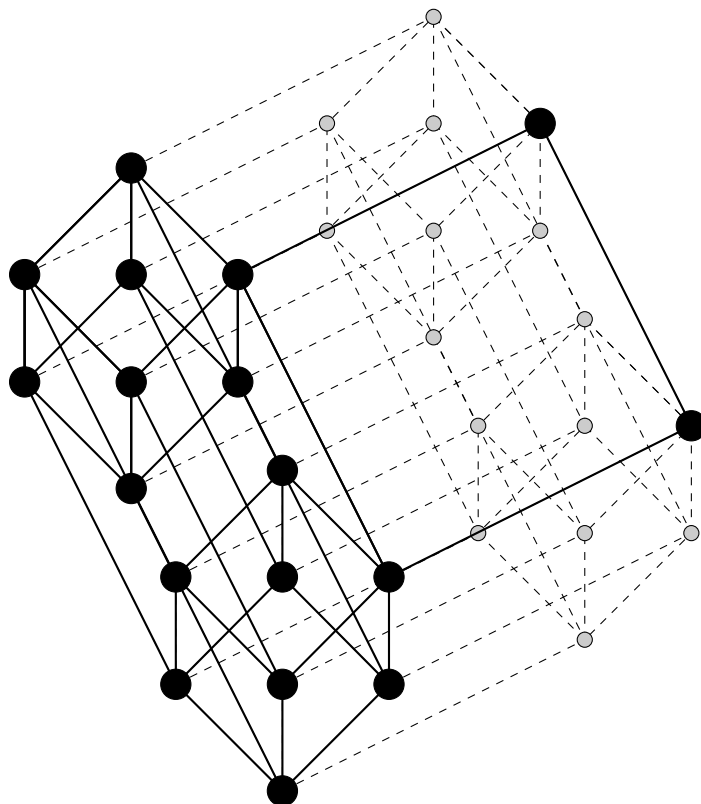


Fig. 5: Representation of the Walsh support of cardinality 18 in $\mathbb{F}_2^5$.